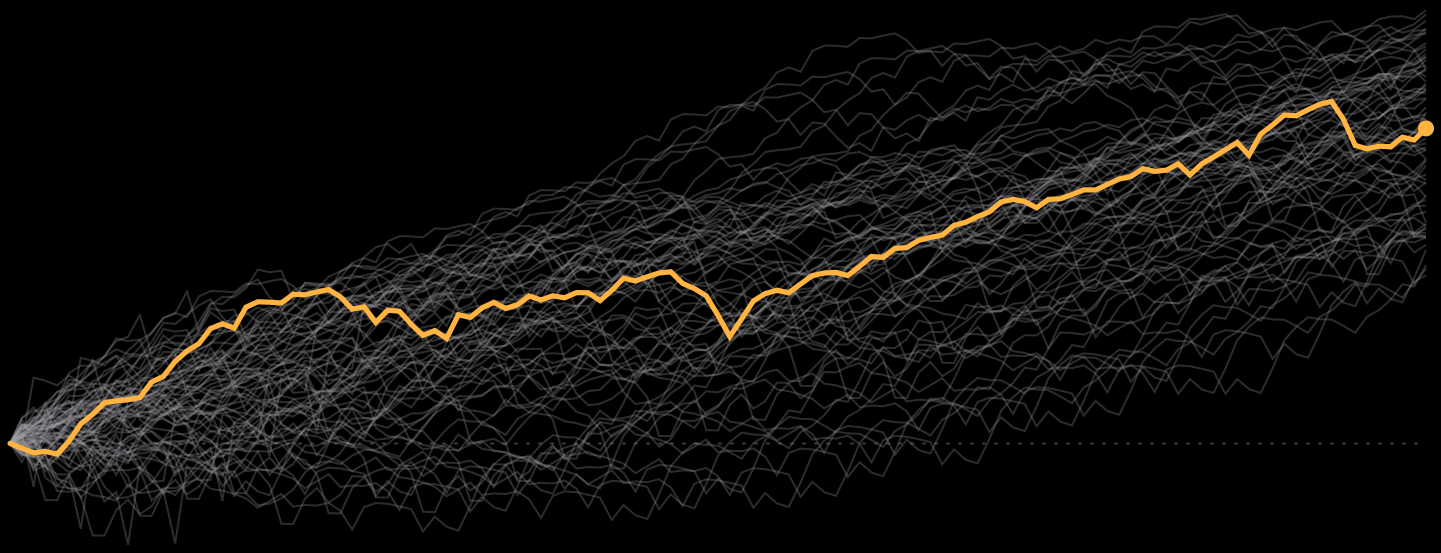
The complete retirement planning toolkit.

A calculator, two guided walkthroughs, fourteen planning tools and 66 lessons, each checked against every market since 1926. Free, and the tools need no account.



Every 30-year stretch since 1926 for a 60/40 portfolio, after inflation. The amber line is the middle one.

CONTENTS

What's inside.

01 The overview 03

- What RetCalc is 04
- How it works 05
- A tool for every stage 06
- Highlights 07
- Every tool at a glance 08
- Learn, and My Dashboard 09

02 Your Dashboard 10

- What the Dashboard is 11
- One profile behind every tab 12
- Setup 13
- Profile and profiles 14
- Dashboard 15
- Plan summary 16
- Net worth 17
- Your home 18
- Income and Spending 19
- Taxes and Savings 20
- Year by year 21
- Calculator: changes ahead 22
- Calculator: the projection 23
- Retirement: drawdown 24
- Retirement: bridge, health cover 25
- Tools 26
- Next steps, Learn and About 27
- The printable report 28
- The report, sheet by sheet 29
- Assumptions, and help 30
- Check-in email and privacy 31
- Phones and tablets 32

03 The deep dive 33

- Basic and Advanced 34
- Advanced: Changes ahead 35
- Drawdown Simulator 36
- Retirement Plan Optimizer 37
- Social Security Calculator 38
- Early Retirement Bridge 39
- Roth Conversion Calculator 40
- Healthcare Cost Planner 41
- FIRE Calculator 42
- Portfolio Backtest 43
- College Savings 44
- Rent vs. Buy 45
- Income Tax 46
- Budget 47
- Debt Payoff 48
- Mortgage Calculator 49
- Retirement Readiness Guide 50
- The plan, on paper 51
- Paycheck to Retirement 52
- Learn 53
- One household, start to finish 54
- From claiming to drawdown 55

04 Under the hood 56

- The stack 57
- The calculation engine 58
- Hosting and delivery 59
- Accounts and privacy 60
- Where the numbers come from 61

Calculators · Retirement income · Saving and investing · Everyday money · Guided walkthroughs · Learn · Putting it together

PART 1

The overview.

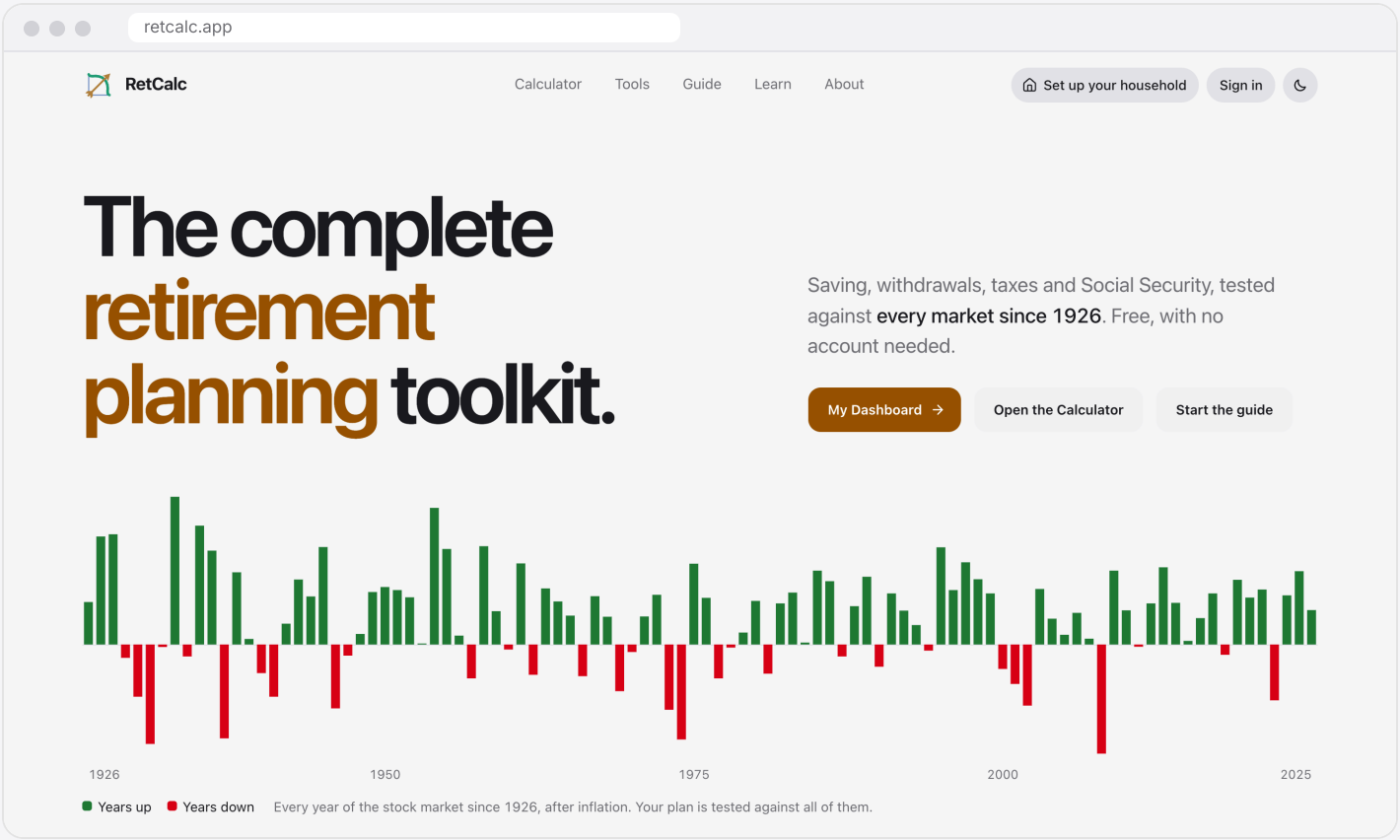
Free, private, and tested against a hundred years of real markets.

What RetCalc is	04	Highlights	07
How it works	05	Every tool at a glance	08
A tool for every stage	06	Learn, and My Dashboard	09

WHAT IT IS

A whole retirement plan, worked out in your own browser.

RetCalc is a free retirement planning toolkit at **retcalc.app**. It takes someone from their first paycheck to their last withdrawal: what to save, where to put it, whether it will last, what it will cost in tax and health insurance, and when to claim Social Security. Every projection is tested against real market history, month by month, back to 1926. The tools, the guides and the lessons work without an account; a free account adds My Dashboard, which keeps a household's whole picture in one place.



14 planning tools in three groups	**2** calculators: Basic and Advanced	**2** guided walkthroughs	**66** lessons in 8 tracks	**100** years of monthly market history	**$0** no ads, no paywall

For someone starting out

What a paycheck really takes home, a budget that adds up, and how today's saving grows into a retirement income.

For someone getting close

When to claim, how to cover the years before 59½ and 65, whether to convert to Roth, and what the best plan looks like.

For someone already retired

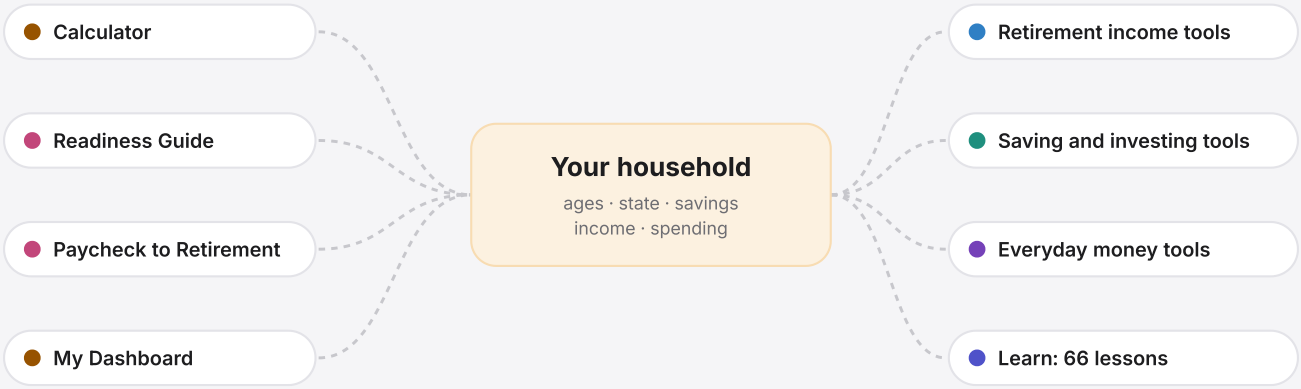
How much is safe to spend, which withdrawal rule to follow, and how the plan would have fared in the worst markets on record.

HOW IT WORKS

Three ideas run through every page.

1 One household, every tool.

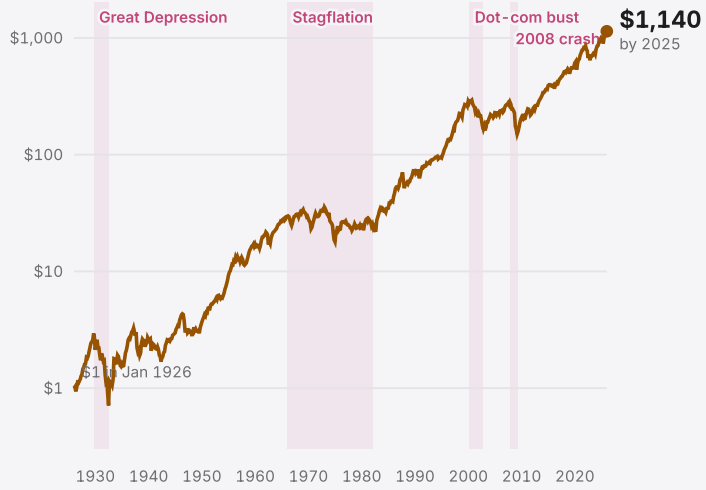
Ages, state, savings, income and spending are entered once and fill in everywhere. Results also carry forward: the Calculator hands its balance to the Drawdown Simulator, Income Tax hands its paycheck to the Budget, and the Budget's savings split goes to the Advanced calculator. Each lesson in Learn ends with the tool that puts it to work.



2 Tested against history, not one average.

A single average return hides the order returns come in, and that order decides whether a retirement lasts. RetCalc replays every real starting month since January 1926: the Depression, 1970s stagflation, 2000 and 2008. Where history runs out, it runs 5,000 Monte Carlo simulations per redraw.

One dollar in US stocks in January 1926 grew to about **$1,140** after inflation by the end of 2025, in the same data the tools use. The shaded years are when it didn't feel that way.



US stocks after inflation, dividends reinvested. Log scale. Shiller (Yale) and BLS CPI-U.

3 Private by default.

Every calculation runs in the visitor's own browser, and nothing they type is sent anywhere. For the tools an account is optional: signing in with a one-time email code or Google keeps a household, saved scenarios, Guide answers and lessons read in step across devices. As the site puts it, *"Nothing leaves your browser unless you sign in to save it."* My Dashboard is the one area that needs a free account, and it too saves to the browser first.

WHERE TO START

A tool for every stage of the road.

The tools follow the order life brings the questions. Most people start with the Calculator or a guided walkthrough, which opens the right tools along the way; the Guide tab offers both.

1 2 3 4

Today

Get the basics right

- Income Tax
- Budget
- Debt Payoff
- Mortgage Calculator
- Rent vs. Buy

Saving years

Grow it

- Calculator
  Basic · Advanced
- FIRE Calculator
- Portfolio Backtest
- College Savings

Getting close

Plan the handoff

- Social Security Calculator
- Healthcare Cost Planner
- Early Retirement Bridge
- Roth Conversion Calculator
- Retirement Plan Optimizer

Retired

Make it last

- Drawdown Simulator
- Roth Conversion Calculator
- Healthcare Cost Planner

GUIDED WALKTHROUGH

Retirement Readiness Guide

"Am I on track?" A ten-minute Quick check or a full walkthrough that finds the number to aim for, tests it against history, scores readiness out of 100 and ends in a full-color report.

GUIDED WALKTHROUGH

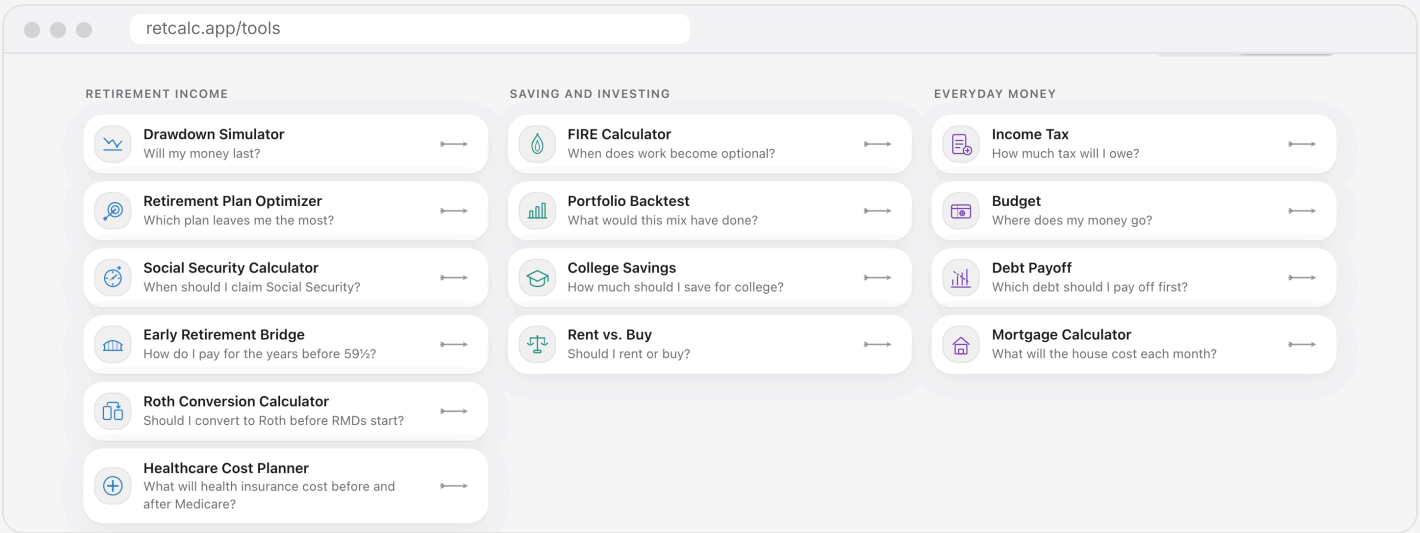
Paycheck to Retirement

"Where does my paycheck go?" Five steps, about fifteen minutes: take-home pay, a budget, where each paycheck goes (401(k), HSA, IRA, brokerage) and where that leads by retirement.

CALCULATOR

Basic and Advanced

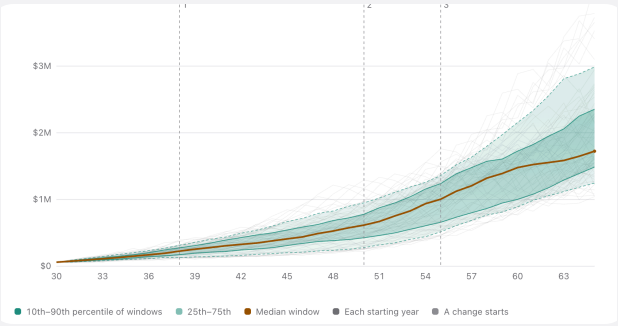
One Calculator with a switch under its title. Basic asks six questions; Advanced adds fees, taxes, account types, a target and the changes you see coming.



The tool list, compact view. All fourteen tools in their three groups, each with the question it answers.

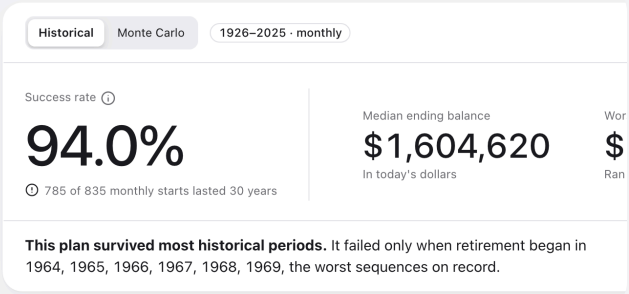
HIGHLIGHTS

Six places to see what it can do.



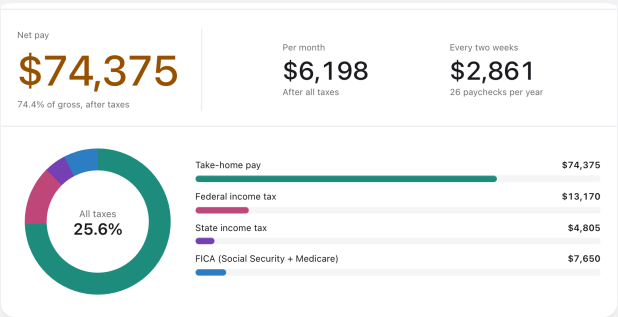
Advanced calculator: Changes ahead

Add what you see coming, such as children, a raise or an inheritance, and see what each does to the balance at retirement. Then two answers: when you could stop saving, and when you could retire.



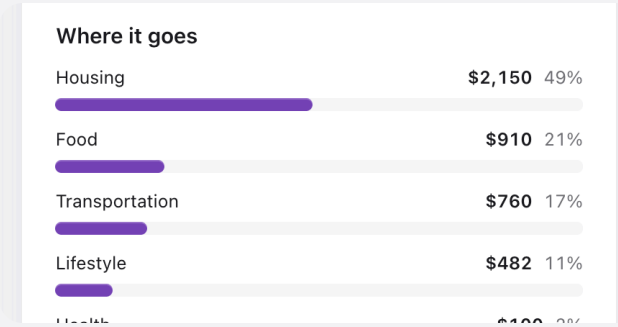
Drawdown Simulator

Fifteen withdrawal strategies replayed through every retirement since 1926, then ranked against each other at the same level of risk.



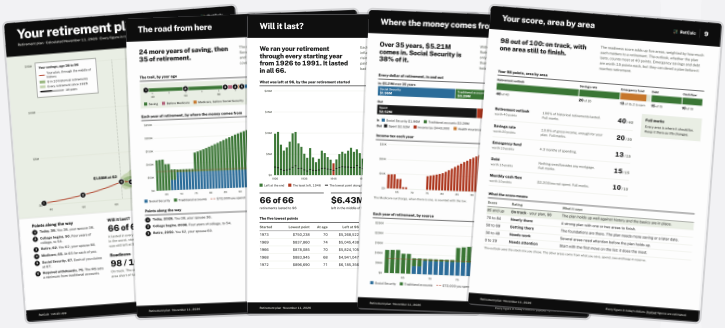
Income Tax

2026 federal, state (all 50 plus DC) and FICA tax on a salary, or a retirement mode that stacks withdrawals, gains and Social Security the way the return really does.



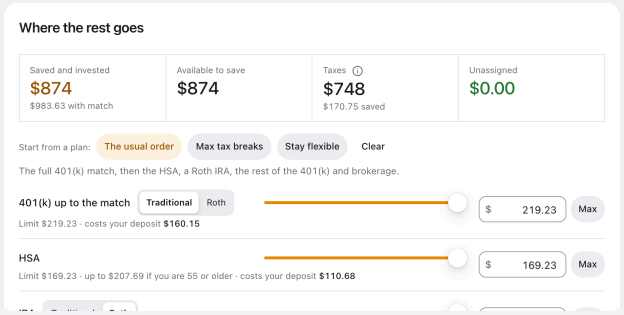
Budget and Paycheck

A budget, then a paycheck that pays it: taxes rerun live as sliders move money into the 401(k) match, HSA, IRA and brokerage, in the order you choose.



Retirement Readiness Guide

A check-up that scores readiness out of 100, tests the plan against history and prints a report of about a dozen pages.



Paycheck to Retirement

Gross pay to retirement in five steps, running the real tools inside it and copying the results into them at the end.

AT A GLANCE

Every tool, and the question it answers.

CALCULATORS

Calculator: Basic	What will my savings grow to?
Calculator: Advanced	Will I reach my target, with the changes I see coming?

RETIREMENT INCOME

Drawdown Simulator	Will my money last?
Retirement Plan Optimizer	Which plan leaves me the most?
Social Security Calculator	When should I claim Social Security?
Early Retirement Bridge	How do I pay for the years before 59½?
Roth Conversion Calculator	Should I convert to Roth before RMDs start?
Healthcare Cost Planner	What will health insurance cost before and after Medicare?

SAVING AND INVESTING

FIRE Calculator	When does work become optional?
Portfolio Backtest	What would this mix have done?
College Savings	How much should I save for college?
Rent vs. Buy	Should I rent or buy?

EVERYDAY MONEY

Income Tax	How much tax will I owe?
Budget	Where does my money go?
Debt Payoff	Which debt should I pay off first?
Mortgage Calculator	What will the house cost each month?

GUIDED WALKTHROUGHS

Retirement Readiness Guide	How ready am I, and what should I do next?
Paycheck to Retirement	Where should each paycheck go?

LEARN AND MY DASHBOARD

Learn	How does a retirement plan work?
My Dashboard	Where does my whole plan stand today?

ALSO ON THE SITE

66 lessons in Learn, and 28 plain-English articles whose worked examples the test suite checks against the calculator, so the numbers in the writing match the numbers in the tools.

TWO MORE WAYS IN

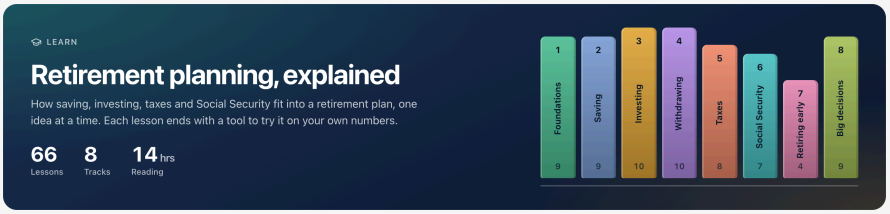
Learn, and My Dashboard.

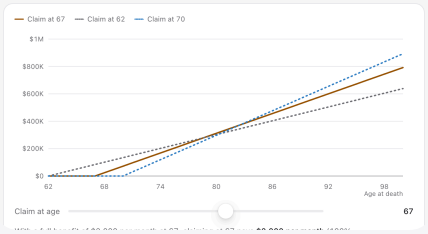
LEARN · RETCALC.APP/LEARN

Retirement planning, explained.

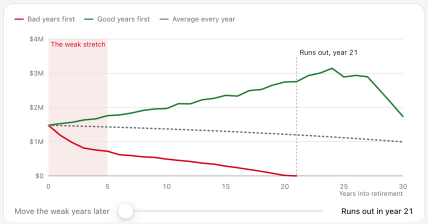
66 lessons in 8 tracks, from Foundations to Big decisions, about 14 hours of reading. 39 of them have a chart to try: move the claiming age, or put the bad years first, and watch what changes.

Each lesson ends with the tool that puts it to work on your own numbers. Lessons can be marked as read, and a signed-in reader's progress follows them to every device.





When to claim. Total collected by age at death, claiming at 62, 67 or 70.



Sequence risk. The same returns in a different order, bad years first or last.

66 lessons	**8** tracks, Foundations to Big decisions	**39** charts to try	**~14 hrs** of reading in all

MY DASHBOARD · RETCALC.APP/DASHBOARD

The whole plan, kept in one place.

A free account adds a personal hub: net worth, pay, taxes, spending, saving and retirement, all worked out from one profile and revisited each month or quarter to update balances.

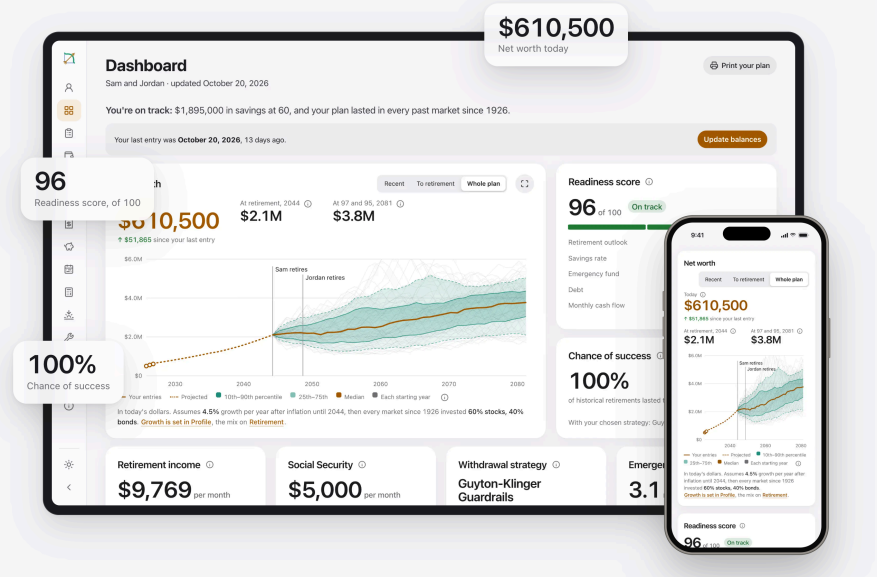
Fifteen tabs, with the site's own tools running inside it and your numbers filled in. No bank logins: you type your own numbers.

PART 2

The Dashboard, tab by tab, starts on page 10.

Your whole plan, in one place.

A free account keeps your accounts, pay, spending and retirement together. No bank logins: you type your own numbers.



The Dashboard on the homepage, with the made-up household of Sam and Jordan.

15 tabs, one profile behind them	**14** tools inside it, with your numbers filled in	**$0** a free account, no ads	**0** bank logins: you type your own numbers

PART 2

Your Dashboard.

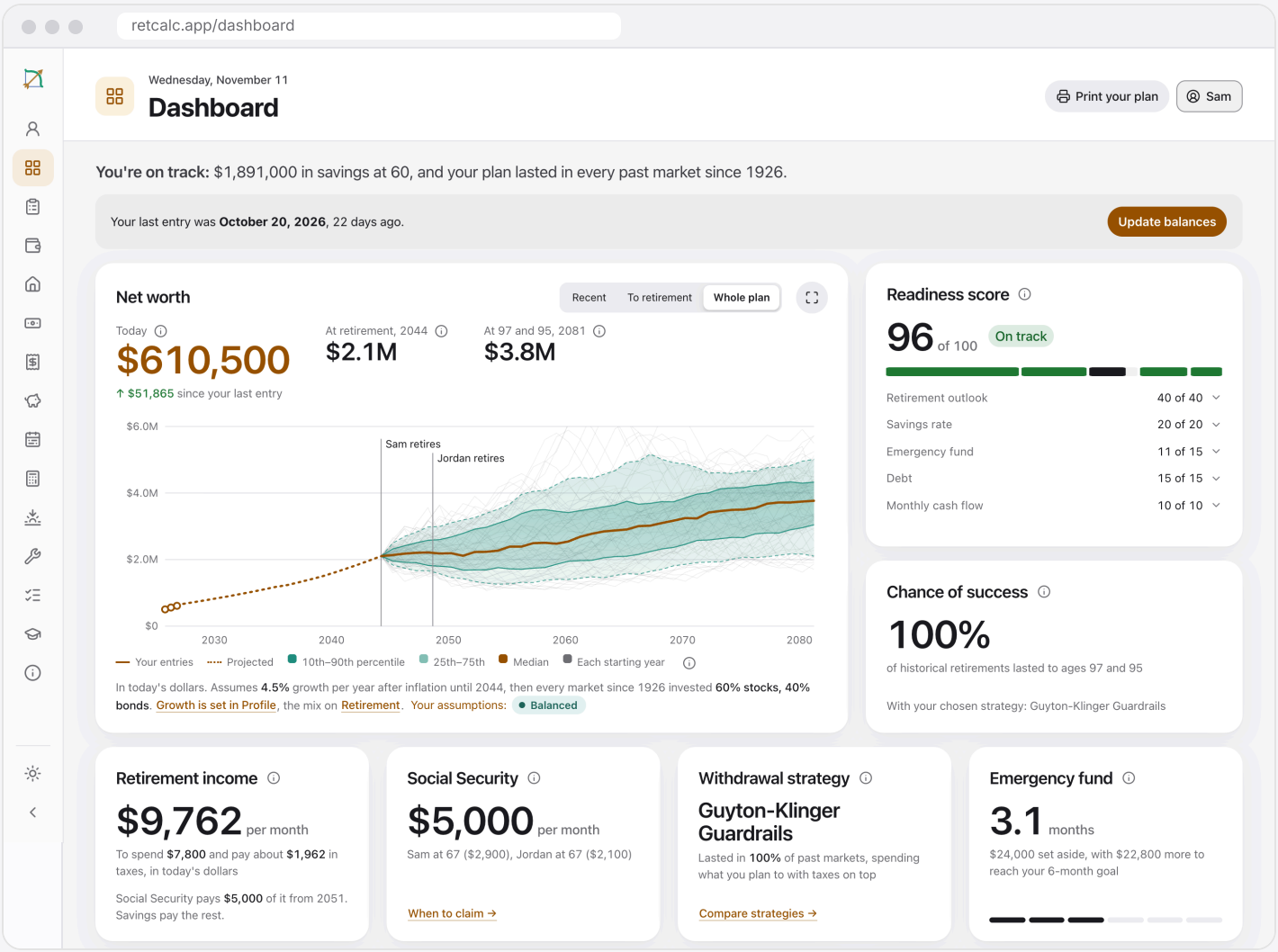
One free account keeps a household's whole picture and turns it into one retirement plan: fifteen tabs, one profile behind all of them.

What the Dashboard is	11
One profile behind every tab	12
Setup	13
Profile and profiles	14
Dashboard	15
Plan summary	16
Net worth	17
Your home	18
Income and Spending	19
Taxes and Savings	20
Year by year	21
Calculator: changes ahead	22
Calculator: the projection	23
Retirement: drawdown	24
Retirement: bridge, health cover	25
Tools	26
Next steps, Learn and About	27
The printable report	28
The report, sheet by sheet	29
Assumptions, and help	30
Check-in email and privacy	31
Phones and tablets	32

MY DASHBOARD

A household's whole financial picture, in one place.

The Dashboard is a personal hub at **retcalc.app/dashboard**. A household enters its numbers once, and fifteen tabs show net worth, pay, taxes, spending, saving and a retirement plan, all worked out from that one profile. It is made to be opened each month or quarter, to update the balances and see where the plan stands.



The Dashboard tab for the sample household: Sam, 42, and Jordan, 40, in Illinois, with their daughter Nora. Unless a caption says otherwise, every screenshot here is theirs.

15 tabs, all worked out from one profile	**14** planning tools that open inside it, filled in	**8** short setup steps at most, each one skippable	**1926** every month since is a test of the plan	**$0** a free account: no bank logins, no ads

Keeping track

Net worth by date, a budget, the paycheck after taxes and where each one goes, updated whenever the balances change.

Planning retirement

Savings at retirement, the chance of success, Social Security, the years before 59½ and 65, and changes ahead.

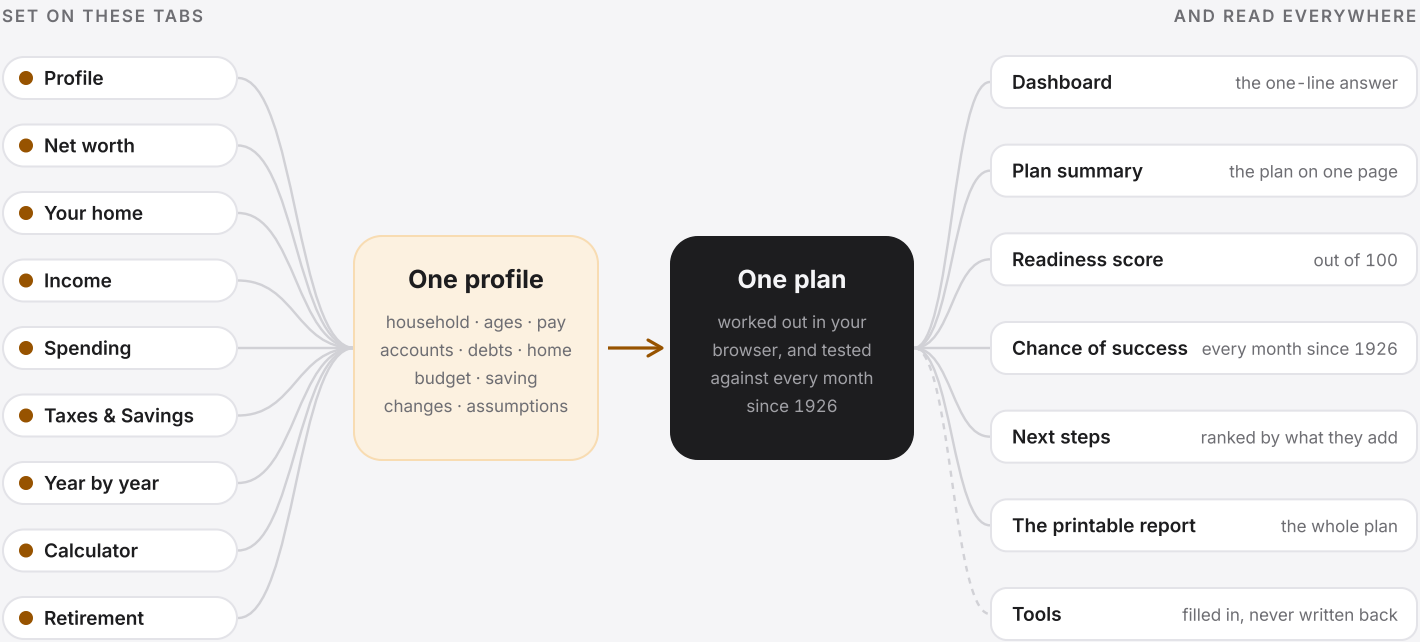
Two of you

Two incomes and two retirement ages, each person's accounts and benefit, and a filter on net worth for whose is whose.

HOW IT WORKS

One profile behind every tab.

Each figure is set in one place. Every other tab lists it as a fact, with a link to where it is set, so the tabs can't disagree. Change a retirement age on Profile and the savings at retirement, the chance of success, the readiness score, the net worth chart and the report all move with it.



A FREE ACCOUNT

Sign in once, on any device

The Dashboard needs an account: sign in with Google, or with a six-digit code sent by email. There are no passwords and no payments. Nothing is linked to a bank; a household types its own numbers. One account can hold several profiles, a couple's plan and a solo one, say.

WHAT'S STORED, AND WHERE

Saved to the browser, synced to the account

Every change is saved in the browser first, then synced so the Dashboard appears wherever its owner signs in, and only its owner can read it. The server stores it but never calculates with it: every figure is worked out in the browser.

WHAT THE ACCOUNT KEEPS

- Each profile, with every tab's choices
- Dated net worth entries
- Lessons read
- The email address, an optional first name and the check-in setting

Signing out removes the synced copy from that browser; the account keeps it. **Download my data** and **Delete my account** are on the account page.

FIFTEEN TABS, IN THE ORDER OF THE RAIL

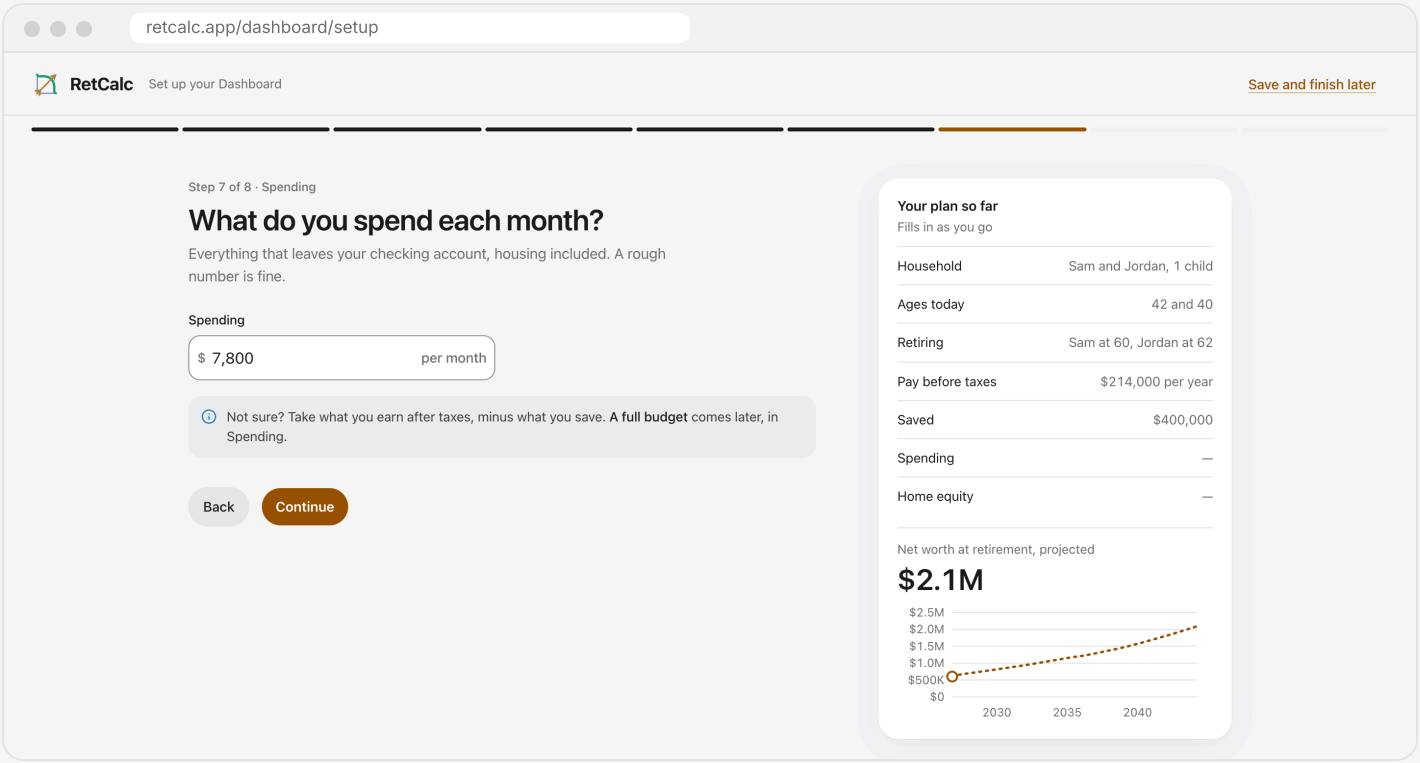
01 Profile	02 Dashboard	03 Plan summary
04 Net worth	05 Your home	06 Income
07 Spending	08 Taxes & Savings	09 Year by year
10 Calculator	11 Retirement	12 Tools
13 Next steps	14 Learn	15 About

A rail of icons runs down the left of every tab, with a dot on any tab that still has something to fill in. On a phone it becomes a tab bar along the bottom.

GETTING STARTED

Setup: up to eight short steps.

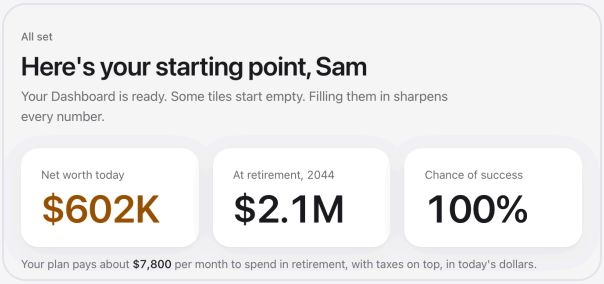
A new Dashboard starts with a guided setup. Each step asks one thing, saves the answer as it is typed and can be skipped. Beside the questions, **Your plan so far** fills in, and once savings and a retirement age are in it draws the projection.



Step 7 of 8, Spending. The household, ages, retirement ages, pay and savings are in; the projection already reads $2.1M at retirement.

#	STEP	WHAT IT ASKS
1	Household	Just me, or me and a partner; children who depend on you, and any planned
2	About you	First name, birth month and year for each person, so ages stay right
3	Children	An optional name and birth month and year for each child
4	Retirement	The age each person hopes to retire
5	Pay	Where you live, and each person's pay, before or after taxes
6	Savings	What's saved all together, and what goes in
7	Spending	One number per month; a full budget comes later
8	Home	Own (worth, left on the mortgage) or rent (rent per month)

Children drops out for a household without children. **Save and finish later** counts the steps not yet reached as skipped.



The finish screen. Net worth today, savings at retirement and the chance of success; the plan pays about $7,800 per month to spend in retirement, with taxes on top.

THEN

Open my Dashboard lands on the Dashboard tab, which welcomes a new user with a *Profile N of 10* bar. Tiles with nothing to show yet are dashed, each offering the one action that fills it.

SKIPPED A STEP?

Profile shows a dot and Next steps lists *Finish setting up* until every step is answered.

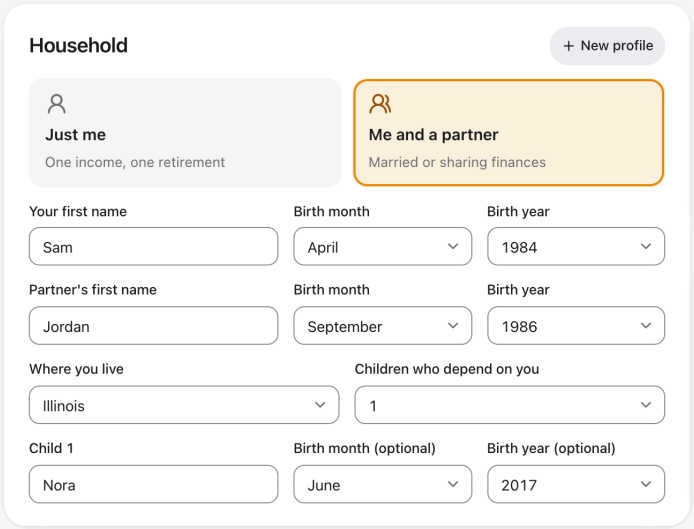
Profile

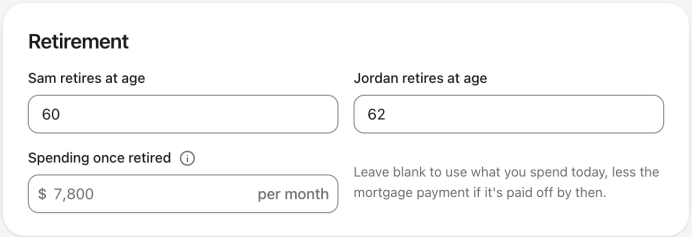
What does the plan know about us?

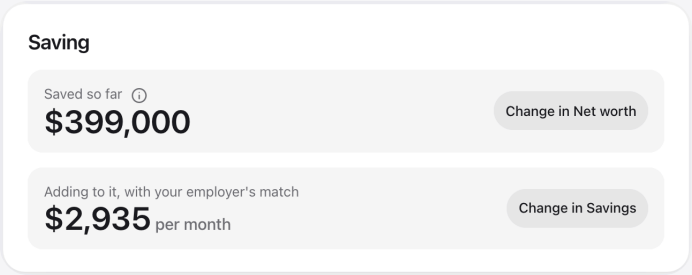
Every setup answer on one page, editable where it stands, plus the assumptions every figure uses. Edits save as they are made; a bar at the top says what changed and what it did to the plan, with Keep and Undo.

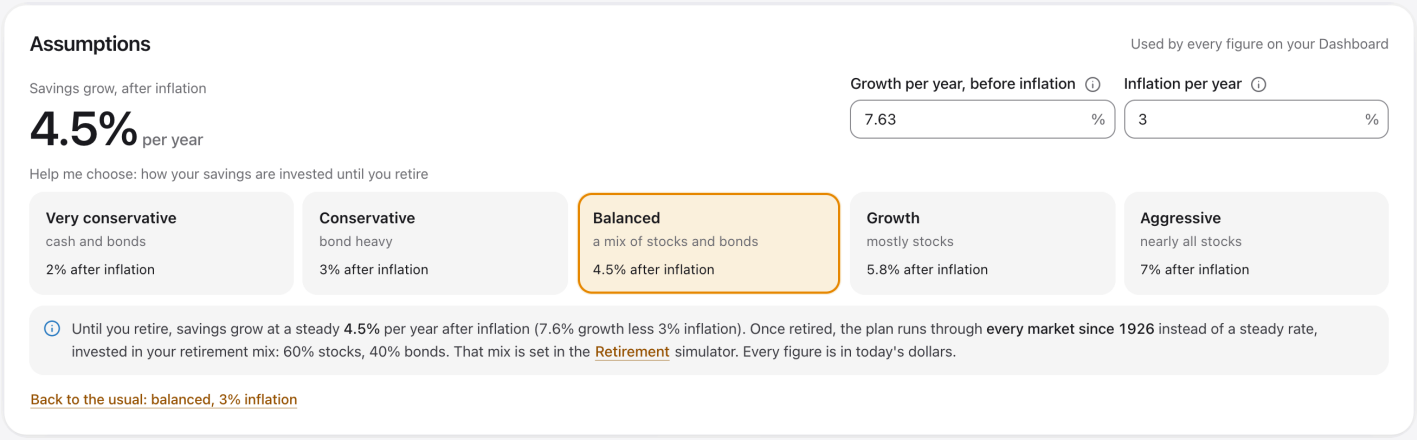
01

retcalc.app/dashboard/profile









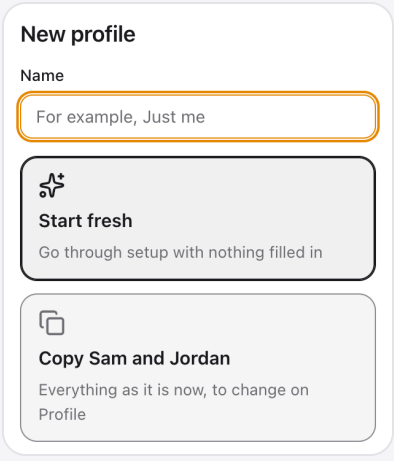
Assumptions: savings grow 4.5% per year after inflation until retirement; five presets run from 2% to 7%. Once retired, the plan runs through every market since 1926.

ALSO ON PROFILE

- **Pay**, and **Home**: own or rent
- **Children ahead:** children planned, childcare, help with college and the kind of school
- **Email:** the monthly check-in, on or off
- **Reset**, asked twice; the rest of RetCalc is untouched

PROFILES

One account can hold several plans: a couple's and a solo one, or a plan and a what-if. **New profile** starts fresh, through setup, or as a copy of the open one. Each can be renamed or deleted, and the one that's open is remembered on each device.



Dashboard

Where do we stand today?

The home tab answers in one line, then shows the pieces: net worth across the whole plan, the readiness score, the chance of success, retirement income, Social Security, the withdrawal strategy, the emergency fund, the next steps and a timeline.

02

retcalc.app/dashboard

You're on track: $1,891,000 in savings at 60, and your plan lasted in every past market since 1926.

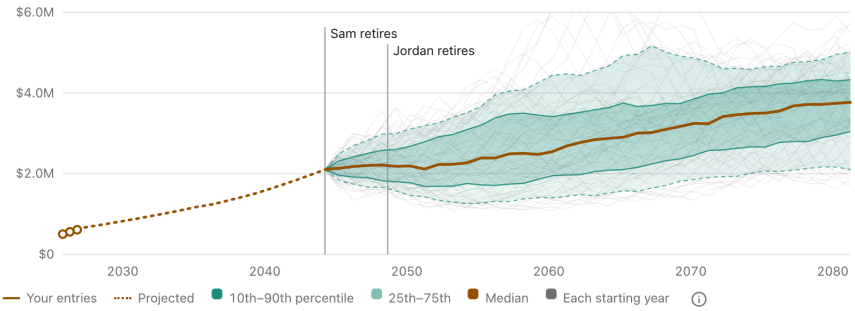
Net worth

Recent | To retirement | Whole plan

Today
$610,500
↑ $51,865 since your last entry

At retirement, 2044
$2.1M

At 97 and 95, 2081
$3.8M



Your entries · Projected · 10th–90th percentile · 25th–75th · Median · Each starting year

In today's dollars. Assumes **4.5%** growth per year after inflation until 2044, then every market since 1926 invested **60% stocks, 40% bonds**. Growth is set in Profile, the mix on Retirement. Your assumptions: Balanced

Readiness score

96 of 100 On track

Retirement outlook	40 of 40
Savings rate	20 of 20
Emergency fund	11 of 15
Debt	15 of 15
Monthly cash flow	10 of 10

Chance of success

100%

of historical retirements lasted to ages 97 and 95

With your chosen strategy: Guyton-Klinger Guardrails

Retirement income

$9,762 per month

To spend **$7,800** and pay about **$1,962** in taxes, in today's dollars

Social Security pays **$5,000** of it from 2051. Savings pay the rest.

Social Security

$5,000 per month

Sam at 67 ($2,900), Jordan at 67 ($2,100)

When to claim →

Withdrawal strategy

Guyton-Klinger Guardrails

Lasted in **100%** of past markets, spending what you plan to with taxes on top

Compare strategies →

Emergency fund

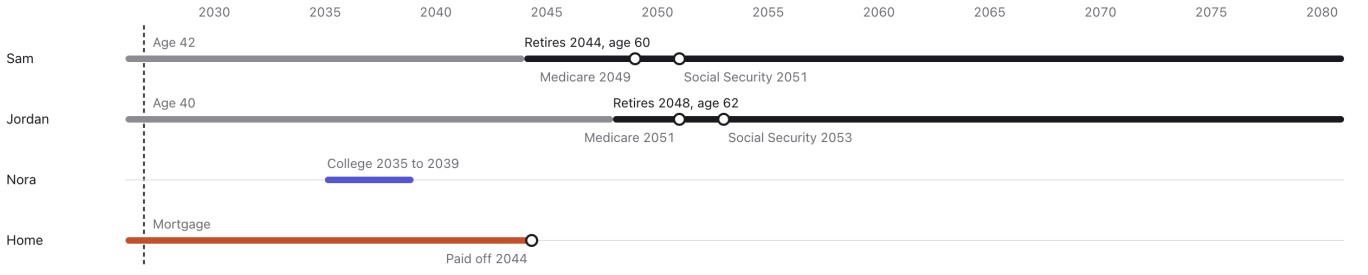
3.1 months

$24,000 set aside, with $22,800 more to reach your 6-month goal

Net worth today, $610,500, up $51,865 since the last entry; $2.1M at retirement in 2044. After retirement the chart is the Retirement tab's own run through history: the median, the 25th to 75th and 10th to 90th percentiles, and a faint line for each starting year. Readiness: 96 of 100, on track. A tile with nothing to show yet is dashed and offers the one action that fills it.

Your timeline

Today is the dashed line



Your timeline. A lane for each person (work, retirement, Medicare, Social Security), one for Nora's college years and one for the mortgage. Today is the dashed line.

Plan summary

What does the whole plan come to?

The plan on one page: savings at retirement, what it pays and the tax on top, where each year's pay goes, health cover before Medicare, the budget and net worth. Folds open the climb to retirement, what the children cost, the readiness score and the timeline.

03

retcalc.app/dashboard/summary

Your plan

Savings at retirement, 2044	You plan to spend	Taxes, on top	Social Security, from 2051	Chance of success
$1,891,000	$7,800 per month	$1,962 per month	$5,000 per month	100%

Your plan lasted to ages 97 and 95 in **every past market since 1926**, drawing down with **Guyton-Klinger Guardrails**. Social Security pays part of your spending and taxes once it starts, and savings pay the rest. In today's dollars.

Where your pay goes

$214,000 of gross pay per year. Your employer adds $6,420 on top.

Year | Month

Taxes	$48,333	Spending	$93,600	Saving and investing	$28,800	Left over	$43,267
Federal income tax	$23,084	Custom	$67,800	401(k) up to the match	$12,840	Not yet assigned	$43,267
Illinois income tax	$8,878	Housing	$25,800	401(k) beyond the match	$15,960		
Social Security and Medicare	$16,371						

Where your pay goes: $214,000 of pay per year: $48,333 in taxes, $93,600 of spending, $28,800 saved and invested, and $43,267 not yet assigned. The employer adds $6,420.

Health cover before Medicare

Plan health cover

With no job's plan	Your plan counts
3 years	$1,212 per month
2048 to 2051	for marketplace cover, on top of your spending

You both stop work by 2048. Sam reaches Medicare in 2049 and Jordan in 2051: 3 years of health cover to buy yourselves. Until Medicare starts, cover is a marketplace plan priced on your income, and it's often one of the biggest costs of retiring early.

Your budget

$7,800 per month

Custom	$5,650	72%
Housing	$2,150	28%

Emergency fund goal: **6 months, $46,800**. You have $24,000.

Net worth today

$610,500

Assets	Debts
$943,000	$332,500

Retirement accounts · Investments · Cash · Home · Other · Mortgage · Loans · Credit cards

PRINT YOUR PLAN

The button on this tab and the Dashboard tab builds the printable report of the whole plan (see *The printable report*).

HOW YOUR ASSUMPTIONS COMPARE

A fold on this page rates the plan's guesses against balanced ones.

Readiness score
96 of 100 Show

A fold, closed. Saving to retirement, the readiness score and the timeline each open in place.

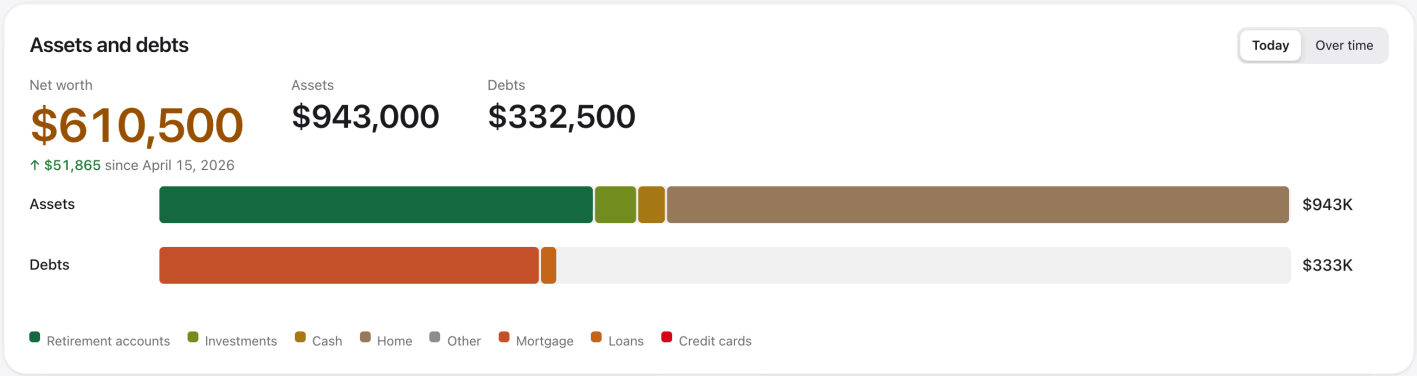
Net worth

What do we own and owe, and how is it changing?

Balances are typed in, never linked to a bank, and saved as dated entries: each **Update balances** adds a point to the net worth line. The plan starts from the latest one.

04

retcalc.app/dashboard/accounts



When you can spend it

Can be spent today · 21%
$90,000

Opens later
$333,000

When, and what opens	Amount
Now Roth contributions, Brokerage, Cash	$90,000
Sam turns 59½ · 2043 401(k) or 403(b), Roth growth	$223,000
Jordan turns 59½ · 2046 401(k) or 403(b)	$98,000
Sam turns 65 · 2049 HSA	$12,000

Today's balances, before any growth. An HSA pays medical costs tax-free at any age. Your home ($520,000) isn't counted here.

How it's taxed

Taxed when spent · $318,000
75%

Comes out tax-free · $105,000
25%

Most of your savings will be taxed when you spend it. Most of what's taxed is pre-tax money, taxed as income. Roth savings, or Roth conversions in low-income years, spread that tax out.

Pre-tax Taxed as income when you take it out	$310,000	73%
Roth Tax-free in retirement	$41,000	10%
Brokerage Only the gain is taxed, at capital-gains rates	$36,000	9%
HSA Tax-free for medical costs	$12,000	3%
Cash Already taxed	$24,000	6%

Of the brokerage's $36,000, $8,000 is gain: the only part taxed when you sell.

When you can spend it, and how it's taxed: $90,000 can be spent today without the 10% penalty, the rest at each owner's 59½ or, for the HSA, 65. 75% will be taxed when spent.

Assets

$943,000

Name	Type	Amount
Sam's 401(k)	401(k) or 403(b) · Sam	$212,000
Jordan's 403(b)	401(k) or 403(b) · Jordan	$98,000
Roth IRA	Roth IRA or Roth 401(k) · Sam	$41,000 (+$11,000)
Brokerage	Brokerage · Shared	$36,000 (+$8,000)
HSA	HSA · Sam	$12,000
Emergency savings	Checking or savings · Shared	$24,000
Home	Home · Shared	$520,000
Total		**$943,000**

IT TRACKS

- Whose each account is, with a filter for everyone, either of a couple, or what's shared
- Nine kinds of assets, from a 401(k) or 457(b) to a home or a car
- Five kinds of debt, each with its rate
- A brokerage cost basis and Roth contributions, for the tax

OVER TIME

A switch turns the bars into the net worth line, with milestones and the growth per year. Entries download as CSV.

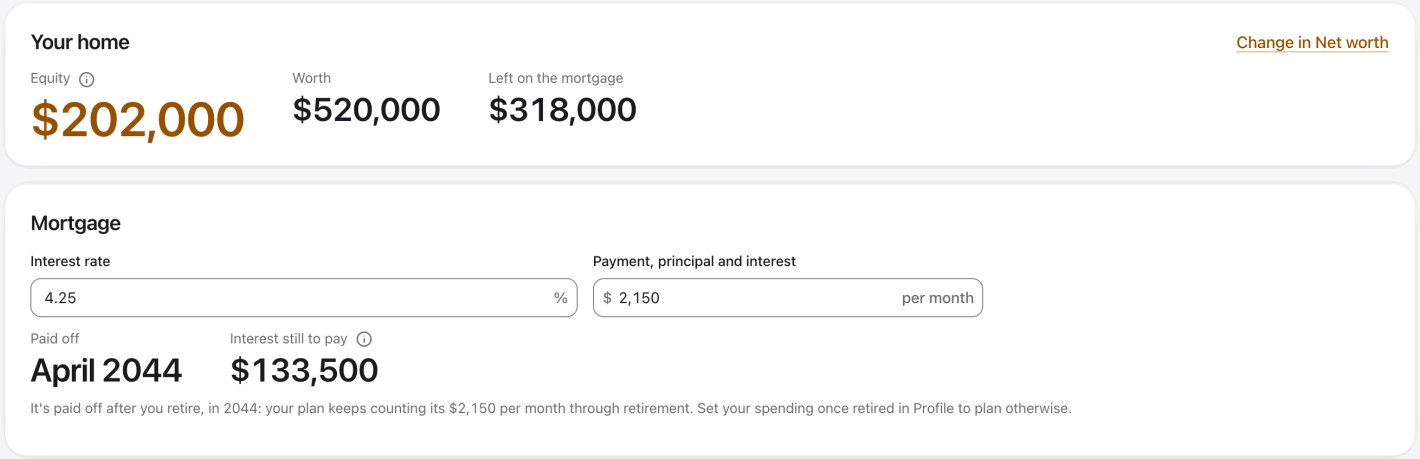
Your home

What does the house really cost?

For owners: the equity, when the mortgage is paid off and the interest still to pay, then the Mortgage Calculator run on their own loan, where property tax, insurance, upkeep and utilities become the budget's housing lines. Renters get their rent and the Rent vs. Buy Calculator, seeded with it.

05

retcalc.app/dashboard/home



The mortgage: $318,000 left at 4.25%, $2,150 per month, paid off in April 2044 with $133,500 of interest still to pay. It ends after Sam retires, so the plan keeps counting the payment through retirement.

WHAT OWNING COSTS

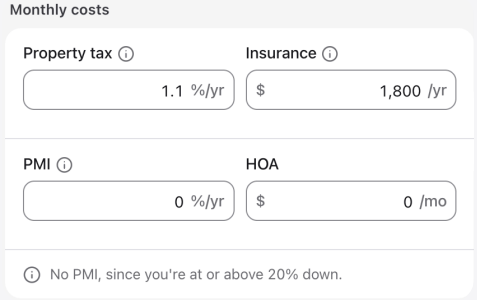
The Mortgage Calculator with the loan locked in from Net worth and the mortgage above. What is set here (property tax, insurance, PMI, HOA, upkeep, utilities) becomes the housing lines on Spending.

TRY THINGS

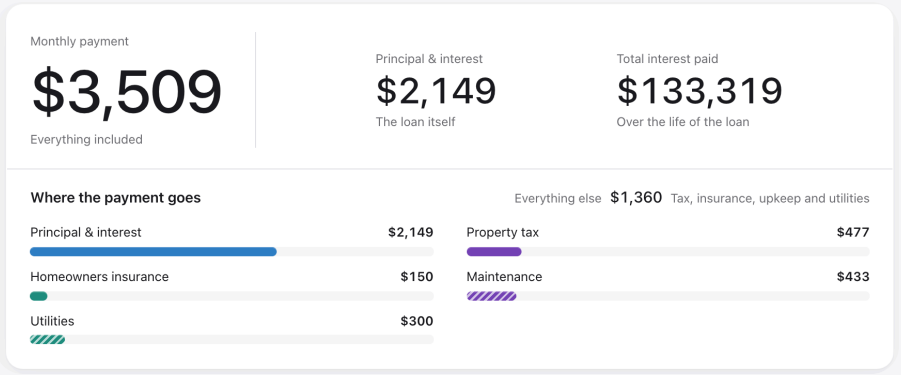
- Extra payments, each month or once
- Recasting after a lump sum
- A refinance, with its before and after

RENTING INSTEAD

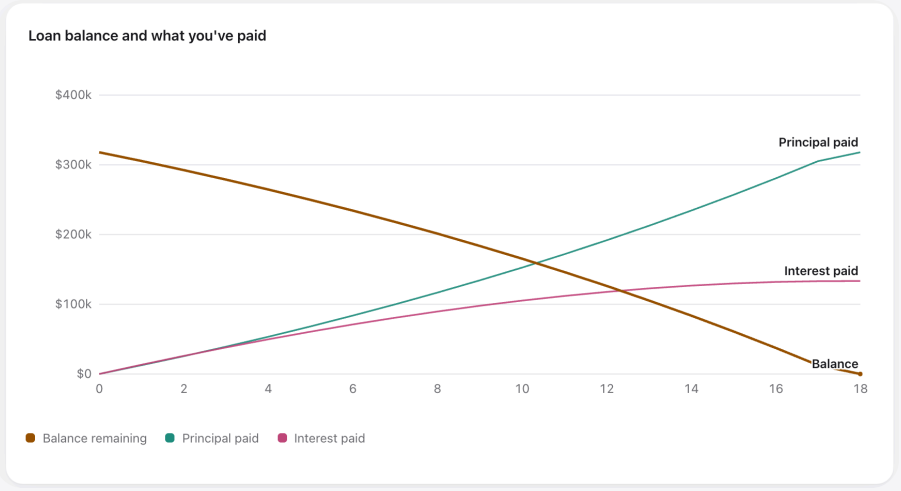
Rent and renter's insurance per month, both budget lines, and Rent vs. Buy on your own rent.



Monthly costs, beside the loan.



$3,509 per month, everything included: $2,149 for the loan itself and $1,360 for tax, insurance, upkeep and utilities.

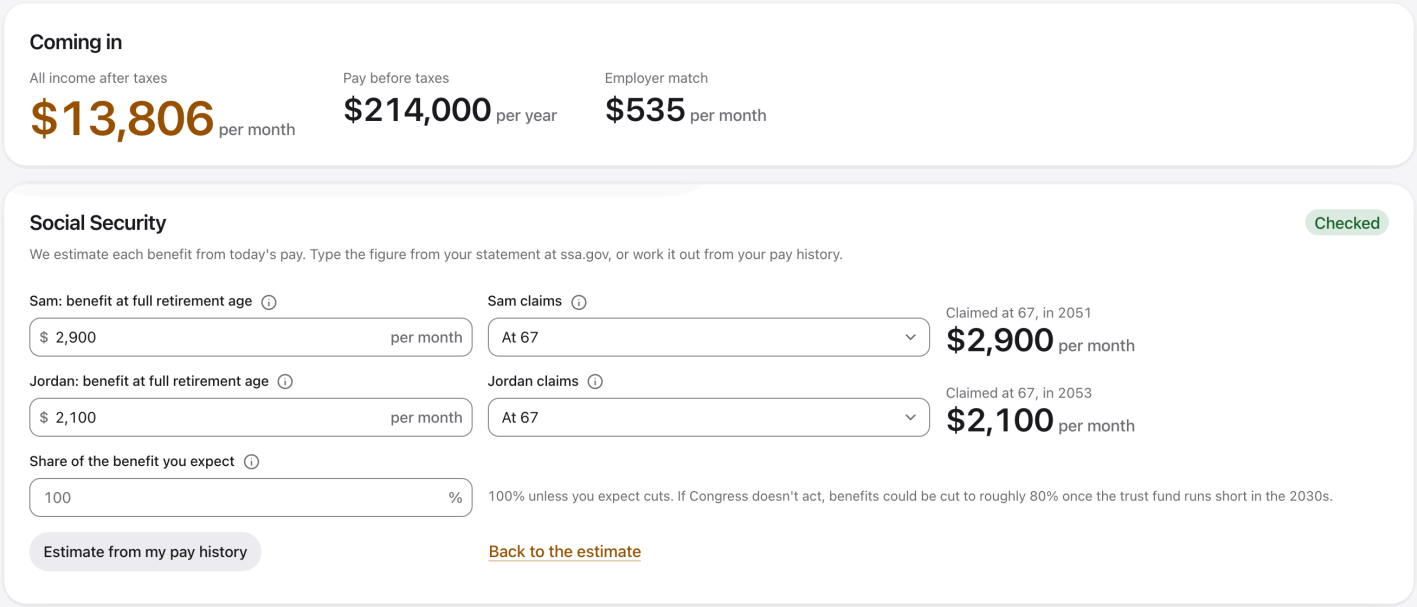


Loan balance and what you've paid, year by year to April 2044.

06 Income

What comes in, and what will Social Security pay?

Each person's pay, the employer's 401(k) match, any other income, and Social Security: a benefit typed from the ssa.gov statement, estimated from today's pay or worked out from a pay history, with the age each of you claims and the share you expect.

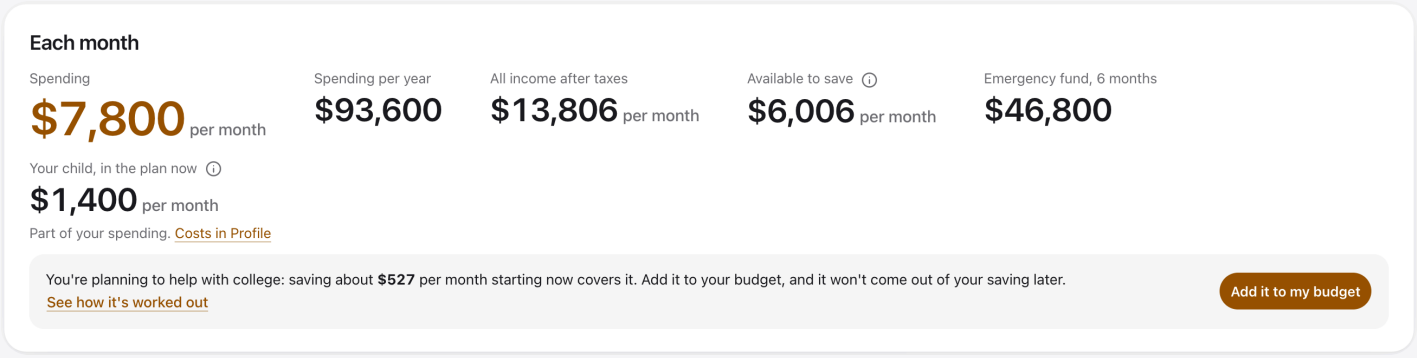


Social Security, checked against a statement: Sam $2,900 per month claimed at 67 in 2051, Jordan $2,100 at 67 in 2053. Claim ages are shared with Retirement, and the share expected scales every benefit in the plan.

07 Spending

What do we spend, and what's left to save?

The Budget tool on the Dashboard's numbers: its total is spending on every tab. Setup's one number starts as a single line; housing lines come from Your home. It also sizes the emergency fund and can build a leaner emergency budget beside the normal one.



Each month: $7,800 of spending against $13,806 of income after taxes leaves $6,006 per month available to save. Nora's costs, $1,400 per month now, are part of it, and saving $527 per month from today would cover the share of college they plan to help with.

EMPLOYER MATCH

Each employer's match and the share of pay it covers, with a line saying whether the household saves enough to get all of it. Both of theirs: 50% of the first 6%.

YOUR BUDGET

Lines in groups (housing, transportation, health, food, lifestyle, debt), each per month or per year, with CSV. Housing lines from Your home are locked and marked.

EMERGENCY FUND

Six months of spending is typical: $46,800 here. A leaner emergency budget can sit beside the normal one, and the fund can be sized on either.

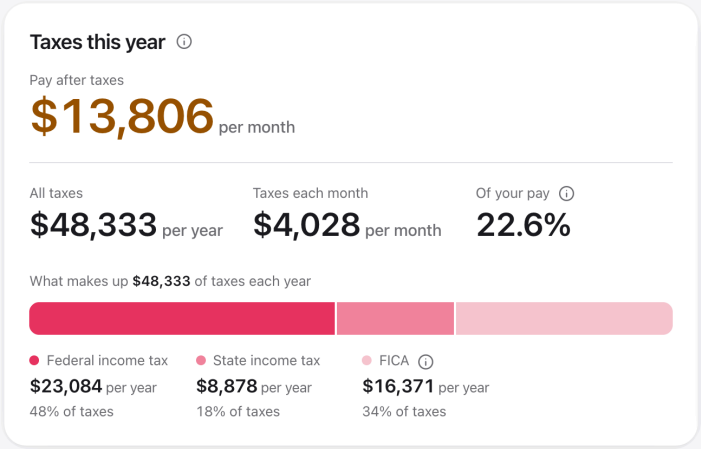
Taxes & Savings

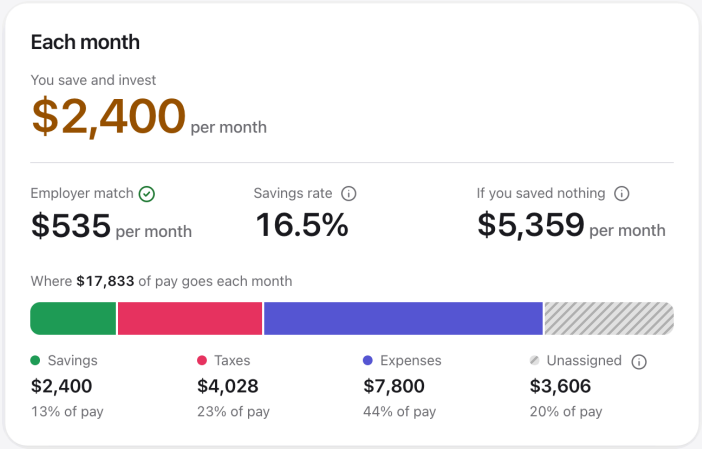
What do we pay in tax, and where should each paycheck go?

The paycheck, worked out line by line with 2026 federal, state and FICA tax, is the household's one tax calculator: its pay after taxes is the figure every tab uses. Below it, each dollar left after spending gets a place: the 401(k) match first, then an HSA, an IRA, more 401(k) or a brokerage account.

08

retcalc.app/dashboard/savings





Taxes this year, $48,333: $23,084 federal, $8,878 Illinois and $16,371 FICA, 22.6% of pay. They save $2,400 per month, plus $535 from their employers: a savings rate of 16.5%.

START FROM A PLAN

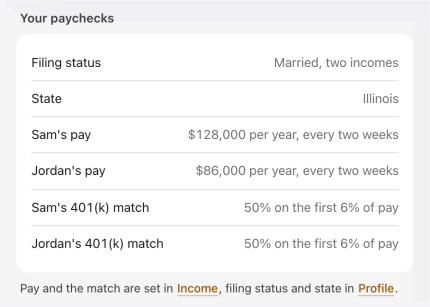
- **The usual order:** the match, an HSA, an IRA, more 401(k), then brokerage
- **Max tax breaks:** pre-tax first
- **Stay flexible:** money reachable before retirement

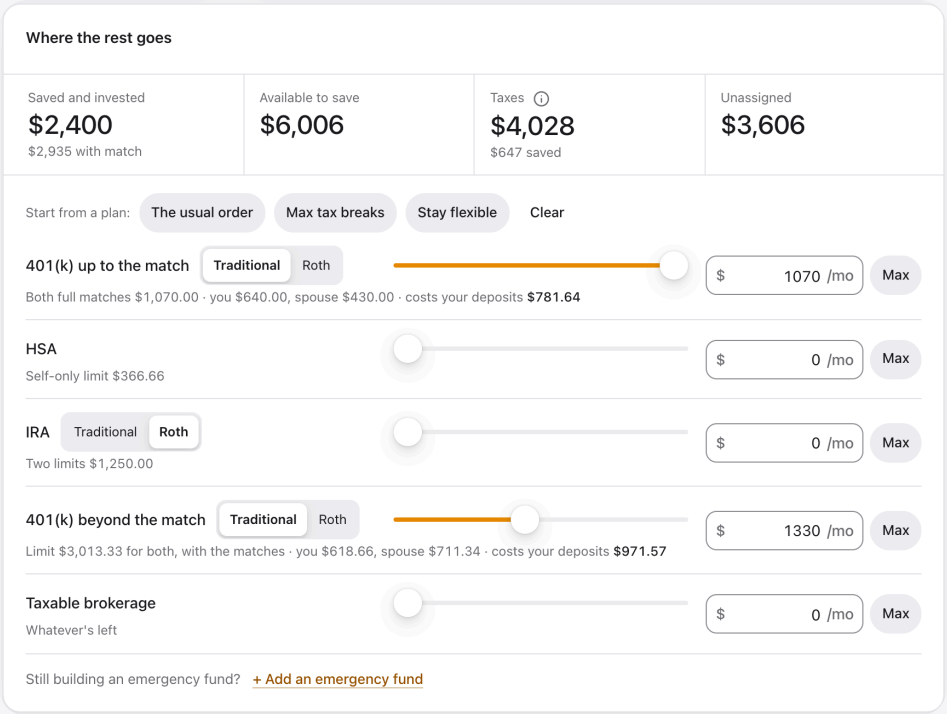
THEN ADJUST

A slider per line, each capped at its 2026 limit, with Max. Pre-tax lines lower the taxes above as they move; with two incomes the lines split between partners.

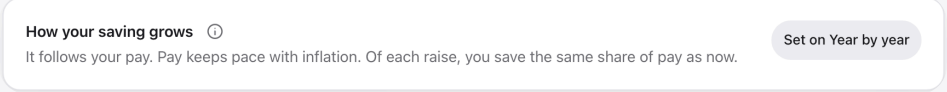
ALSO HERE

The deduction, standard or itemized; a local income tax in the 14 states that have one; an itemized paycheck per check, month and year.





Where the rest goes. Both 401(k)s up to the match and $1,330 per month beyond it; $3,606 per month still unassigned.



Year by year

How do the years to retirement add up?

Every year from now to retirement, worked out from pay with raises, taxes from the paycheck, spending, and saving by account up to each limit, with the children's costs and college taken out of saving in their years. Savings at retirement on every tab are built from these years.

09

retcalc.app/dashboard/years

What shapes the years

Savings at retirement **$1,891,305**

Income this year **$214,000**

Saved this year, with the match **$35,220**

Income in 2043 **$214,000**

Sam's raise per year: 0 %

Jordan's raise per year: 0 %

How much of each raise you save: The same share of pay as now

Raises are beyond inflation. Taxes come from your paycheck's own figures each year, and saving follows take-home pay as set here, each account up to its limit with the rest in the next. What's spent is what's left. Children's costs and college come out of saving, as on Profile.

Changes ahead

None yet. A promotion, part-time years, a bigger house: add one here, or click a pay or spending figure in the table. + Add a change

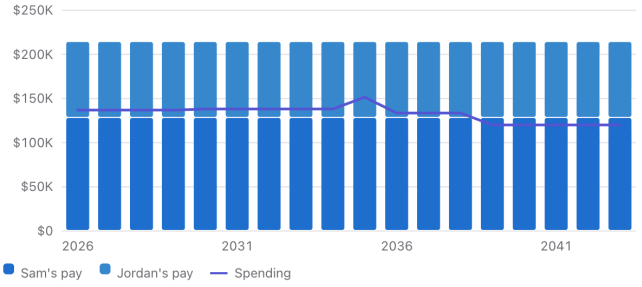
Every year to retirement

Today's dollars | Future dollars | CSV

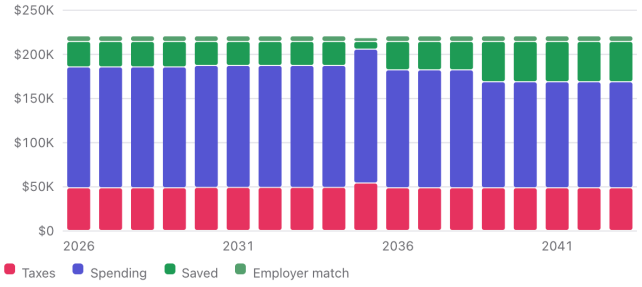
		Coming in			Going out		Saved					
Year	Ages	Sam's pay	Jordan's pay	Income	Taxes	Spending	Traditional	Roth and HSA	Brokerage	Employer match	All saved	What's going on
2026	42 · 40	$128,000	$86,000	**$214,000**	$48,333	$136,867	$28,800	$0	$0	$6,420	**$35,220**	
2027	43 · 41	$128,000	$86,000	**$214,000**	$48,333	$136,867	$28,800	$0	$0	$6,420	**$35,220**	
2028	44 · 42	$128,000	$86,000	**$214,000**	$48,333	$136,867	$28,800	$0	$0	$6,420	**$35,220**	
2029	45 · 43	$128,000	$86,000	**$214,000**	$48,333	$136,867	$28,800	$0	$0	$6,420	**$35,220**	
2030	46 · 44	$128,000	$86,000	**$214,000**	$48,776	$138,068	$27,156	$0	$0	$6,420	**$33,576**	A teenager: children $1,200 more than today
2031	47 · 45	$128,000	$86,000	**$214,000**	$48,776	$138,068	$27,156	$0	$0	$6,420	**$33,576**	
2032	48 · 46	$128,000	$86,000	**$214,000**	$48,776	$138,068	$27,156	$0	$0	$6,420	**$33,576**	
2033	49 · 47	$128,000	$86,000	**$214,000**	$48,776	$138,068	$27,156	$0	$0	$6,420	**$33,576**	
2034	50 · 48	$128,000	$86,000	**$214,000**	$48,776	$138,068	$27,156	$0	$0	$6,420	**$33,576**	
2035	51 · 49	$128,000	$86,000	**$214,000**	$53,756	$151,568	$8,676	$0	$0	$4,338	**$13,014**	College starts
2036	52 · 50	$128,000	$86,000	**$214,000**	$48,333	$133,567	$28,800	$0	$3,300	$6,420	**$38,520**	Empty nest
2037	53 · 51	$128,000	$86,000	**$214,000**	$48,333	$133,567	$28,800	$0	$3,300	$6,420	**$38,520**	

Every year to retirement, in today's dollars (or each year's own). In 2030 Nora becomes a teenager and saving dips; in 2035 college starts. Click a pay or spending figure to type your own, for one year or until retirement. CSV download.

Income and spending



Where each year's income goes



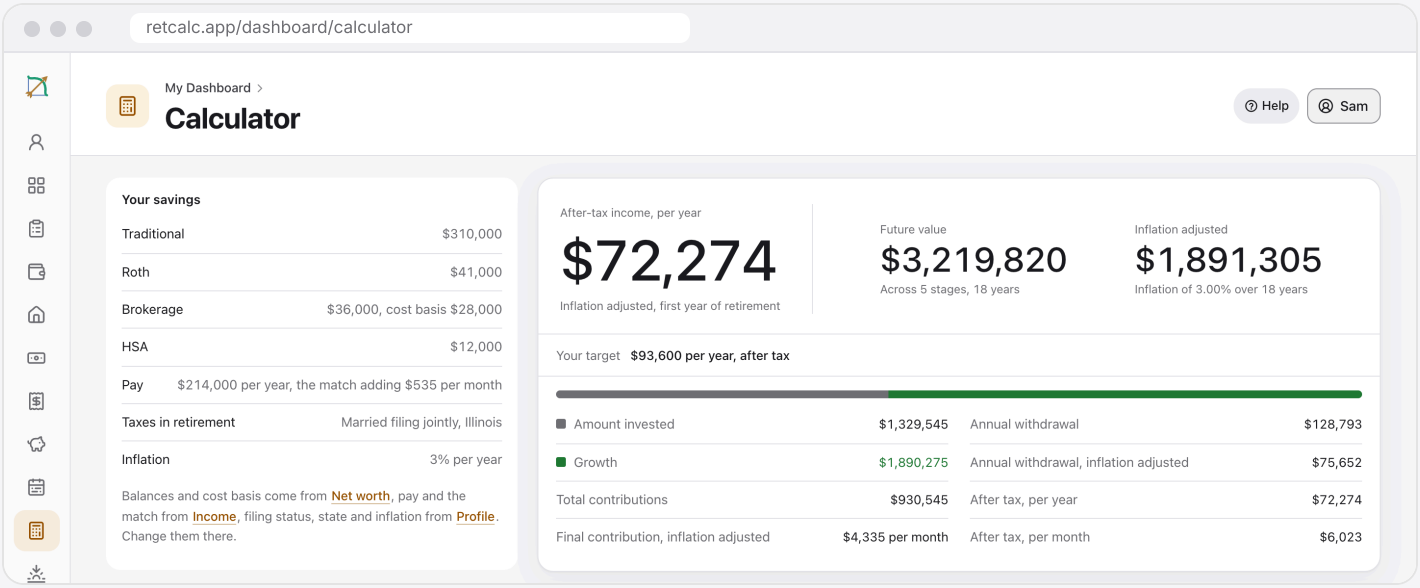
Calculator

What will our savings grow to, with the changes we see coming?

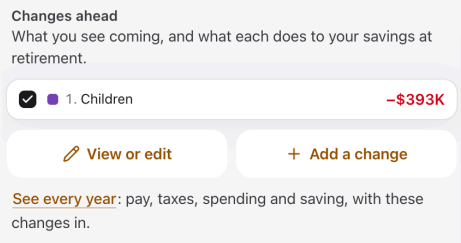
The Advanced calculator, run on the Dashboard's numbers. Balances, pay, taxes and inflation are listed with links to where they are set. What is set here (the changes ahead, fees, a goal, the mix) is saved with the plan, and the result becomes savings at retirement on every tab.

10

retcalc.app/dashboard/calculator



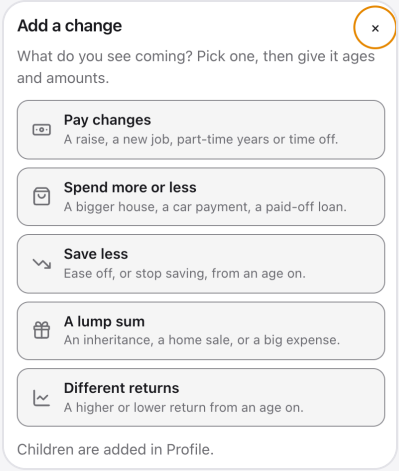
$1,891,305 at 60 in today's dollars, about $72,274 per year after tax, across five stages and 18 years.



Changes ahead. One so far: Nora, whose costs take $393K off savings at retirement.

HOW CHANGES WORK

Each change is plain English on screen and compiles to stages behind it, so every tab follows the plan with its changes in. A switch turns one off without deleting it.



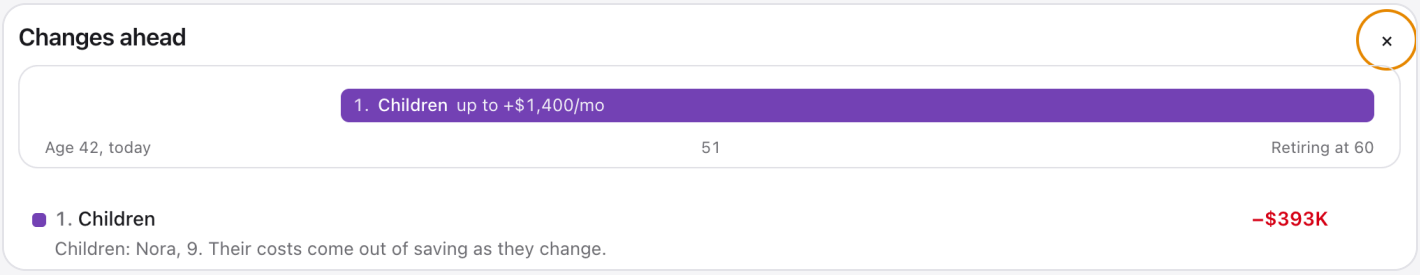
Add a change: five kinds, each a form with a live sentence and its effect.

FIVE KINDS

- **Pay changes:** a raise, a new job, part-time years or time off
- **Spend more or less**
- **Save less,** or stop, from an age on
- **A lump sum,** in or out of an account, on the day you turn an age
- **Different returns** from an age on

TODAY'S DOLLARS

Every change, like every figure on the Dashboard, is in today's dollars.

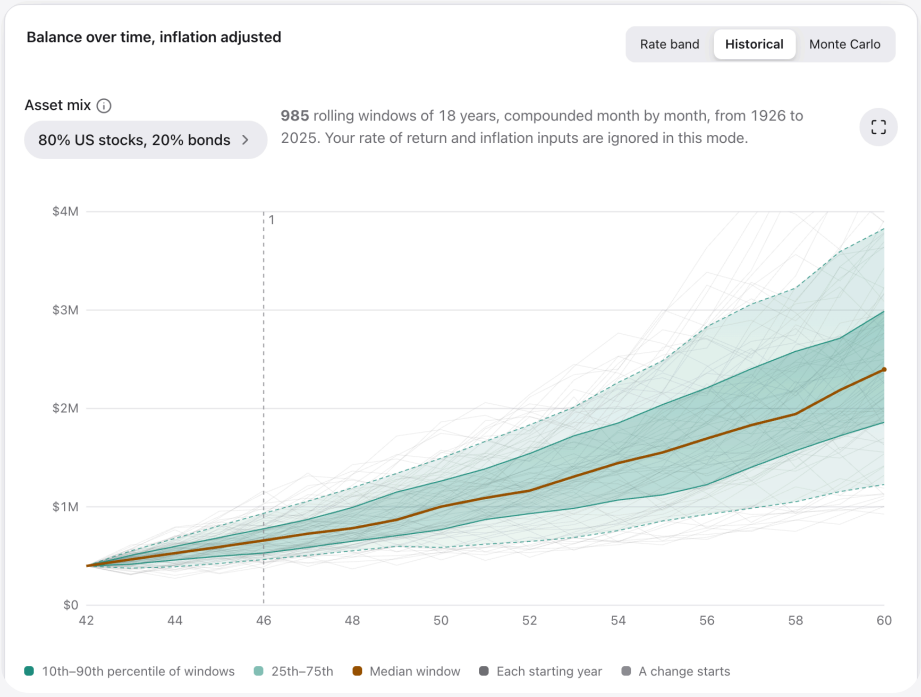


View or edit: every change on a lane by age, from today to retirement.

CALCULATOR · THE PROJECTION

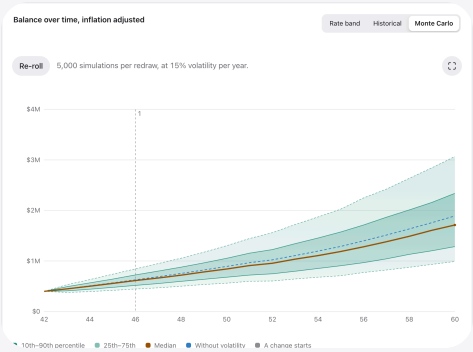
Every year to retirement, tested three ways.

The same projection as the Advanced calculator on the site: a band of better and worse returns, every rolling window since 1926 with a mix of US stocks, small-cap value, bonds and cash, or 5,000 Monte Carlo simulations per redraw. A dashed line marks where each change starts.

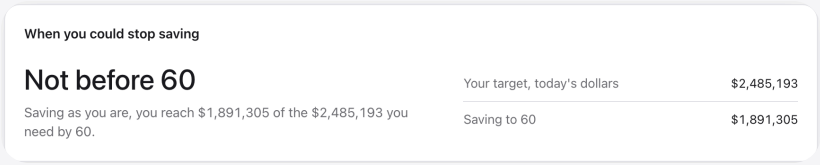


Historical: 985 rolling 18-year windows since 1926, each starting year drawn in gray.

Rate band: every stage 2% higher and lower.



Monte Carlo: 5,000 runs at 15% volatility per year.



WHEN YOU COULD STOP SAVING

The earliest age saving could stop outright, the match included, and still reach the target. When there is one, **Add to my changes** makes it part of the plan. Sam and Jordan need to keep saving to 60.

By account type

at retirement, in today's dollars

CSV

Account	Balance	Share	First-year withdrawal	Tax	After tax
Traditional 401(k) / IRA	$1,589,483	84%	$64,483	$3,378	$61,105
Roth 401(k) / IRA	$90,548	5%	$3,673	$0	$3,673
Taxable brokerage	$184,772	10%	$7,496	$0	$7,496
HSA	$26,502	1%	$0	$0	$0
Total	**$1,891,305**		**$75,652**	**$3,378** (4.5%)	**$72,274**

Note

Taxed with 2026 married filing jointly brackets and Illinois state tax. **45%** of the brokerage balance is growth by then, so only that share of each sale is taxed. The HSA is kept for health care, tax-free, so the first year's withdrawal comes from the other accounts. Your employer puts in **$143,248** over the 18 years, before growth. Social Security and other income aren't included here and would raise the tax; the Income Tax tool can add them.

BY ACCOUNT TYPE

What each account holds at retirement, its share, the first year's withdrawal and the tax on it, with 2026 married-filing-jointly brackets and Illinois tax.

ALSO ON THIS SCREEN

- **Milestones:** when growth outpaces what goes in, and each round number
- **Stage by stage:** years, return, contributions and growth
- A year-by-year table
- CSV downloads, and full screen for the chart

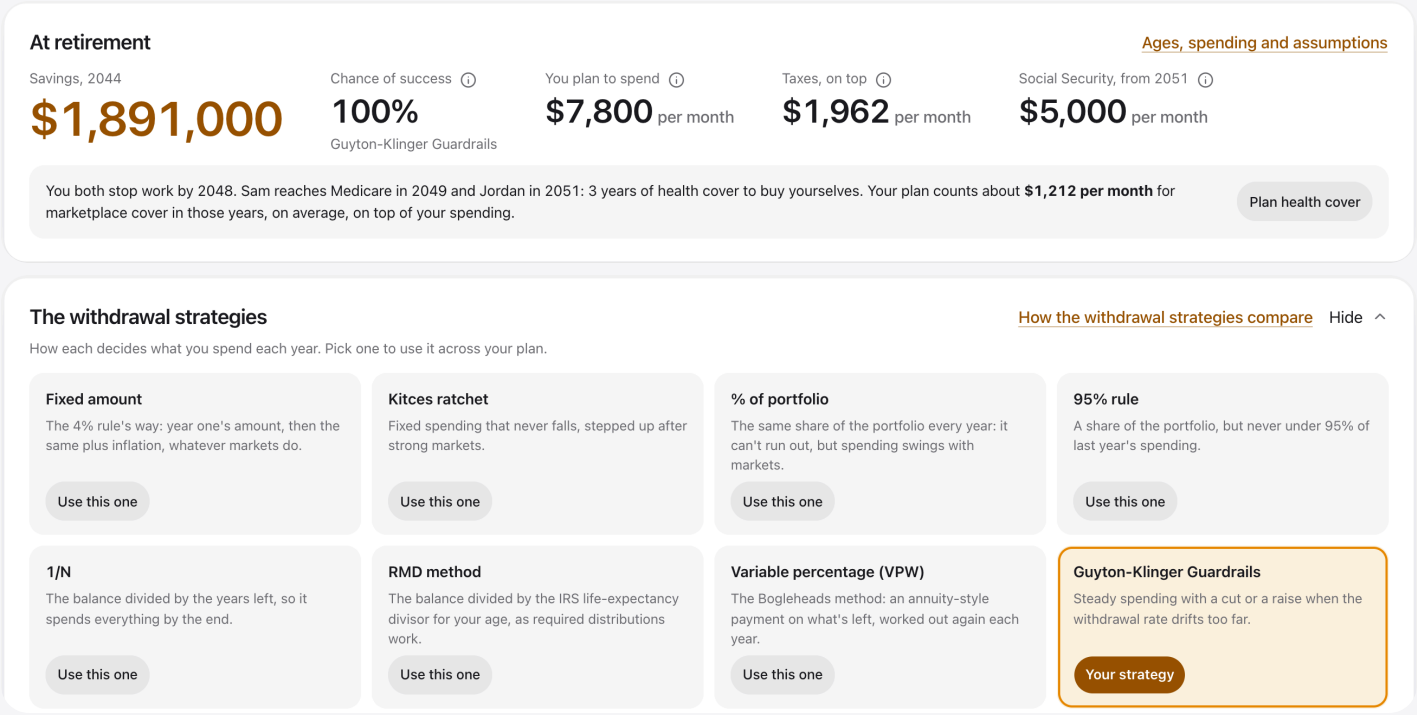
Retirement

Will our money last, and how should we draw it down?

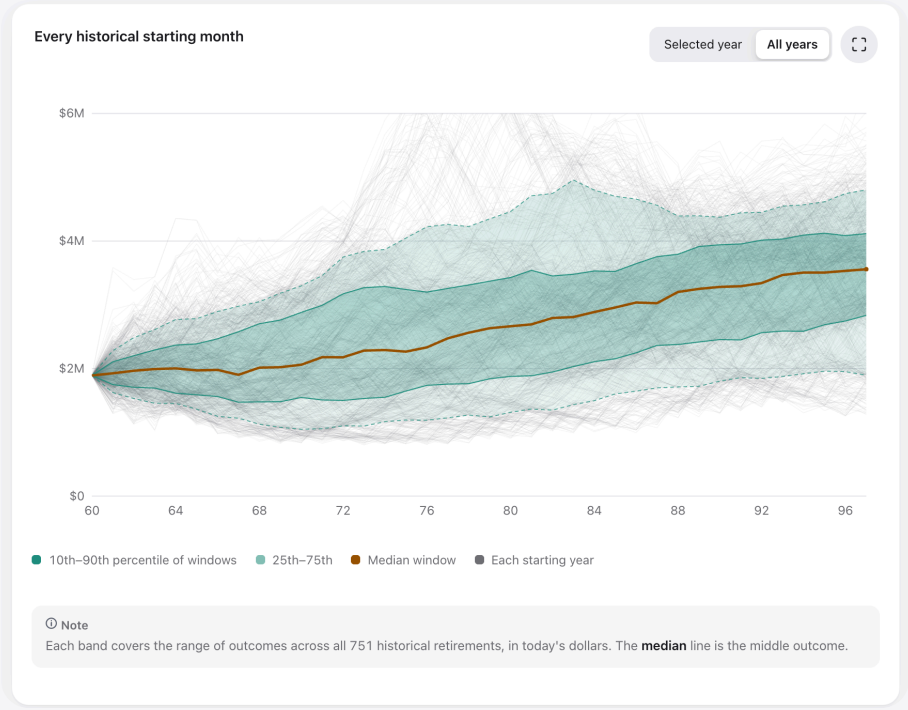
The Drawdown Simulator, run on the plan: savings at retirement from the Calculator, ages from Profile and Social Security from Income are listed, not typed. The chosen strategy's run through every historical starting month is the chance of success and the net worth chart on every tab.

11

retcalc.app/dashboard/retirement



The withdrawal strategies, open: eight of the fifteen, each with a line on how it decides what to spend. Sam and Jordan use Guyton-Klinger Guardrails; **How the withdrawal strategies compare** sets all fifteen side by side.

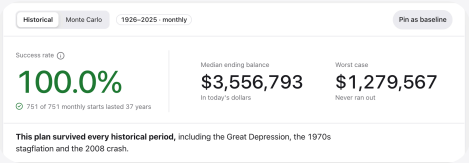


YOUR DRAWDOWN

Every historical starting month: 751 of 751 since 1926 lasted 37 years, to ages 97 and 95. The median left $3,556,793; the worst, $1,279,567, never ran out.

CHOSEN HERE, SAVED TO THE PLAN

- The strategy and its settings: starting rate, guardrails, floors and ceilings
- Years in retirement
- The mix once retired: US stocks, small-cap value, bonds and cash, with an optional glide
- The spending path

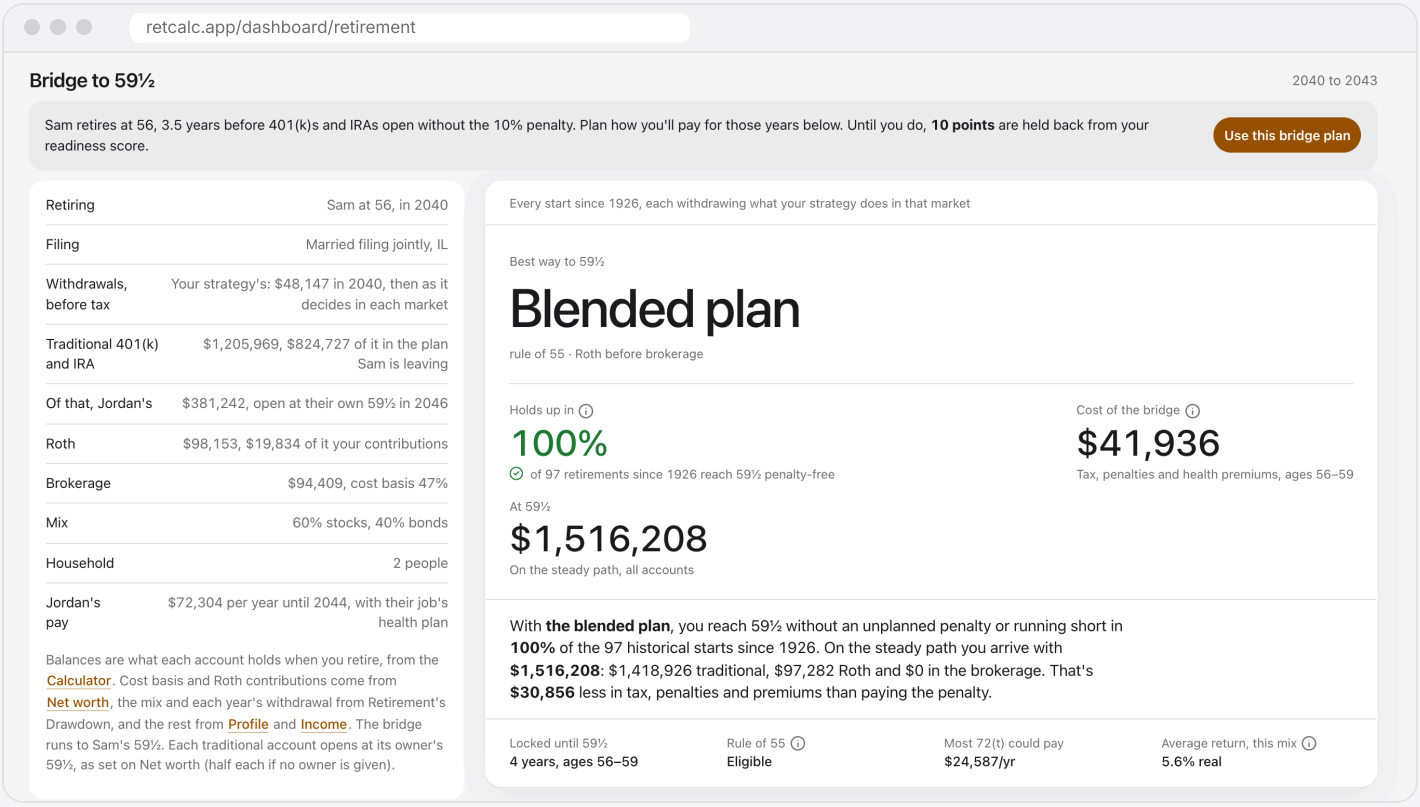


Historical or Monte Carlo, and a baseline to pin and compare against.

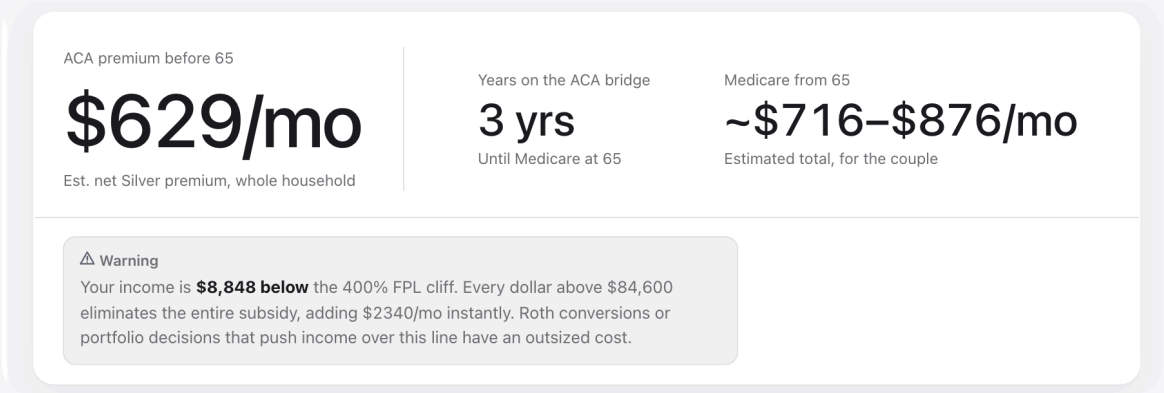
RETIREMENT · TWO MORE VIEWS

The years before 59½, and before Medicare.

A switch under **At retirement** adds two views when they apply. **Bridge to 59½** appears when someone retires before 401(k)s and IRAs open without the 10% penalty; **Health cover** prices the years between the last paycheck and Medicare. To show the bridge, the household below has Sam retiring at 56 and Jordan at 58.



Bridge to 59½: the Early Retirement Bridge on the plan, each year withdrawing what the chosen strategy does. The blended plan (the rule of 55, then Roth before brokerage) reaches 59½ in 100% of 97 starts since 1926, for $41,936 in tax and premiums, with $1,516,208 left at 59½. **Use this bridge plan** saves it to the plan.



HEALTH COVER

The Healthcare Cost Planner on the plan. For the original household (work stops in 2048): about $629 per month for an ACA plan for 3 years, then Medicare at about $716 to $876 per month for the two of them.

ROUTES TO 59½

- A Roth conversion ladder
- 72(t) payments
- The rule of 55, or a 457(b)
- Brokerage, then Roth contributions
- Part-time work, or paying the 10% penalty

FIVE MARKETS

A picker follows the plan through the 90th, 75th, median, 25th and 10th percentile markets from history, ranked by what the strategy withdraws.

PRICED ON THE PLAN

ACA premiums and the subsidy before 65 use the plan's income that year; Medicare Part B, Part D and any IRMAA surcharge use its income at 65.

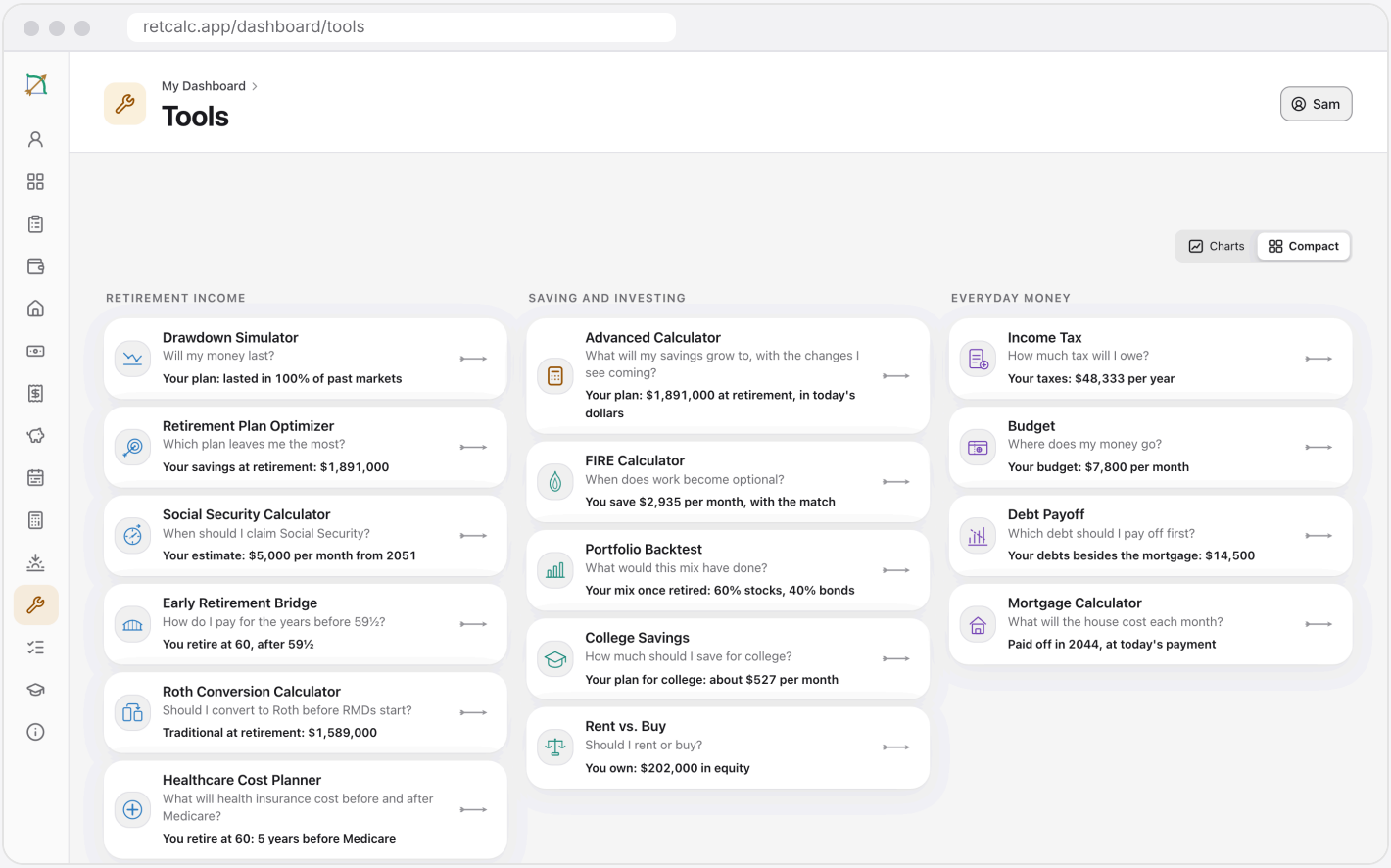
Tools

What would the other tools say, with our numbers?

The site's tool list, led by the Advanced Calculator, with a line on each card from the plan. A tool opens inside the Dashboard with everything known filled in. It is a copy to try things on: changes there never change the plan.

12

retcalc.app/dashboard/tools



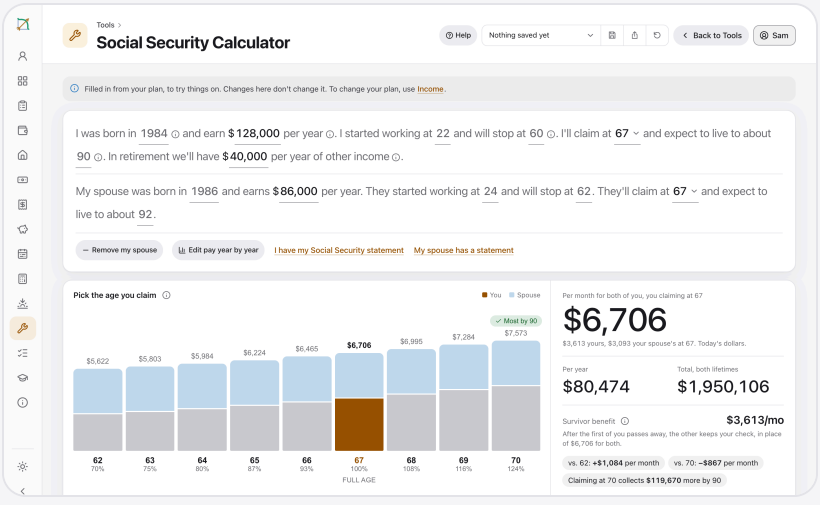
All 14 tools and the Advanced Calculator, compact view, each with its question and a line from the plan: "Your plan: lasted in 100% of past markets", "Your taxes: $48,333 per year".

OPENED FROM HERE

- Filled in from the plan, with the rail kept and a way back to the tab you came from
- Saved scenarios, shared with the same tool on the site
- Reset returns to the plan's figures

ONE THAT WRITES BACK

College Savings has **Apply to my plan**, which writes each child's cost, start and years into the profile.

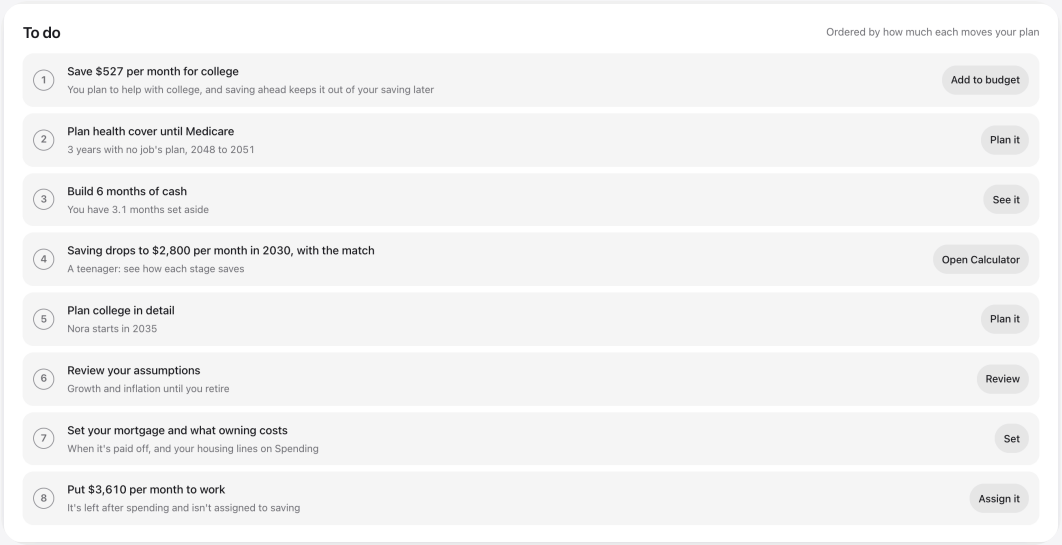


The Social Security Calculator, opened from Tools. Birth years, pay and retirement ages come from the plan; the banner says where to change the plan itself.

13 Next steps

What should we do next?

A to-do list ranked by how much each step would raise the readiness score, each with a button to the tab that does it. Done steps move below; the top four also sit on the Dashboard tab.



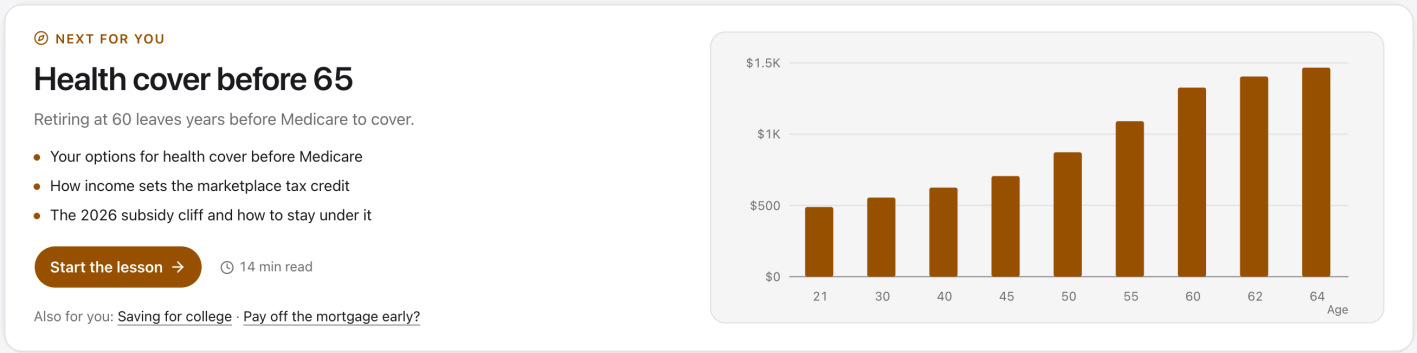
STEPS THAT CAN APPEAR

- Split your savings into accounts
- Get the full employer match
- Build months of cash
- Pay down debt at 8% or more
- Save for college
- Check your Social Security
- Choose a withdrawal strategy
- Plan your bridge to 59½
- Plan health cover until Medicare
- Update your balances

14 Learn

What does all this mean?

The same 66 lessons as retcalc.app/learn, in 8 tracks, inside the Dashboard. **Next for you** picks a lesson from what the plan needs, and lessons show the household's own numbers beside the idea.

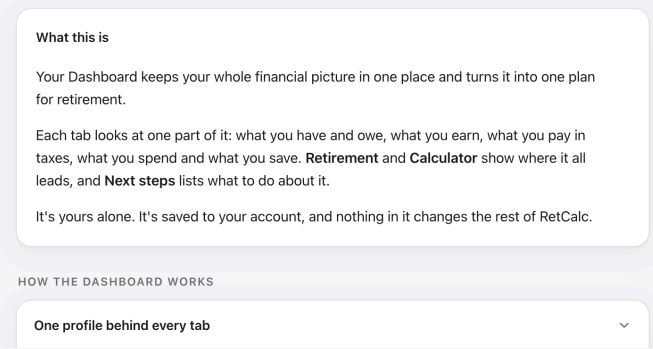


Next for you: retiring at 60 leaves years before Medicare, so the lesson on health cover before 65 comes first.

15 About

How does it work?

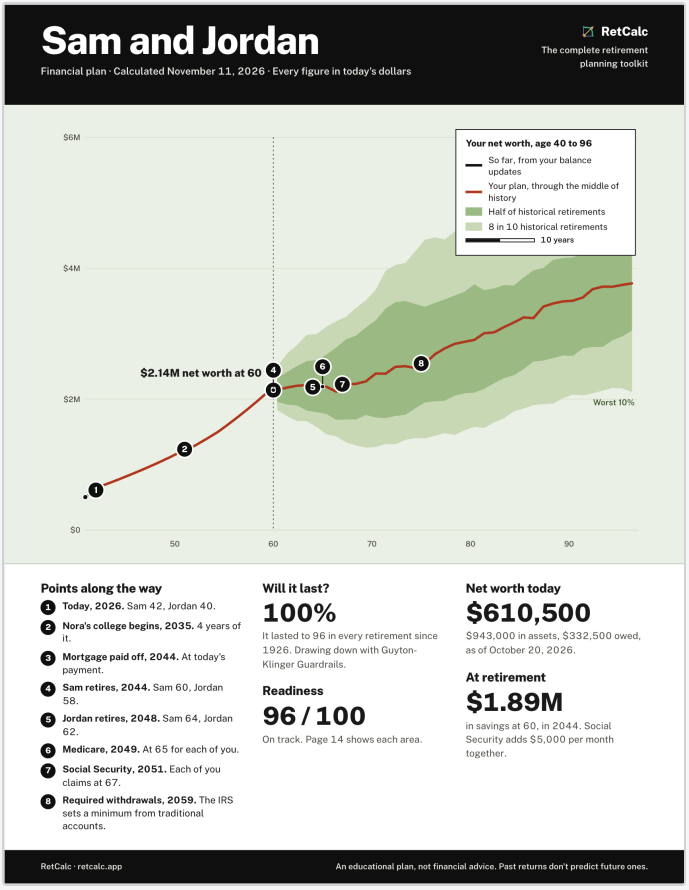
How the Dashboard works, tab by tab; the Calculator and how the projection runs; each tool; the methods, historical and Monte Carlo, and where every number comes from; then the fine print.



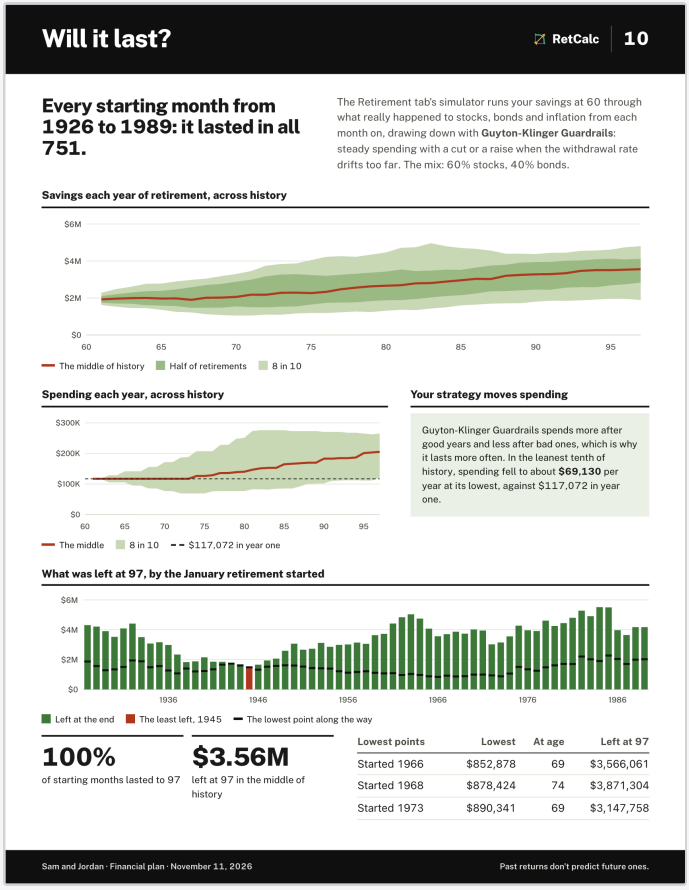
ACROSS THE DASHBOARD

The plan, printed.

Print your plan, on the Dashboard and Plan summary tabs, builds US Letter sheets in the browser from the Dashboard as it stands (nothing is sent anywhere to make it), to print or save as a PDF. Fourteen sheets are always there; up to seven more appear when they apply, each saying why it is there.



The cover: the net worth line to the end of the plan, with the points along the way.



Will it last? Every starting month from 1926 to 1989, and what was left at 97.

14
sheets every report has

7
more that appear when they apply

19
sheets in Sam and Jordan's report

0
figures sent anywhere to build it

ALWAYS THERE

- Cover, with contents
- Where you stand
- Net worth
- Income and taxes
- Spending
- Saving and investing
- The road from here
- Will it last?
- Where the money comes from
- Your readiness score
- Next steps
- Assumptions and method
- Saving year by year
- Retirement year by year

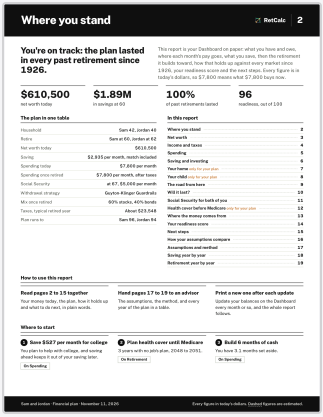
WHEN THEY APPLY

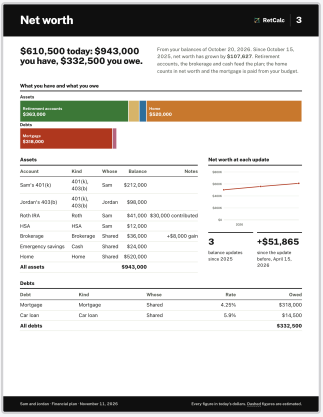
- Your home, or Renting
- Your children
- Changes ahead
- Getting to 59½
- Social Security, for both of you
- Health cover before Medicare
- How your assumptions compare

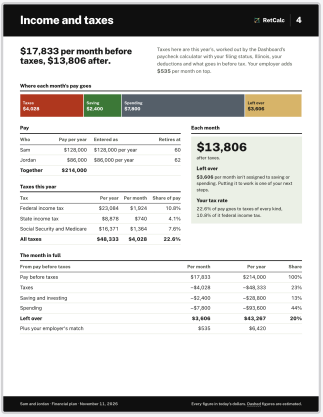
Every figure is in today's dollars. The two year-by-year tables run over as many sheets as they need, crowded sheets zoom to fit, and a couple's report shows whose each asset and debt is.

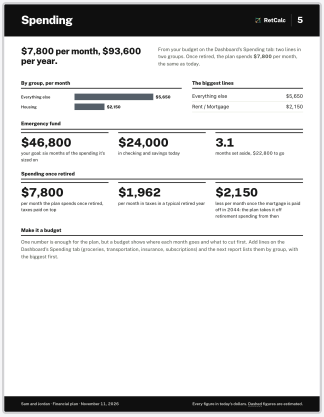
Sam and Jordan's report, sheet by sheet

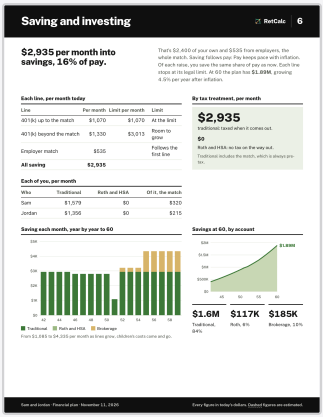
Theirs runs to 19 sheets. The cover and Will it last? are on the page before; here are 16 more.

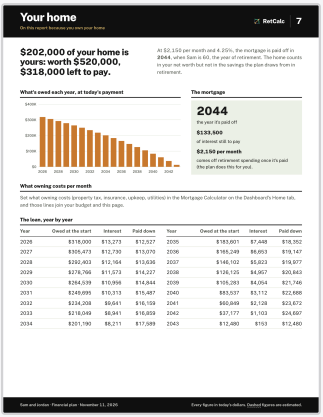


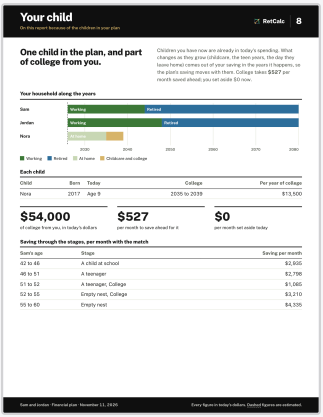


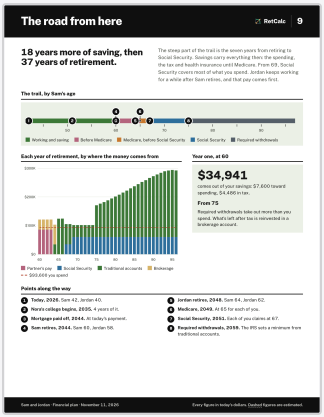


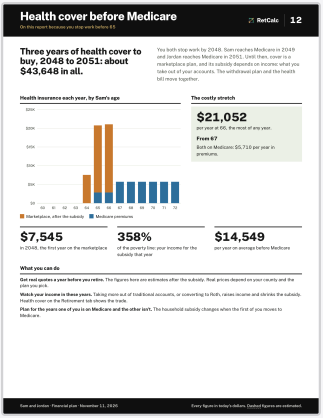


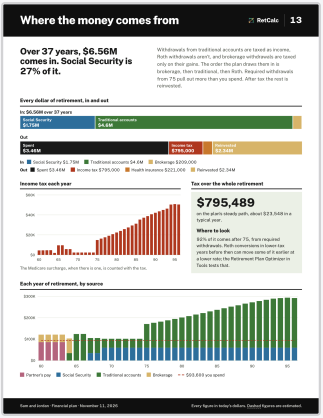


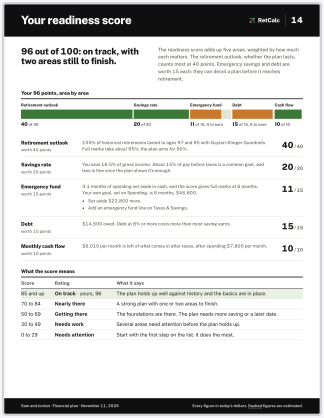


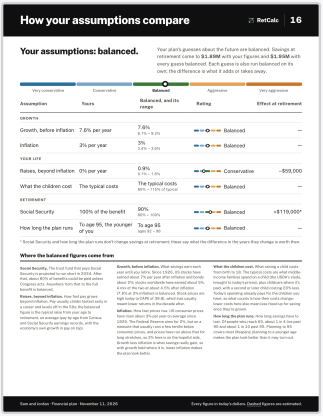


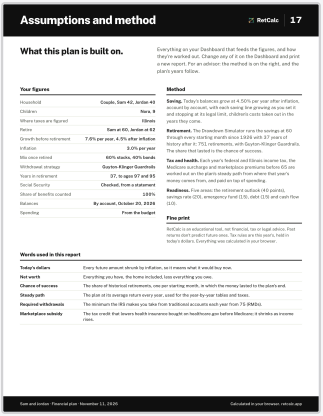


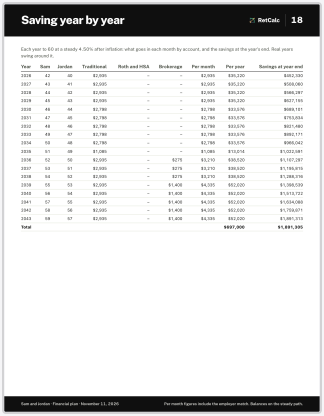








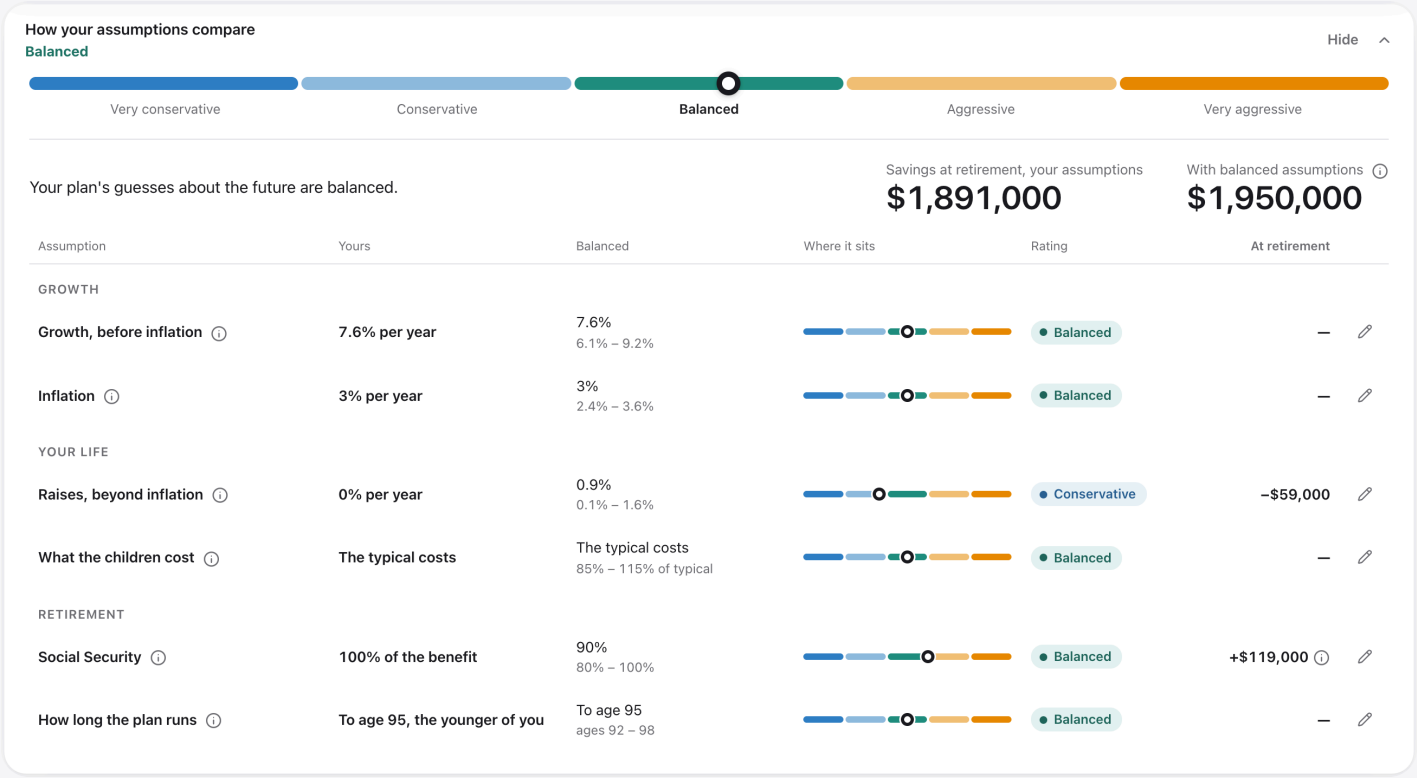




ACROSS THE DASHBOARD

How your assumptions compare.

Every projection rests on guesses about the future. This rates each one the plan uses against a balanced middle, on a five-band scale from very conservative to very aggressive, with what it does to savings at retirement. There is no score out of 100, only the bands. It is a fold on Plan summary, a chip on the Dashboard tab and a sheet in the report.



Sam and Jordan's guesses are balanced overall: $1,891,000 at retirement on their assumptions, $1,950,000 on balanced ones. Raises of 0% beyond inflation rate as conservative (−$59,000); counting 100% of Social Security rather than 90% adds $119,000.

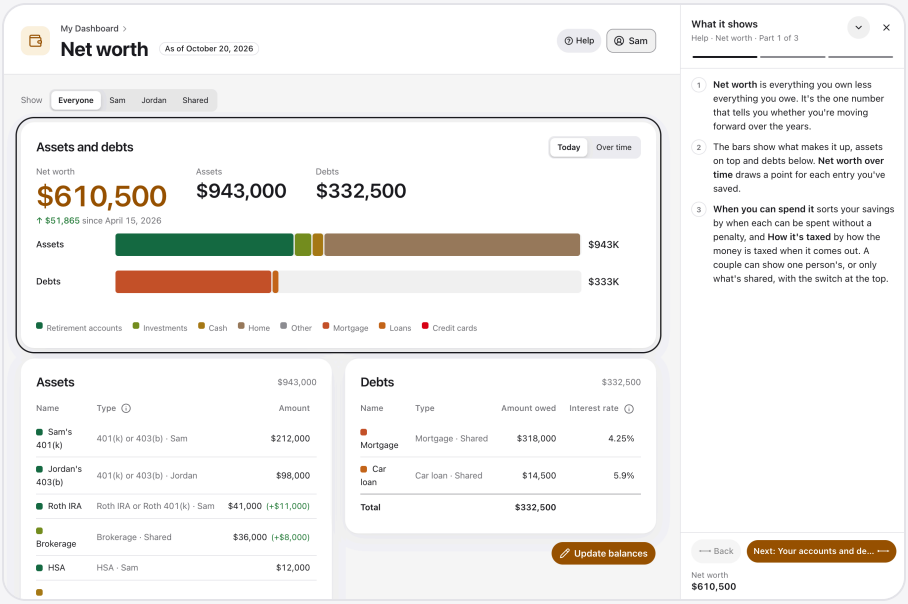
Help on every tab

Help in the title bar opens a step-by-step guide in a side panel that scrolls to each part of the page, highlights it and ticks off the tasks.

24
guides: 10 for Dashboard tabs, 14 for the tools

53
plain-language tips

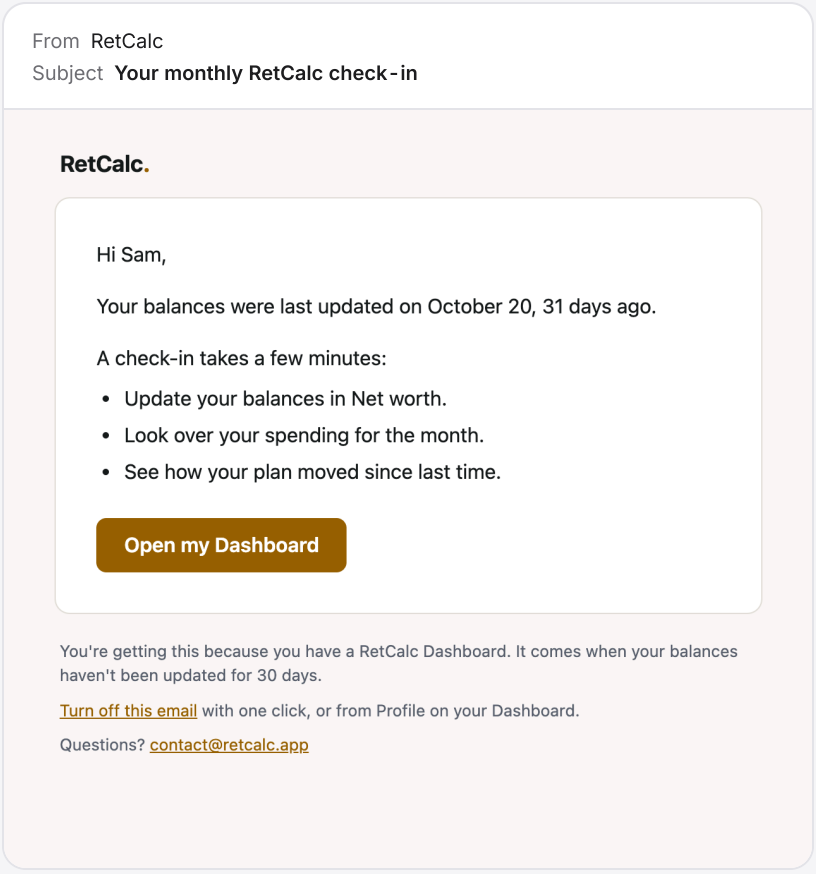
The "?" tips say what a figure is, why it is there and how the plan uses it.



The Net worth guide, part 1 of 3, with the tile it describes outlined on the page.

ACROSS THE DASHBOARD

A monthly check-in, and what stays private.



The check-in email

A short reminder to update balances, look over the month's spending and see how the plan moved. It goes out only when the latest balances are 30 or more days old, and never more than once in 28 days.

WHAT IT SAYS

- When balances were last updated
- Three things to check
- A button to open the Dashboard

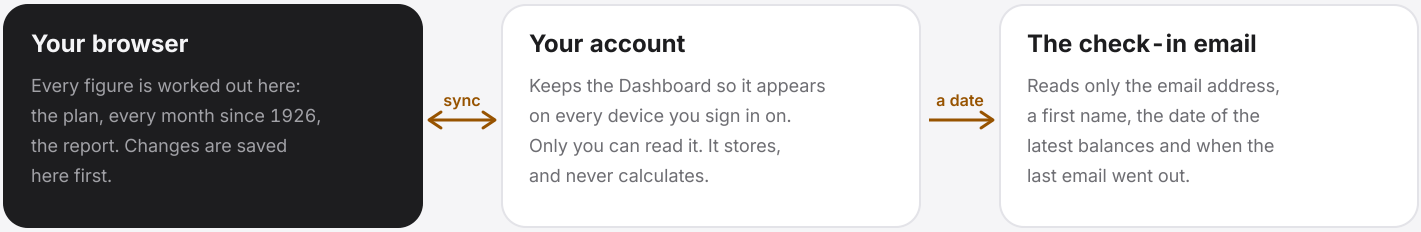
WHAT IT NEVER HAS

Dollar figures or plan results. Those are worked out in the browser, so the server that sends the email doesn't have them.

TURNING IT OFF

It is on for every Dashboard that is set up. It turns off from Profile, or with one click from any email.

Privacy



WHERE THE WORK HAPPENS

Every calculation runs in the browser. The account stores the Dashboard (profiles, entries, choices), lessons read, the email preference, the email address and an optional first name.

WHO CAN READ IT

Only the person signed in: the database lets each account reach its own data and no one else's. Signing out removes the synced copy from that browser.

NEVER SHARED

- Not sold or shared with advertisers
- No ads
- No bank connections: balances are typed in

YOURS TO TAKE OR ERASE

Download my data saves everything as a file; **Delete my account** signs out every device and erases it all.

ERROR REPORTS

Errors only, with addresses cut to their path and no inputs, cookies or user details, sent through the site's own server. The numbers you type never reach them.

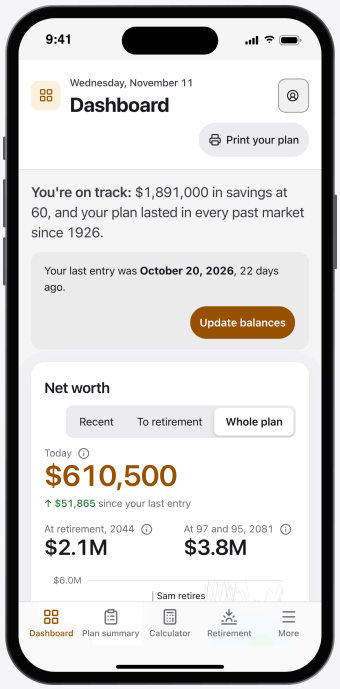
ANALYTICS

Cookieless page counts only.

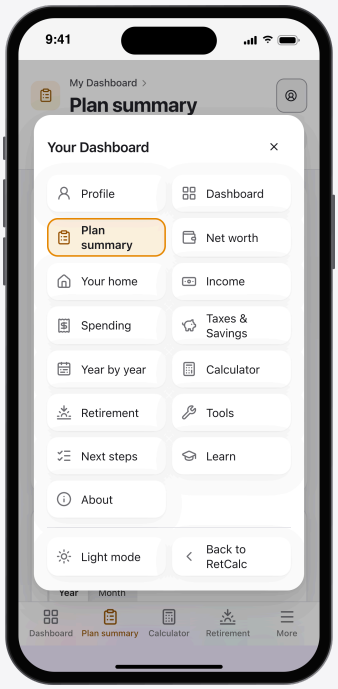
ACROSS THE DASHBOARD

On a phone, every tab.

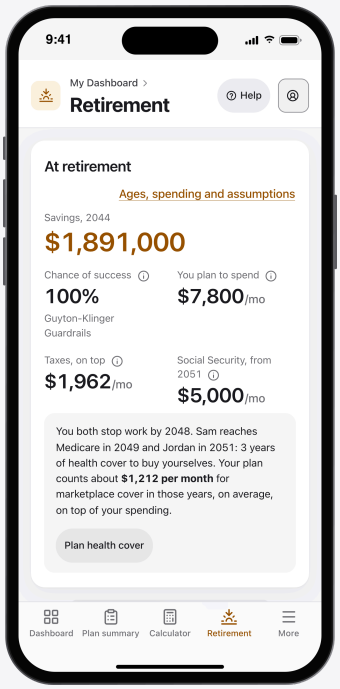
Nothing on the Dashboard is desktop-only except the full-screen chart buttons. On a phone, and on a tablet held upright, the rail becomes a tab bar; held sideways, a tablet keeps the rail with its labels. The Dashboard can be added to the home screen and opens full screen, like an app.



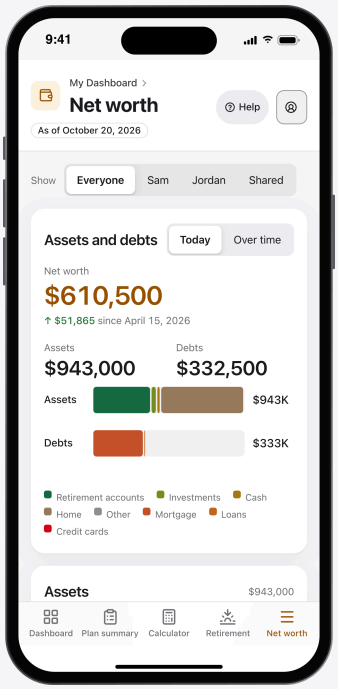
Dashboard. Figures shrink to fit, switching to /mo and /yr.



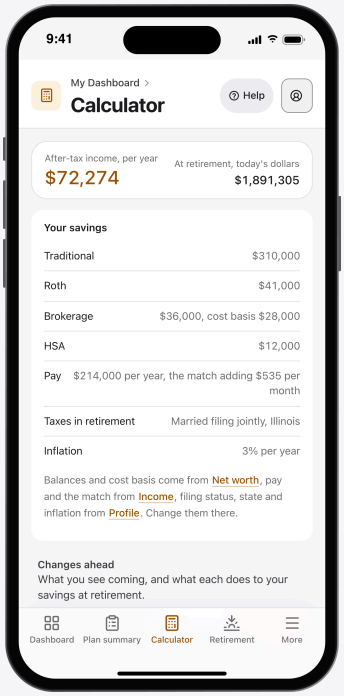
More opens every tab, the theme and the way back.



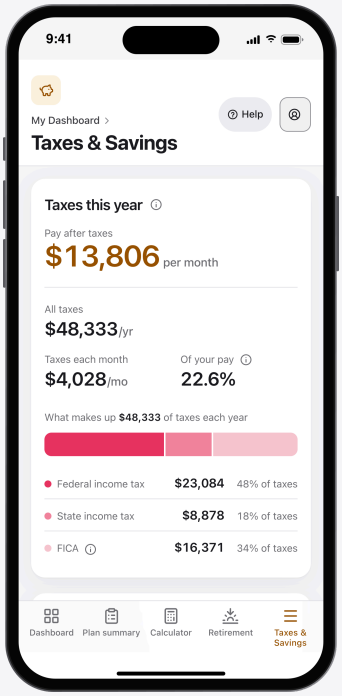
Retirement. Charts read out by tap-and-drag.



Net worth. Rows reorder by touch and hold.



Calculator. Inputs first, then the results.



Taxes & Savings. The paycheck, on a phone.

THE TAB BAR

Dashboard, Plan summary, Calculator, Retirement and More. It slides away while scrolling down and comes back on the way up, like Safari's toolbar, and shows the same dots as the rail.

MADE FOR TOUCH

- Charts read out by tap-and-drag
- The timeline and wide tables scroll sideways inside their box
- Net worth rows reorder by touch-and-hold

LIGHT OR DARK

Light, dark (Apple-style black), or match the device, from the rail or the More menu.

PART 3

The deep dive.

Every tool on the site, what it asks for and what it gives back, then the guides, Learn and one household from paycheck to drawdown.

Basic and Advanced	34
Advanced: Changes ahead	35
Drawdown Simulator	36
Retirement Plan Optimizer	37
Social Security Calculator	38
Early Retirement Bridge	39
Roth Conversion Calculator	40
Healthcare Cost Planner	41
FIRE Calculator	42
Portfolio Backtest	43
College Savings	44
Rent vs. Buy	45
Income Tax	46
Budget	47
Debt Payoff	48
Mortgage Calculator	49
Retirement Readiness Guide	50
The plan, on paper	51
Paycheck to Retirement	52
Learn	53
One household, start to finish	54
From claiming to drawdown	55

Basic and Advanced

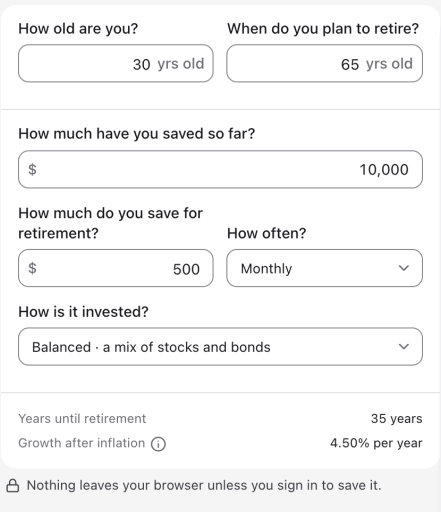
What will my savings grow to, and will it be enough?

One Calculator, with a Basic | Advanced switch under its title. Basic answers in six questions. Advanced adds what a real plan has to carry: fees, inflation, account types, an employer match, 2026 taxes, a target to reach and the changes you see coming.

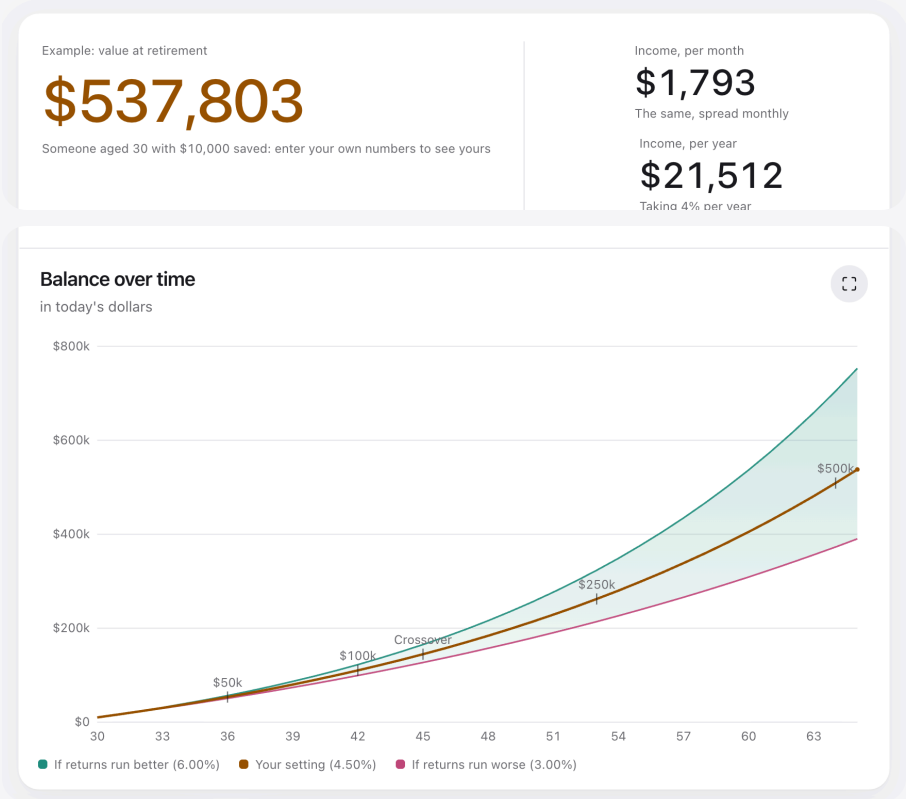
B·A

retcalc.app/calculator

BASIC · SIX QUESTIONS



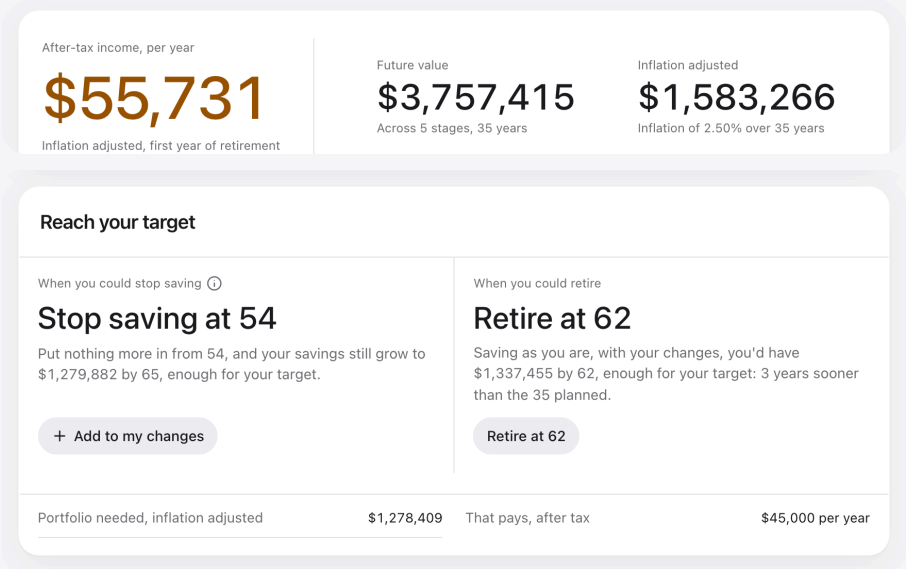
Age, retirement age, savings, how much and how often, and one of five risk levels. Out comes the balance at retirement in today's dollars and the income it could pay. **Open these numbers in Advanced** carries the answers over.



Basic. The answer, what built it, and a band for better and worse returns. Sample: aged 30, $10,000 saved, $500 per month.

ADVANCED · THE WHOLE PLAN

- Traditional, Roth, brokerage and HSA balances, each with its own contribution
- Employer match, contribution growth, fees and a glide path
- Solve for an after-tax income or a portfolio value
- **Changes ahead:** children, a raise, saving less, a lump sum or different returns, each from an age on (next page)
- **Reach your target:** when you could stop saving, and when you could retire, each put into the plan in one click
- A chart three ways: a rate band, every historical window since 1926, or 5,000 Monte Carlo runs, with a full-screen view



Advanced, with three changes ahead. $55,731 per year after tax against a $45,000 target. Reach your target: stop saving at 54, or retire at 62, three years early.

Advanced: Changes ahead

What do the changes I see coming do to my plan?

Plans change along the way: a child, a raise, a few years of saving less, an inheritance. Changes ahead adds each one from an age on, shows what it does to the balance at retirement, and turns them into the stages the calculator runs.

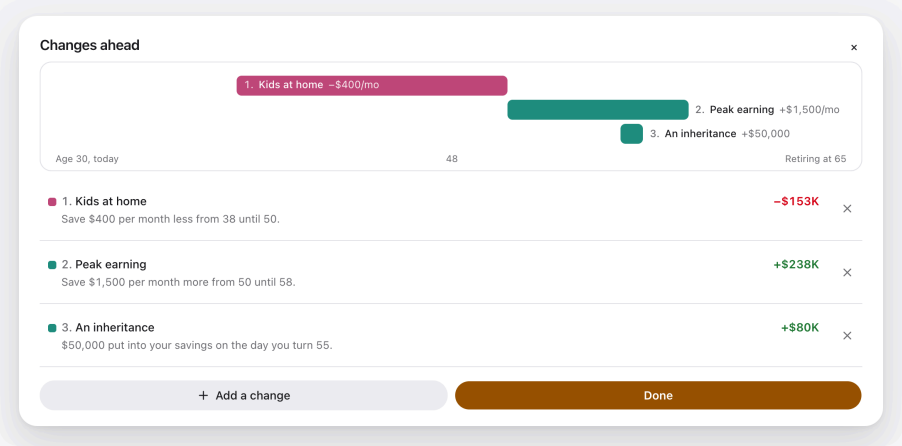
A

retcalc.app/advanced

Four kinds of change

- **Save more or less**, from an age on
- **A lump sum:** an inheritance, a home sale or a big expense
- **Different returns**, higher or lower from an age on
- **Children:** what they cost as they grow, then which accounts give up the saving

Each change
Gets a number, a switch to try the plan without it, and its effect on the balance at retirement in today's dollars.

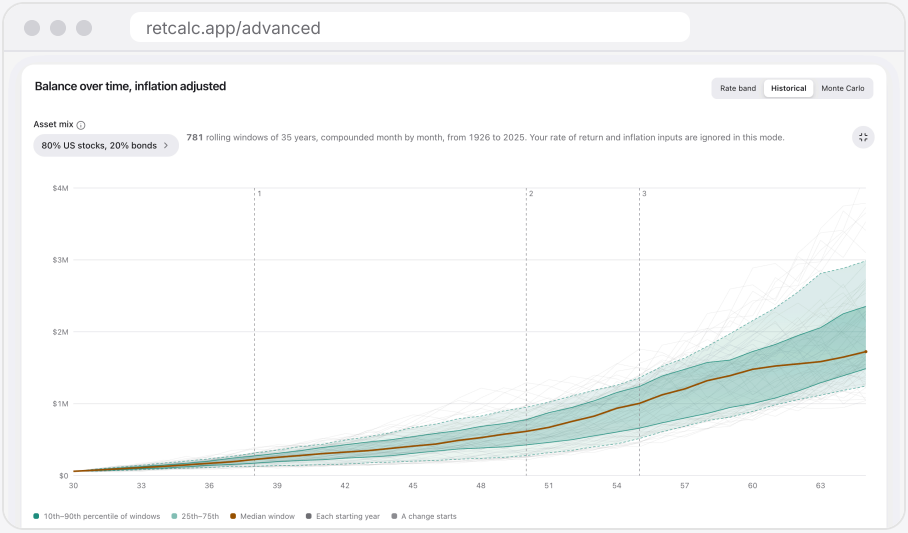


View or edit. Each change as a lane across the ages, with its sentence. Sample: 30 now, retiring at 65, with $60,000 saved and $1,000 per month.

Tested three ways
A rate band, every rolling window since 1926 (781 of them for this 35-year plan) or 5,000 Monte Carlo runs. A dashed line marks where each change starts.

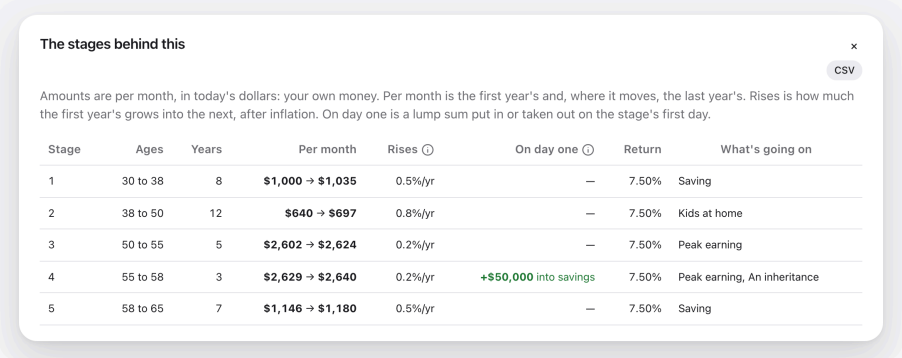
Full screen
The chart opens to the size of the window, here and on Basic, and on charts in six of the tools.

Hands off to
Test withdrawals carries the balance, in today's dollars, into the Drawdown Simulator.



Historical, in full screen. 88.9% of windows reach the $1,278,409 the target needs; the median finishes at $1,724,096.

The stages behind this. The starting figures and the changes, as the five stages the calculator runs: years, saving per month, rises and return, with a CSV.



Stage	Ages	Years	Per month	Rises	On day one	Return	What's going on
1	30 to 38	8	$1,000 → $1,035	0.5%/yr	—	7.50%	Saving
2	38 to 50	12	$640 → $697	0.8%/yr	—	7.50%	Kids at home
3	50 to 55	5	$2,602 → $2,624	0.2%/yr	—	7.50%	Peak earning
4	55 to 58	3	$2,629 → $2,640	0.2%/yr	+$50,000 into savings	7.50%	Peak earning, An inheritance
5	58 to 65	7	$1,146 → $1,180	0.5%/yr	—	7.50%	Saving

Drawdown Simulator

Will my money last?

Most calculators ask how much you will have. This one asks the harder question: once you start spending it, will it last? It replays a plan through every real retirement since 1926, month by month, with Social Security, pensions, annuities and one-time costs.

01

retcalc.app/drawdown

Historical | Monte Carlo | 1926–2025 · monthly | Pin as baseline

Success rate

94.0%

785 of 835 monthly starts lasted 30 years

Median ending balance

$1,604,620

In today's dollars

Worst case

$0

Ran out in 50 of 835 retirements

This plan survived most historical periods. It failed only when retirement began in 1964, 1965, 1966, 1967, 1968, 1969, the worst sequences on record.

15 STRATEGIES, FIVE FAMILIES

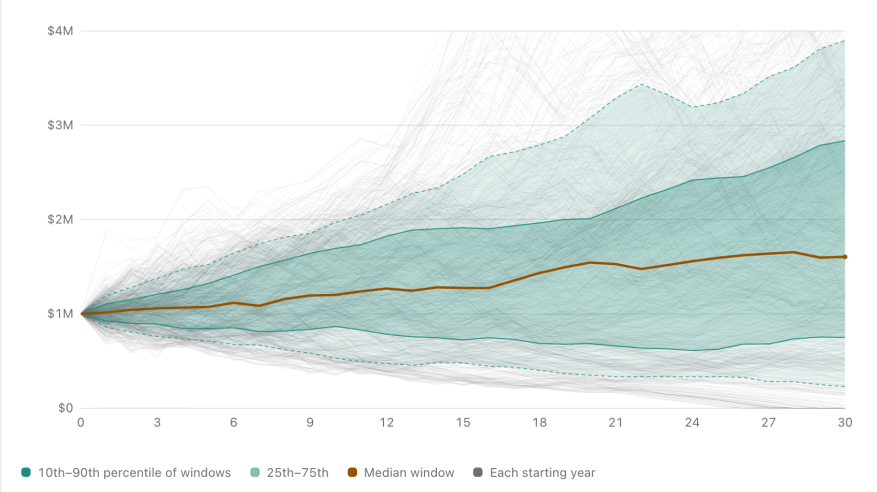
Steady income
Fixed, rising with inflation (4% rule) · Kitces ratchet

Share of portfolio
Fixed percentage · 95% rule (Clyatt) · 1/N · RMD method · Variable percentage (VPW)

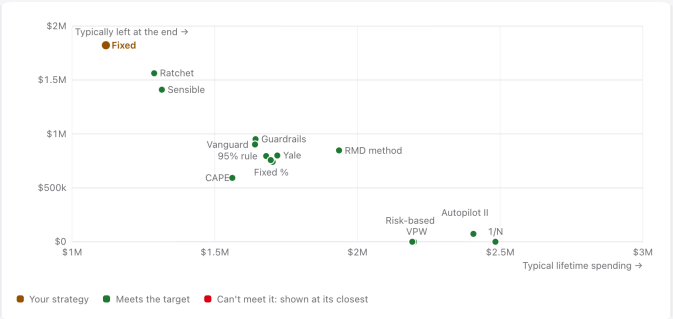
Guardrails
Guyton-Klinger · Risk-based guardrails

Smoothed
Floor and ceiling · Vanguard dynamic spending · Yale endowment · Hebeler Autopilot II · Sensible withdrawals

Valuation
CAPE-based



Every historical starting month. Percentile bands across 835 real 30-year retirements, each start drawn in gray.



Strategy showdown. All 15 tuned to the same risk, then compared on lifetime spending and what is left.

Rate \ Stocks	0%	10%	20%	30%	40%	50%	60%	70%	80%	90%	100%
2.50%	under	under	under	under	under	under	under	under	under	under	under
2.75%	under	under	under	under	under	under	under	under	under	under	under
3.00%	under	under	under	under	under	under	under	under	under	under	under
3.25%	✕ 69	94	✓ 100	✓ 100	✓ 100	✓ 100	✓ 100	✓ 100	✓ 100	✓ 100	100
3.50%	✕ 51	✕ 75	96	100	✓ 100	✓ 100	✓ 100	✓ 100	✓ 100	100	99
3.75%	✕ 47	✕ 60	✕ 81	95	97	99	100	✓ 100	✓ 100	100	99
4.00%	✕ 42	✕ 49	✕ 68	✕ 85	91	94	**94**	95	96	96	95
4.25%	✕ 39	✕ 43	✕ 56	✕ 72	✕ 86	✕ 89	91	92	93	92	91
4.50%	✕ 37	✕ 39	✕ 52	✕ 63	✕ 74	✕ 83	✕ 86	✕ 87	✕ 88	✕ 88	✕ 88
4.75%	✕ 36	✕ 37	✕ 43	✕ 56	✕ 67	✕ 75	✕ 81	✕ 83	✕ 83	✕ 84	✕ 84
5.00%	✕ 29	✕ 36	✕ 39	✕ 52	✕ 62	✕ 70	✕ 75	✕ 79	✕ 80	✕ 80	✕ 80
5.25%	✕ 25	✕ 31	✕ 36	✕ 46	✕ 57	✕ 65	✕ 70	✕ 73	✕ 76	✕ 77	✕ 78
5.50%	✕ 23	✕ 28	✕ 32	✕ 39	✕ 51	✕ 59	✕ 66	✕ 69	✕ 72	✕ 74	✕ 75

Success grid. Withdrawal rate against stock share. Click a cell to use it.

Retirement Plan Optimizer

Which plan leaves me the most?

Thousands of combinations of Social Security claiming ages, withdrawal order, Roth conversions and income guards, each run through every market since 1926, to find the plan that leaves the most, lasts the longest or spends the most.

02

retcalc.app/optimizer

Left for you and your heirs after tax, in a typical market

$4.01M

+$375k vs. $3.63M the usual way

Lifetime tax

$167,402

was $340,554 ↓ better

Lasted in

100%

was 100%

Social Security at

70 & 70

was 67 & 67

Tried **every one of 4,941 plans** in all **66 historical retirements** since 1926: 326,106 retirements simulated.

How it searches

Every pair of claiming ages from 62 to 70 for a couple (81 pairs), crossed with how far to fill each bracket with conversions and whether to stay under the IRMAA and ACA income lines. In the sample above that came to **4,941 plans** in **66 historical retirements**: 326,106 retirements simulated.

Three goals

- Leave the most
- Make it last
- Spend the most

What you get

- How much each kind of change adds
- A year-by-year roadmap in plain words
- Tax and premiums against the usual way
- Close runners-up, so the choice isn't a black box

What makes the difference

median left after tax

Claim Social Security at 70 for both of you, instead of 67 for both of you +$207k left

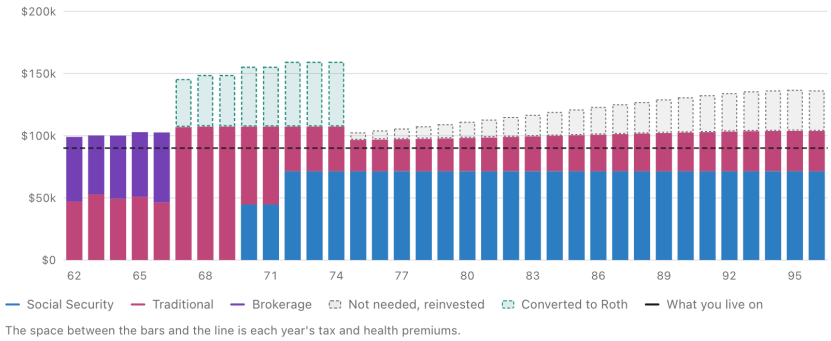
Every year you wait past full retirement age adds 8% to the check, for life, and it rises with inflation. Your savings carry the years in between, which usually costs less than the bigger check pays back over a long retirement.

Draw traditional money first, up to the top of the 12% bracket, and convert what you don't spend to Roth until 74, and keep income under the ACA subsidy cliff before 65 +$168k left

Traditional money is taxed whenever it comes out. Taking it in the lower-income years, at 12%, beats taking it later, when required distributions and Social Security stack up and push it into higher brackets. Roth money then grows tax-free for you and your heirs.

Where each year's money comes from

your roadmap, average path



The space between the bars and the line is each year's tax and health premiums.

Where each year's money comes from. Social Security, traditional, brokerage, and what is converted to Roth.

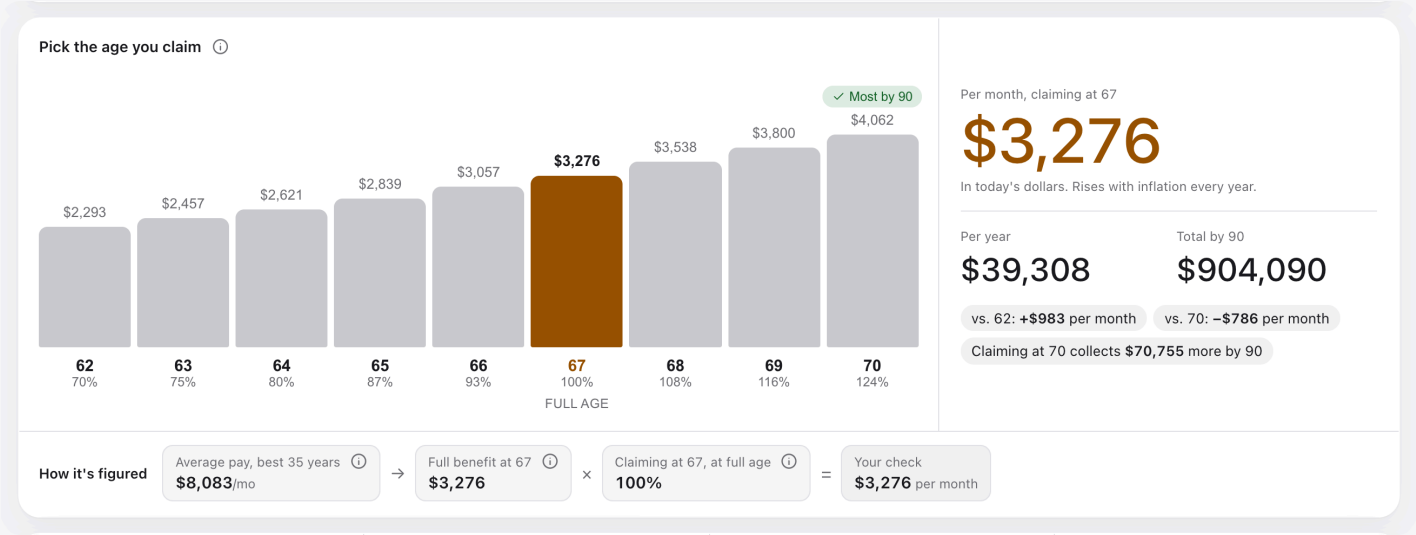
Social Security Calculator

When should I claim Social Security?

What the check will be at every age from 62 to 70, how it is figured, and which age pays off best depending on how long you live. Written as a sentence to fill in; works for a single person or a couple.

03

retcalc.app/social-security

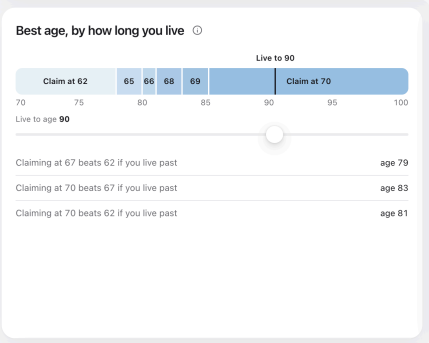


You enter

- Year born and pay, or pay year by year
- A spouse, if there is one
- Life expectancy
- Other retirement income, to work out how much of the benefit is taxed

You get

- A staircase of checks by claiming age: click a step to pick it
- Best age by lifespan, and the break-even
- Survivor benefit and every pair of ages for couples
- Year-by-year and side-by-side tables



Best age, by how long you live; and when waiting pays off.

Every age, side by side

CSV

Claim at	Per month	% of full benefit	Total by 90	Compared to 70
62	$2,293	70%	$770,442	−$204,403
63	$2,457	75%	$795,992	−$178,853
64	$2,621	80%	$817,612	−$157,233
65	$2,839	87%	$851,679	−$123,166
66	$3,057	93%	$880,505	−$94,340
67 full age	**$3,276**	**100%**	**$904,090**	**−$70,755**
68	$3,538	108%	$933,964	−$40,881

Year by year

CSV

Your age	Your benefit	Added up
67	$39,308	$39,308
68	$39,308	$78,617
69	$39,308	$117,925
70	$39,308	$157,233
71	$39,308	$196,541
72	$39,308	$235,850
73	$39,308	$275,158

Every age, side by side; and year by year. Sample: born 1990, earning $85,000.

Early Retirement Bridge

How do I pay for the years before 59½?

Retiring before 59½? It compares a Roth conversion ladder, 72(t) payments, the rule of 55, brokerage money and part-time work, with the tax, penalties and ACA premiums each costs, and shows what is left at 59½.

04

retcalc.app/bridge

Ways to 59½

CSV

Plan	Holds to 59½	Tax	Penalties	Health premiums	At 59½
✓ **Blended plan** Best 72(t) of $60,312/yr · Roth before brokerage	⊙ 100%	$22,263	$0	$52,499	$1,936,129
Roth conversion ladder Convert every year and spend each conversion five years later	85%	$23,566	$0	$118,621	$1,814,081
Brokerage, then Roth contributions No conversions; sell taxable investments first	⊗ 53%	$13,234	$0	$15,497	$2,002,631
72(t) payments $60,312 per year from an IRA, locked in through age 59	⊙ 100%	$22,927	$0	$61,290	$1,922,375
Rule of 55 Needs you to leave your job at 55 or later					—
Pay the 10% penalty Withdraw from the 401(k) or IRA as needed and pay the extra 10%	100%	$54,531	$81,776	$81,449	$1,736,040

In Holds to 59½, ⊙ marks 95% or more and ⊗ under 80%; between, no mark.

Routes compared

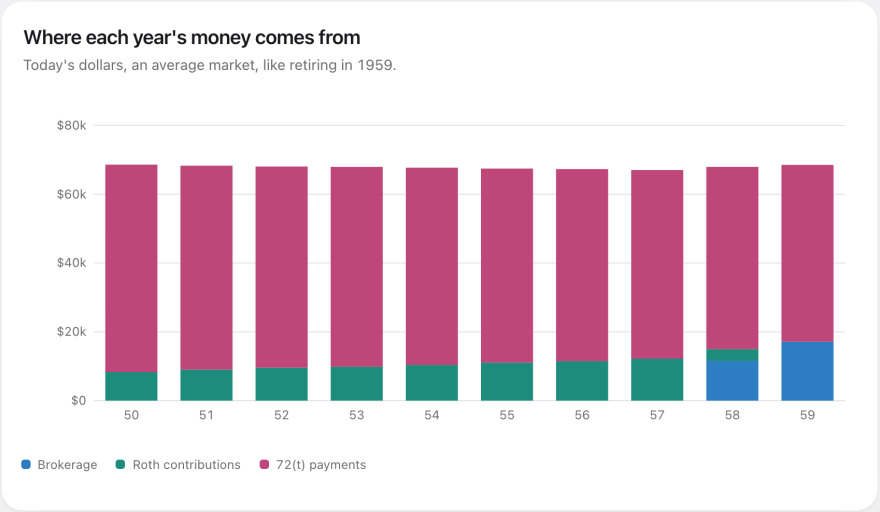
- Roth conversion ladder
- 72(t) / SEPP payments
- Brokerage, then Roth contributions
- Rule of 55
- Part-time work
- Paying the 10% penalty, as a benchmark

Seven conversion fills

From none, through the standard deduction and the tops of the 10% and 12% brackets, to just under the ACA subsidy cliff.

Tested against

Every start since 1926, or Monte Carlo.



Where each year's money comes from, from retirement to 59½.

Roth Conversion Calculator

Should I convert to Roth before RMDs start?

Projects required minimum distributions through any age up to 100, then tests a conversion schedule against doing nothing: lifetime tax, after-tax net worth, IRMAA surcharges and the survivor's higher bracket.

05

retcalc.app/roth

The two measures disagree, so read them together.

Lifetime tax saved
−$1,073
Lifetime tax + IRMAA, discounted at 3.0% real

After-tax net worth vs doing nothing
+$65,102
↑ After-tax, at age 92

Peak RMD, doing nothing
$169,099
Converting trims it to $0

Lifetime tax is what you pay through age 92. After-tax net worth is what is left at that age, with traditional dollars discounted at 24% because they are still owed to the IRS.

RMDs begin	First RMD, converting	Years converting	Average conversion
Age 75	None in this window	9 years	$143,150/yr

When the measures disagree, it says so. In the sample, converting leaves $65,102 more after-tax net worth at 92 and trims the peak RMD to $0, but costs $1,073 more in lifetime tax and IRMAA, discounted at 3% real.

Conversion strategies

- Fill to a tax bracket
- Fill to an IRMAA threshold
- A fixed amount
- A percentage of the balance
- None, for comparison

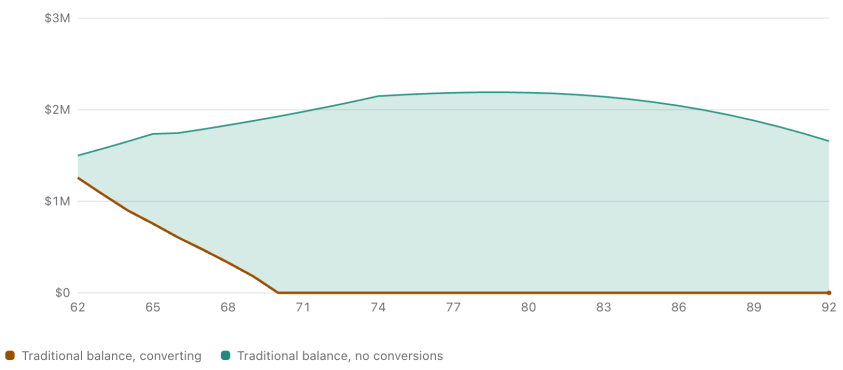
It accounts for

- Social Security and its taxation
- Tax paid from the taxable account
- The survivor filing single after one spouse dies
- Heirs' tax rate on what is left
- A discount rate for present value

Traditional balance

In today's dollars, by age.

Balance | Tax



Traditional balance, converting against doing nothing.

Converting vs. not

Lifetime income tax, converting	$305,186	Total converted	$1,288,352
Lifetime income tax, doing nothing	$494,485	Break-even	Age 62, ahead from the start
Present value, tax + IRMAA	$288,349 vs $287,275	Traditional at the end	$0 vs $1,657,171

Healthcare Cost Planner

What will health insurance cost before and after Medicare?

ACA marketplace premiums and subsidies for the gap before Medicare, then Medicare Part B, Part D, Medigap and IRMAA surcharges by income, on 2026 rules.

06

retcalc.app/healthcare

ACA premium before 65

$2688/mo

Est. net Silver premium, whole household

Years on the ACA bridge

5 yrs

Until Medicare at 65

Medicare from 65

~$716–$876/mo

Estimated total, for the couple

Before 65: the ACA bridge

Your income of $95,000 is **449% of the federal poverty level** for a 2-person household. This ratio drives your subsidy. That income is above the 400% FPL cutoff, so there is no credit and you would pay the Silver plan's full list price. The enhanced credits that removed this cutoff (capping your share at 8.5% of income) expired after 2025.

2026 ACA rules

Benchmark Silver plan (IL est., ages 60 and 59), the reference used to price your credit	$2688/mo
Premium tax credit	none (income above 400% FPL)
Your net premium	**$2688/mo**

If Congress restores the enhanced credits (in place 2021–2025)

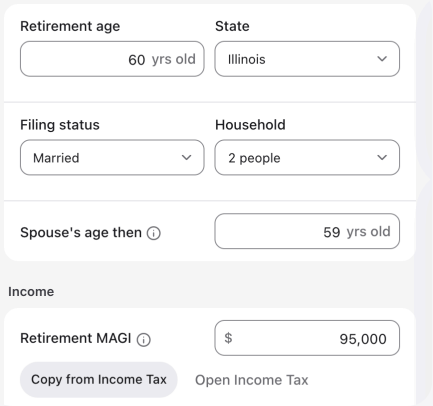
From 65: Medicare

Part B covers doctor visits, outpatient care, and preventive services. **Part D** covers prescription drugs. Many people add a **Medigap supplement** such as Plan G, which covers deductibles and copays that Parts A and B leave unpaid and caps your out-of-pocket costs. Higher incomes trigger IRMAA surcharges that raise the Part B and Part D premiums.

IRMAA Standard: standard premium

Part B premium	$203/mo per person
Part D estimate avg plan, no IRMAA	~$35/mo per person
Medigap Plan G varies by state, insurer & age	~$120–$200/mo per person
Estimated total per person	**~$358–$438/mo**
For the couple both at same IRMAA tier	~$716–$876/mo

Sample couple retiring at 60, with $95,000 of income: just over the subsidy cliff, so the full ACA premium until 65, then Medicare.



You enter

- Retirement age, status and household size
- State and income in retirement (MAGI)
- Social Security, if any, and your own quote

Why it matters

Before 65, the subsidy depends on income, so how a retiree draws money down can change the premium by thousands. The planner shows the household's share of the poverty line, warns near the subsidy cliff, and compares plan tiers. From 65 it prices Part B, Part D and Medigap, with the IRMAA tier income puts you in.

FIRE Calculator

When does work become optional?

Finds the age a portfolio reaches financial independence. Coast FIRE mode finds the moment you could stop contributing and let growth carry you the rest of the way.

07

retcalc.app/fire

Rate band | **Historical** | Your rate of return, and every market since 1926

FIRE age, at your rate of return

66.6

36.6 years from now

Portfolio at FIRE

$1,024,146

In today's dollars

Target portfolio

$1,000,000

Spending $40,000 per year at 4.00%

Historical success rate **50%**

1% (aggressive) 99% (conservative)

FIRE age, in market history

Age 60.3

(30.3 years from now)

50% of historical windows since 1926 show the portfolio reaching the target by age 60.3.

Annual withdrawal	$40,000	Nominal return	8.50%

You enter

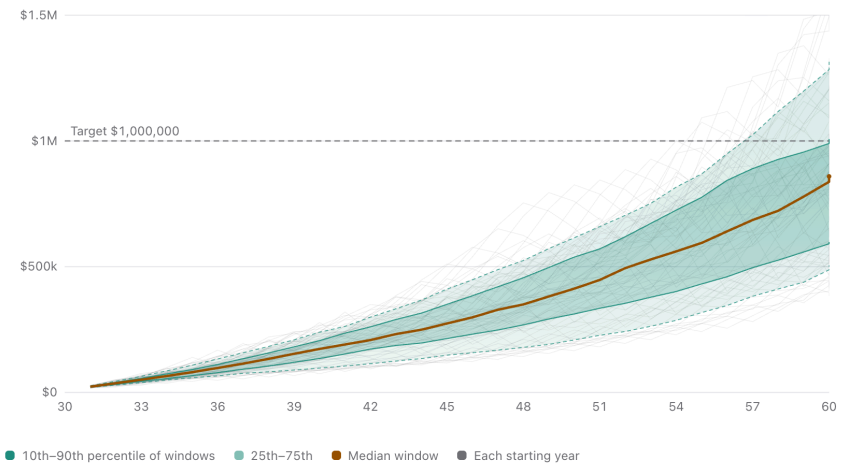
- Age, savings and monthly saving
- Return and inflation
- Spending and withdrawal rate
- FIRE or Coast FIRE

Historical mode

Runs every rolling window since 1926. A success-rate slider then gives a second answer: the age reached in, say, 90% of history.

Portfolio growth

837 rolling windows of 30.33 years, compounded month by month, from 1926 to 2025. Your rate of return and inflation inputs are ignored in this mode.



Portfolio growth, every historical window since 1926.

Portfolio Backtest

What would this mix have done?

Pick a mix of stocks, small-cap value, bonds and cash, then see what it did: return, volatility, the worst year, the deepest fall, and every rolling window back to 1926.

08

retcalc.app/backtest

Return, per year

9.65%

Compound annual growth, 1926 to 2025

After inflation	Volatility	Deepest fall
6.52%	15.26%	-50.87%
Inflation averaged 2.94% per year	Standard deviation of annual returns	1928 to 1932

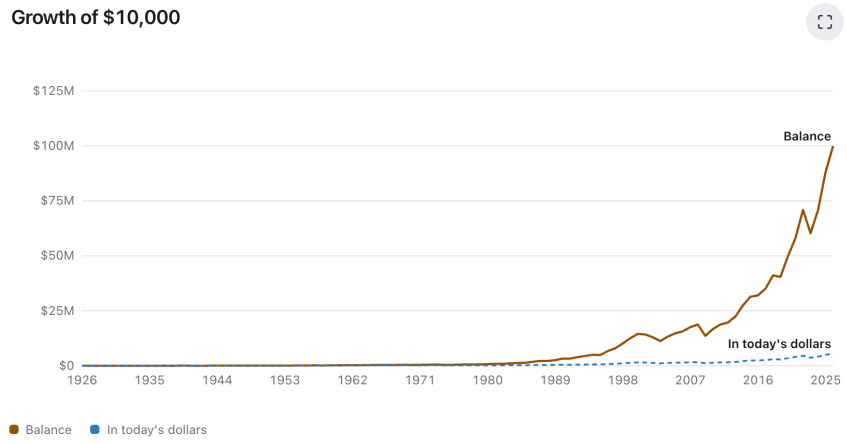
Best year	↑ 44.54% in 1933	Grew to	$99,910,217

You choose

- Stocks, small-cap value, bonds and cash
- Rebalancing yearly, every few years, or by band
- Any span from 1926 to 2025, with one-tap eras

You see

- Return per year, before and after inflation
- Volatility and the deepest fall
- Growth of $10,000
- Rolling returns and how prices multiplied
- Every year in a table



Rolling returns

nominal

Nominal | Real | CSV

Window	Periods	Worst	Median	Best	Positive
1 year	100	↓ -33.24%	13.65%	↑ 44.54%	74%
5 years	96	↓ -8.47%	10.58%	↑ 24.14%	94%
10 years	91	↑ 0.85%	9.61%	↑ 17.22%	100%

Rolling returns. The best, worst and typical result for every holding period, from every start since 1926.

College Savings

How much should I save for college?

How much to save each month to cover tuition, with presets for a public in-state school, a private college and an elite one, and room for several children sharing one account.

09

retcalc.app/college

Save per month

$558

for 18 years at 6.0%

Total projected cost

$232,269

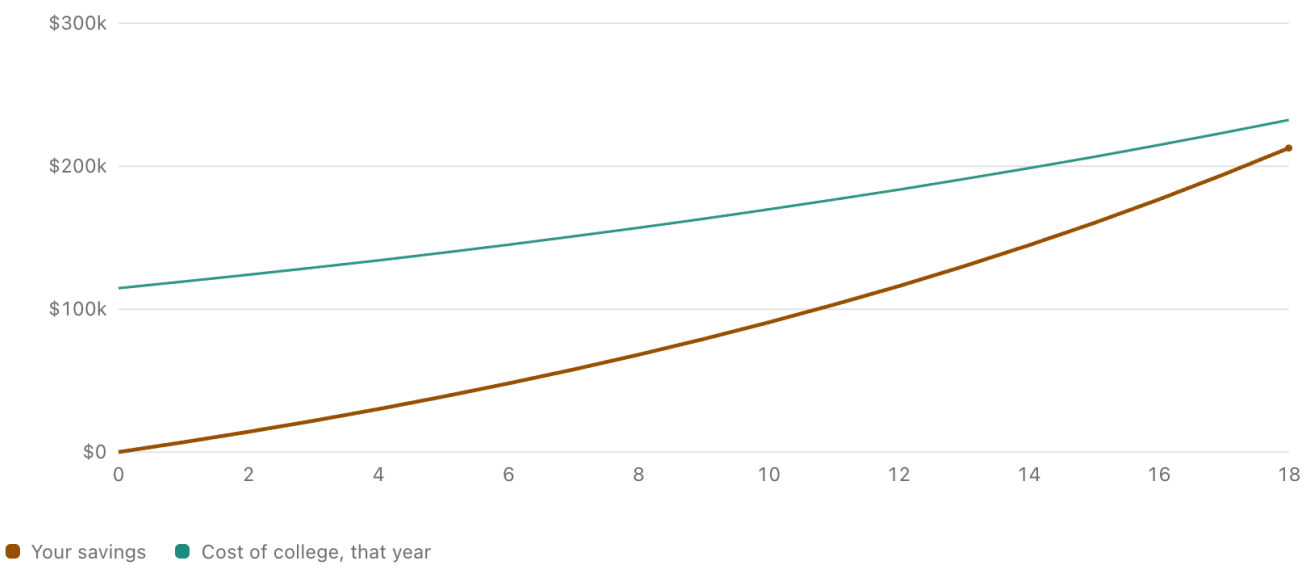
4 years at $27,000/yr, 4.0% inflation

Needed when college starts

$212,674

Less than the $232,269 total, since the money still in the account keeps earning 6.0% while later years' tuition is paid

Savings over time



Savings over time against the cost when college starts. Sample: one child, 18 years away, public in-state preset.

Presets, per year

- Public in-state, about $27,000
- Private non-profit, about $63,000
- Elite, about $82,000
- Or a custom figure

You enter

- Years until college, and years in it
- What is already saved
- Return and tuition inflation
- Each child, if there is more than one

You get

- The monthly saving needed, stepping down for younger children
- The total projected cost
- What is needed today, or when college starts

Source

College Board, Trends in College Pricing, updated each year.

Rent vs. Buy

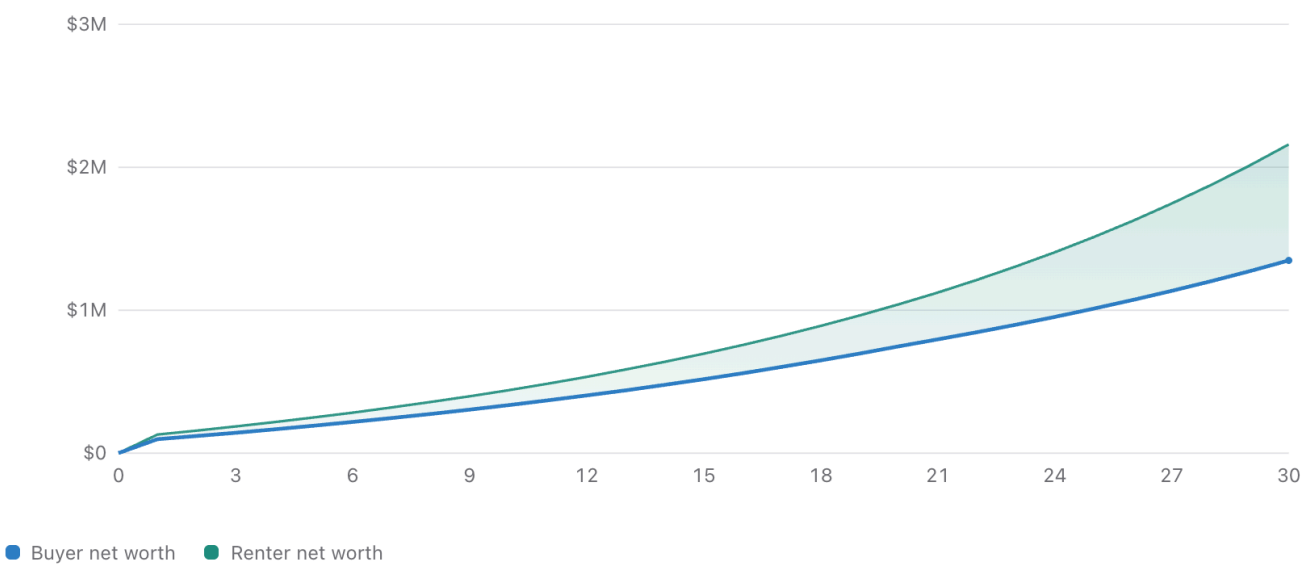
Should I rent or buy?

Compares a buyer's and a renter's net worth over 1 to 40 years, counting what the down payment could have earned invested and what the home gains in value.

10

retcalc.app/rentbuy

Net worth over time



Better after 30 years

Renting

Ahead by $811,712

Break-even

Renting stays ahead throughout

Buyer net worth

$1,347,907

Renter net worth

$2,159,618

Each month, to start		Up front	
Buying, all in	$3,393/mo	Loan amount	$360,000

Each month, to start: the buyer's costs beside the renter's, and which side is ahead.

It counts

- Price, down payment, rate and term
- Property tax, insurance and maintenance
- Closing and selling costs

And

- Rent, and how fast it rises
- Home appreciation and investment return
- Tax on gains

The verdict

Which side is ahead after the chosen number of years, by how much, and the year the lines cross.

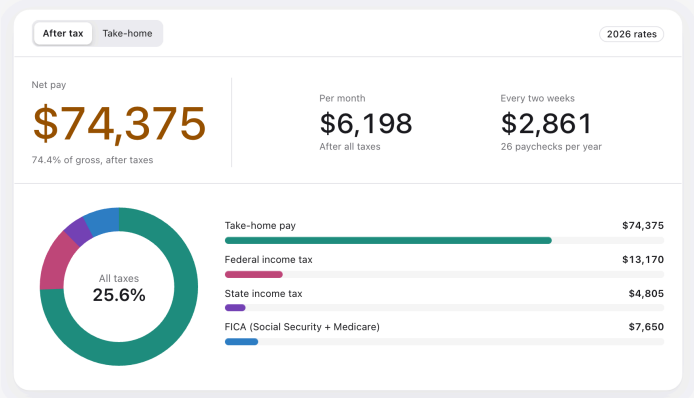
Income Tax

How much tax will I owe?

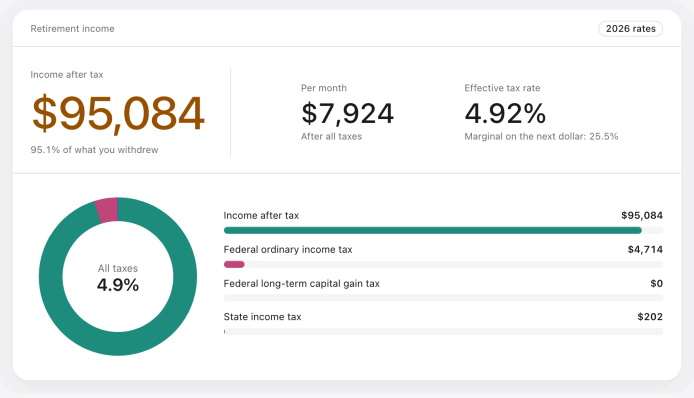
2026 federal, state and FICA tax on a salary, or a retirement mode that shows what a year of withdrawals costs across traditional, Roth, brokerage and Social Security.

11

retcalc.app/incometax



Working. $100,000 salary, single, Illinois.



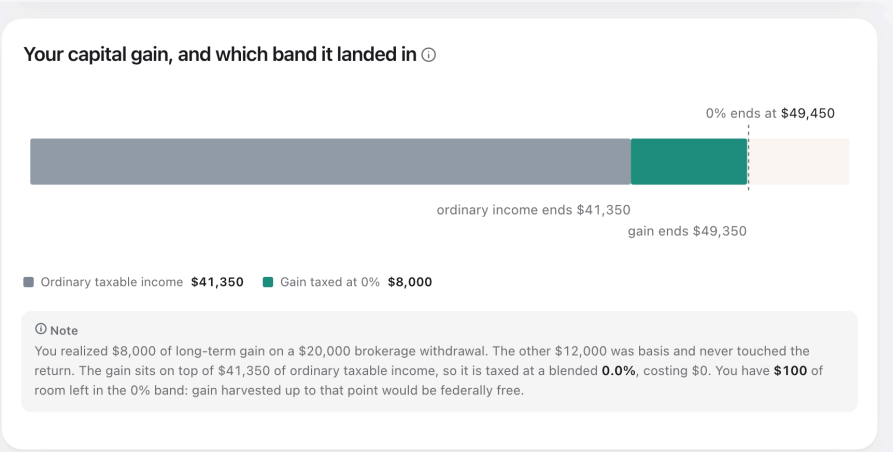
Retired. $100,000 from four sources, 65+.

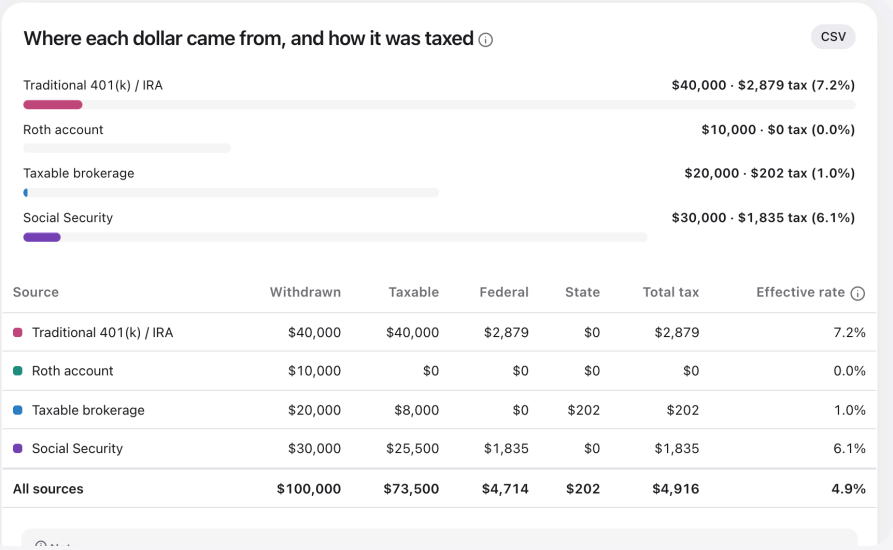
Covers

- Federal brackets, standard or itemized deduction
- Social Security tax per earner up to the $184,500 wage base, Medicare and the 0.9% surtax
- All 50 states plus DC, with their retirement-income rules
- Pre-tax deductions by kind and by earner: 401(k), IRA, HSA, premiums
- Optional local income tax, in the states that have one
- **Your paycheck:** gross pay down to take-home, per check and per month

Retirement mode adds

- Long-term gains stacked into the 0/15/20% bands
- The 3.8% net investment income tax
- Taxable Social Security (the 0/50/85% tiers)
- The 2026 senior deduction
- A true marginal rate, from rerunning the whole return with $1,000 more





Source	Withdrawn	Taxable	Federal	State	Total tax	Effective rate
Traditional 401(k) / IRA	$40,000	$40,000	$2,879	$0	$2,879	7.2%
Roth account	$10,000	$0	$0	$0	$0	0.0%
Taxable brokerage	$20,000	$8,000	$0	$202	$202	1.0%
Social Security	$30,000	$25,500	$1,835	$0	$1,835	6.1%
All sources	**$100,000**	**$73,500**	**$4,714**	**$202**	**$4,916**	**4.9%**

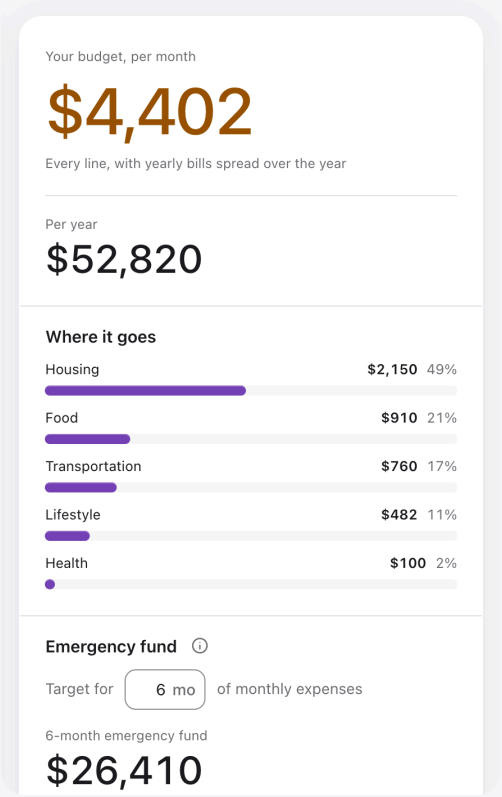
Budget

Where does my money go?

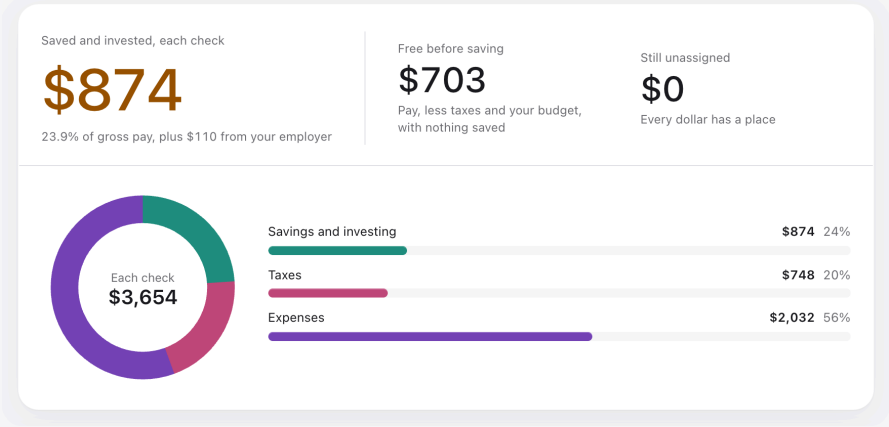
Lay out the expenses, then see how each paycheck covers them and where the rest goes. Two tabs: a Budget, and a Paycheck that turns gross pay into taxes, bills and savings, line by line.

12

retcalc.app/budget



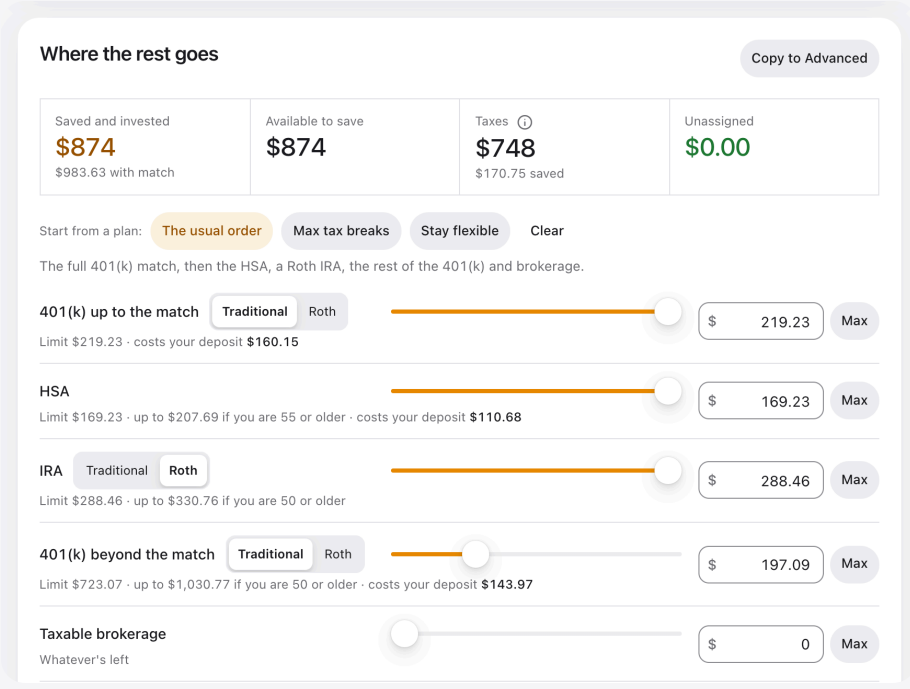
Budget tab. Each line monthly or yearly, grouped by kind, with an emergency fund sized from it.



Paycheck tab. Each check split into saving, taxes and expenses.

Three one-click plans

- **The usual order:** match, emergency fund, HSA, Roth IRA, more 401(k), brokerage
- **Max tax breaks:** everything pre-tax first
- **Stay flexible:** money reachable before retirement



Where the rest goes. Six lines, each capped at its 2026 limit; taxes rerun as the sliders move.

Two incomes

Married with two incomes plans by the month, taxes the couple jointly, and gives each spouse their own pay schedule, 401(k) match and HSA limit.

Connected

Opens in Income Tax; **Copy to Advanced** sends the savings split, by account type, to the Advanced calculator.

Debt Payoff

Which debt should I pay off first?

List what you owe, then compare avalanche (highest rate first) and snowball (smallest balance first) on payoff date and total interest, against paying only the minimums.

13

retcalc.app/debt

Debt-free

Oct 2030

3y 11m from now

Total interest

$8,155

on $49,400 borrowed, $57,555 paid in all

Saved vs. minimums

$10,167

6y 4m sooner

You get

- The debt-free date and total interest
- What it saves against paying minimums
- Avalanche against snowball, with a verdict
- The payoff order
- A blended rate, minimums that don't cover interest, and what $100 more per month would do

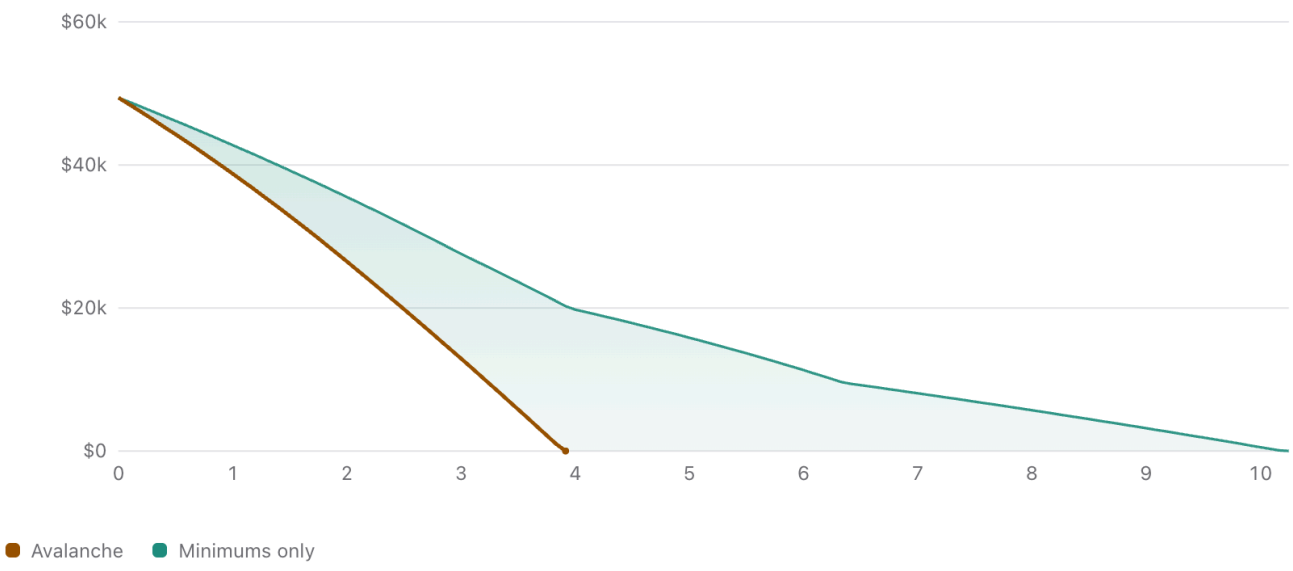
Avalanche vs. snowball

CSV

Avalanche costs $424 less in interest and finishes 1 month sooner. Snowball clears your first debt 1y 5m earlier, at a cost of $424 in extra interest. Your blended rate is 9.7% across 4 debts, and you're putting $1,230 per month at it. Another $100 per month would clear it 4 months sooner and save $929 more.

Approach	Debt-free	How long	Total interest	First debt gone
Avalanche, highest rate first ✓	Oct 2030	3y 11m	$8,155	1y 11m
Snowball, smallest balance first	Nov 2030	4 years	$8,578	6 months
Minimums only, no extra	Feb 2037	10y 3m	$18,321	3y 1m

What you owe, month by month



What you owe, month by month. Sample: a credit card, a store card, a car loan and a student loan, with $300 extra per month.

Mortgage Calculator

What will the house cost each month?

The full monthly payment, how the balance falls, and amortization year by year, with extra payments, a recast or a refinance to compare against the loan as it stands.

14

retcalc.app/mortgage

Extra payments and refinancing

Extra payments, recasting, or refinancing

Yes, show these options

Paying extra each month or once, recasting after a lump sum, and refinancing, each with its before and after.

Paying extra

Extra toward principal

$ 300 /mo

One-time extra payment

$ 10,000

In month

24

After that payment

Same payment, finish early

Refinancing

New rate

6.25 %

New length

30 years

Closing costs

$ 4,500

With the extra payments

Payoff

year 20, month 5

9y 7m sooner

Interest saved

$201,179

Refinancing to 6.25% for 30 years

New payment

$2,217/mo

Principal & interest, from $2,456/mo

Monthly change

Saves −$239/mo

Breaks even on closing costs

1y 7m

Over the life of the loan

Saves $39,855

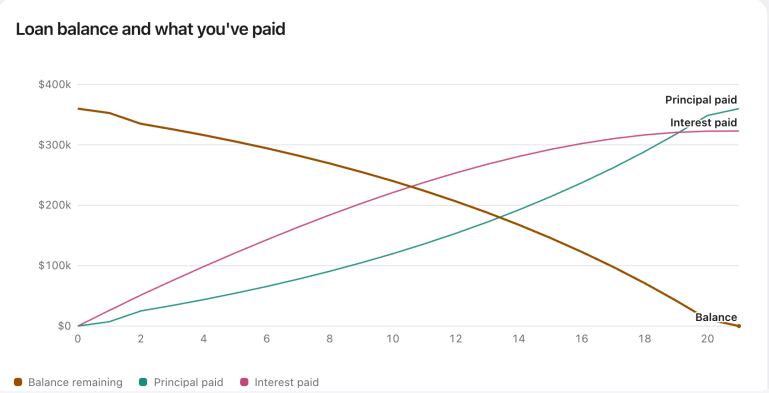
Sample: $450,000 at 7.25%, plus $300 per month, $10,000 in month 24, and a refinance to 6.25%.

What-ifs

- Extra every month, or a lump sum
- Recast, or keep the payment and finish early
- Refinance, with the break-even

Defaults

The rate starts at Freddie Mac's weekly survey, checked each quarter.



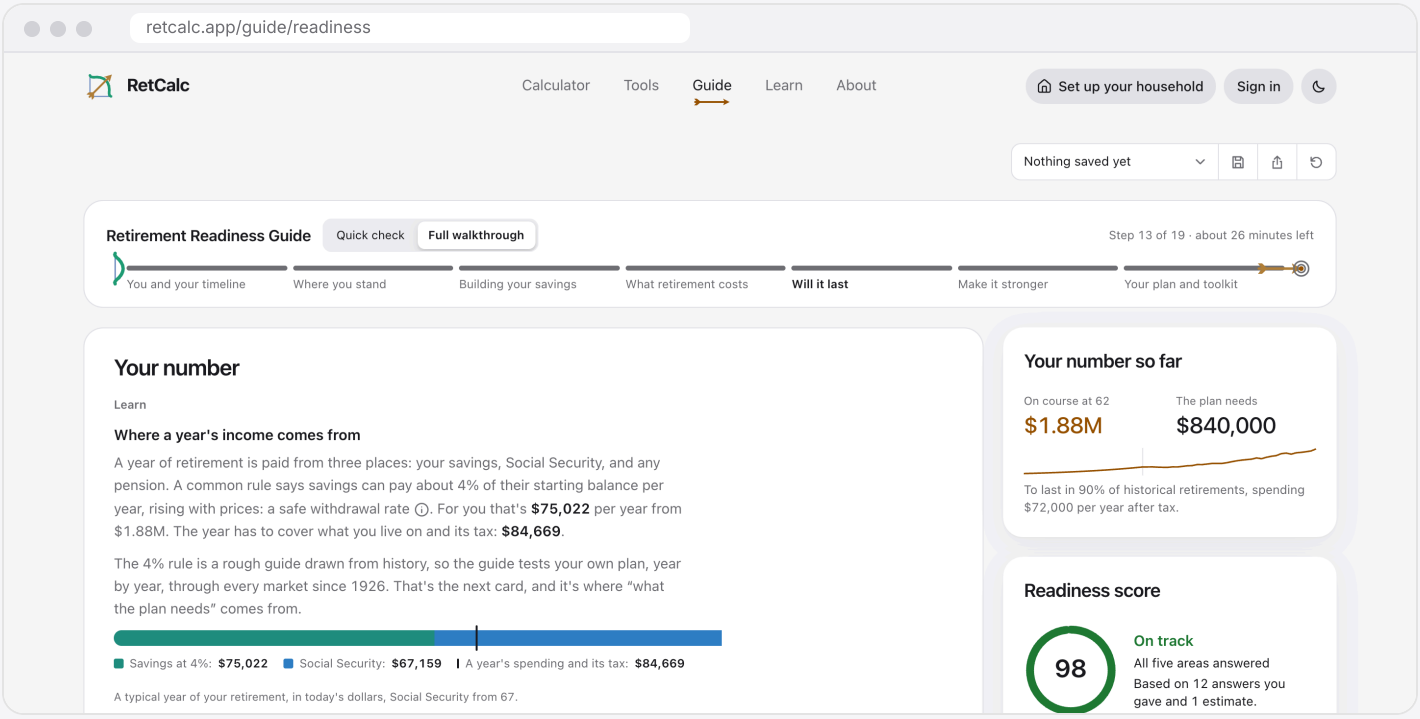
Retirement Readiness Guide

How ready am I, and what should I do next?

A guided check-up. It finds the number to aim for, tests the plan against every market since 1926, scores readiness out of 100 and lists the moves that matter, in the order that matters.

G1

retcalc.app/guide/readiness



Two paces

QUICK CHECK · ABOUT 10 MIN

Twelve short cards, three questions at most on each. The number by the sixth card, then a test of whether it lasts. Defaults stand in for the details, and every one says so.

FULL WALKTHROUGH · AN HOUR OR MORE

Every card, with trips into the tools that find the numbers you don't know: children, raises, a paid-off mortgage. It saves as you go.

Both start from the Guide tab, which opens on "Two ways through your plan": this check-up, or Paycheck to Retirement. Someone already retired can **Start from what you have**.

Seven chapters

01 You and your timeline
02 Where you stand
03 Building your savings
04 What retirement costs
05 Will it last
06 Make it stronger
07 Your plan and toolkit

The readiness score

Retirement outlook 40
Full marks when the plan lasted in 95% of history

Savings rate 20
15% of gross income, or a plan that already lasts

Emergency fund 15
Six months of spending in cash

Debt 15
Nothing owed at 8% or more, besides a mortgage

Monthly cash flow 10
Spend no more than 80% of take-home

On track at 85 and up, then Nearly there, Getting there, Needs work and Needs attention. A confidence line says how many answers were entered, estimated or defaulted.

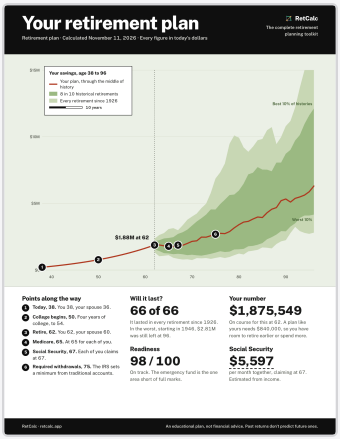
The plan, on paper

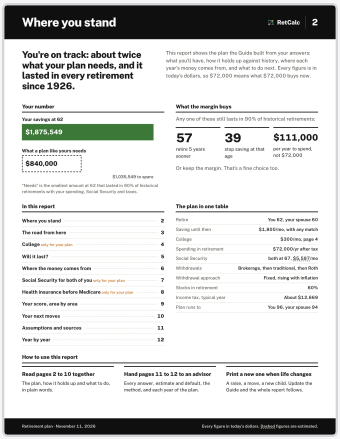
A full-color report in your own numbers.

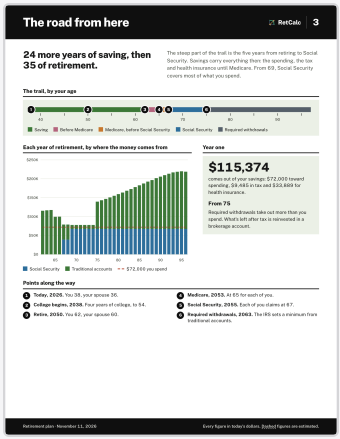
The Guide ends in a report of about a dozen US Letter pages, built in the browser and saved as a PDF: what you will have, how it holds up against every market since 1926, Social Security, taxes and health insurance, and the next moves. Below, the report for a sample couple, 38 and 36, retiring at 62.

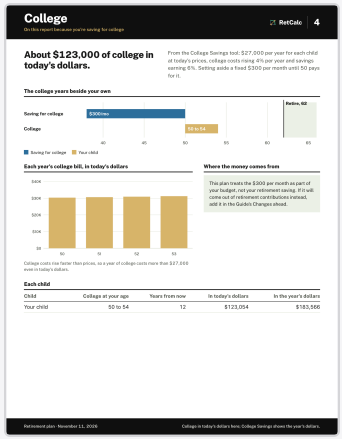
G1

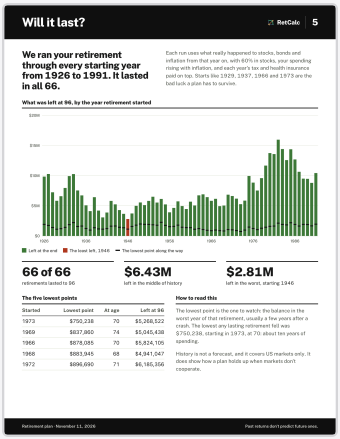
retcalc.app/guide/report

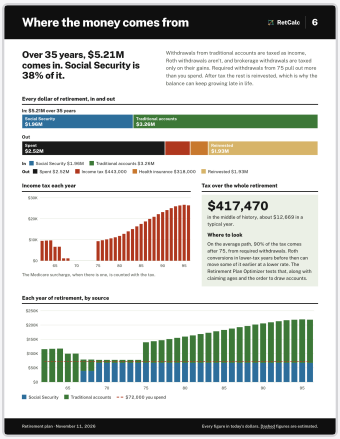


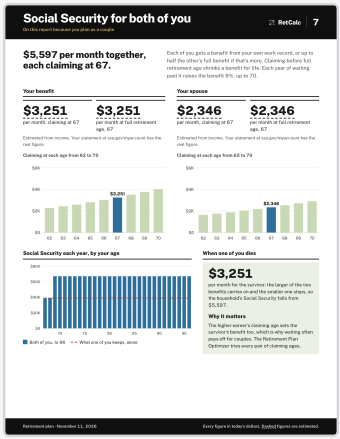


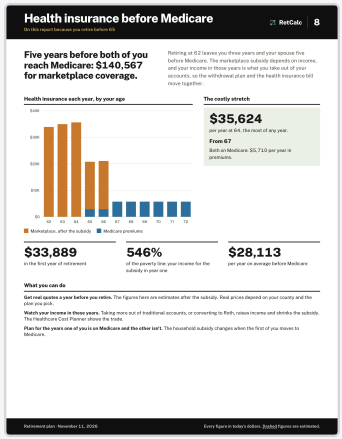


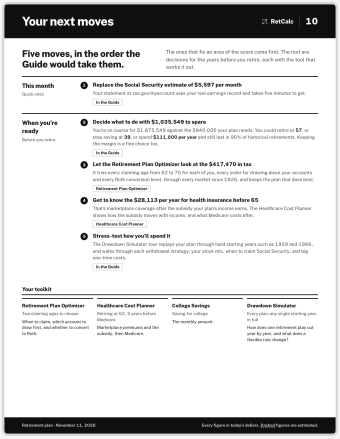


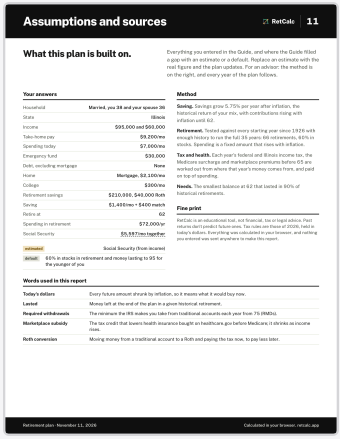


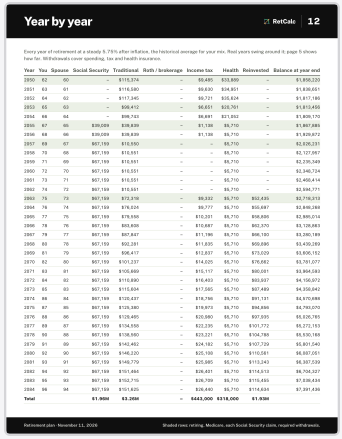












Paycheck to Retirement

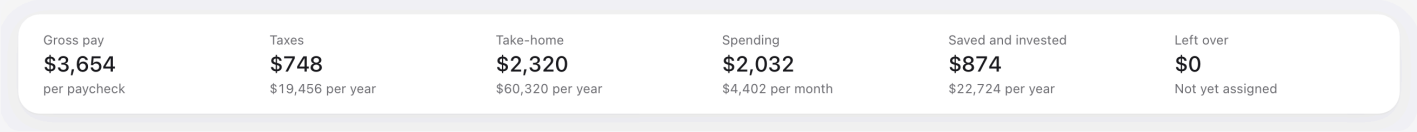
Where should each paycheck go?

For anyone who would rather get straight to the numbers. Five steps, about fifteen minutes: take-home pay after tax, a budget, where each paycheck goes, and where that saving leads by retirement, with any changes you see coming.

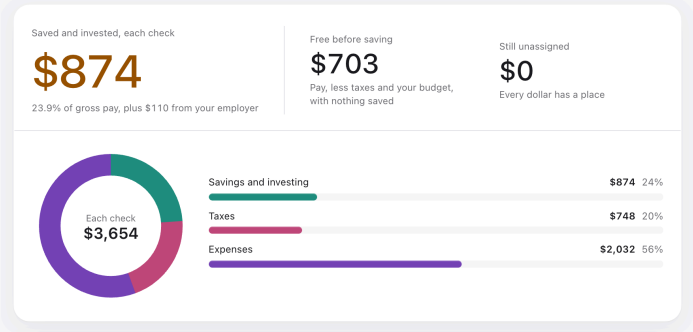
G2

retcalc.app/guide/paycheck

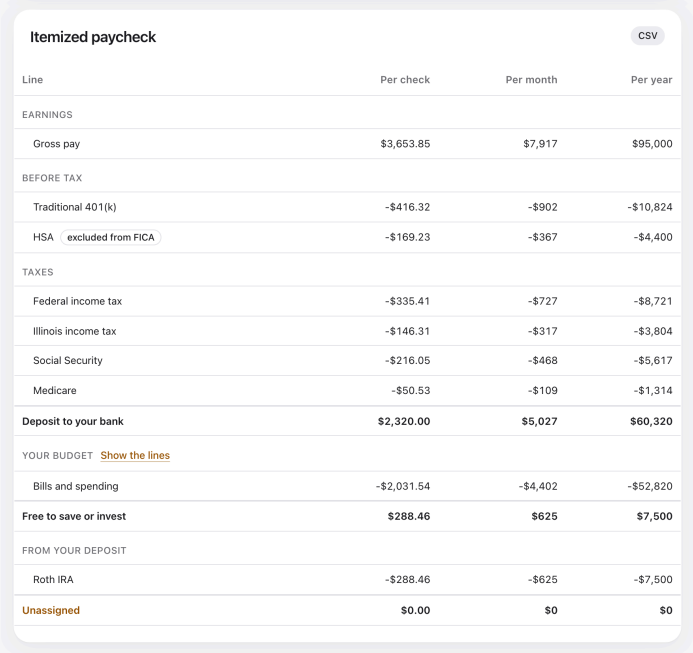
START	STEP 1	STEP 2	STEP 3	STEP 4	STEP 5
Start	**Budget**	**Paycheck**	**Your plan**	**Changes ahead**	**Summary**
Ages, filing status, pay and state	Spending, line by line	Tax, then 401(k), HSA, IRA, brokerage	The Advanced calculator, seeded	Optional: kids, a raise or a bigger house	Copy into my tools



The running totals stay pinned above every step: pay per check, taxes, spending, saving and what is left.



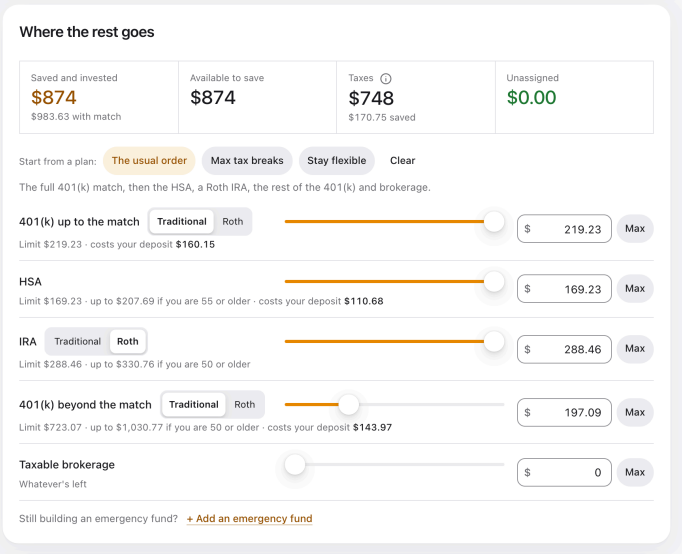
Step 2, Paycheck. Each check, split.



Itemized paycheck — CSV

Line	Per check	Per month	Per year
EARNINGS			
Gross pay	$3,653.85	$7,917	$95,000
BEFORE TAX			
Traditional 401(k)	-$416.32	-$902	-$10,824
HSA excluded from FICA	-$169.23	-$367	-$4,400
TAXES			
Federal income tax	-$335.41	-$727	-$8,721
Illinois income tax	-$146.31	-$317	-$3,804
Social Security	-$216.05	-$468	-$5,617
Medicare	-$50.53	-$109	-$1,314
Deposit to your bank	**$2,320.00**	**$5,027**	**$60,320**
YOUR BUDGET Show the lines			
Bills and spending	-$2,031.54	-$4,402	-$52,820
Free to save or invest	**$288.46**	**$625**	**$7,500**
FROM YOUR DEPOSIT			
Roth IRA	-$288.46	-$625	-$7,500
Unassigned	$0.00	$0	$0

The itemized paycheck, per check, per month and per year.



Where the rest goes. Pick a plan or move the sliders.

It runs the real tools

Budget and Advanced, with Income Tax's math, run inside the walkthrough on its own copy of the numbers, so nothing in the standalone tools changes until **Copy into my tools** sends the results to Income Tax, Budget and Advanced.

It ends in a report

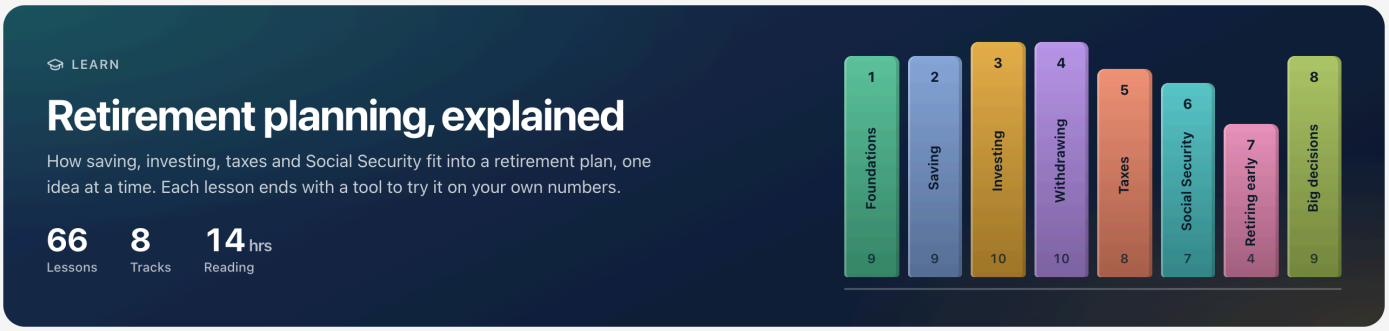
Cover, the paycheck, the spending, the saving, the road to retirement, the changes ahead if used, and the assumptions.

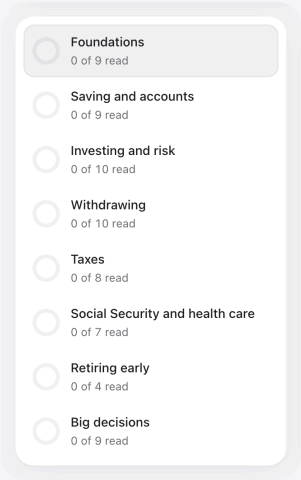
Learn

How does a retirement plan work?

Short lessons on how saving, investing, taxes and Social Security fit into a retirement plan, one idea at a time, about 14 hours of reading in all. Each lesson ends with a tool to try it on your own numbers.

retcalc.app/learn





The eight tracks, with what has been read.

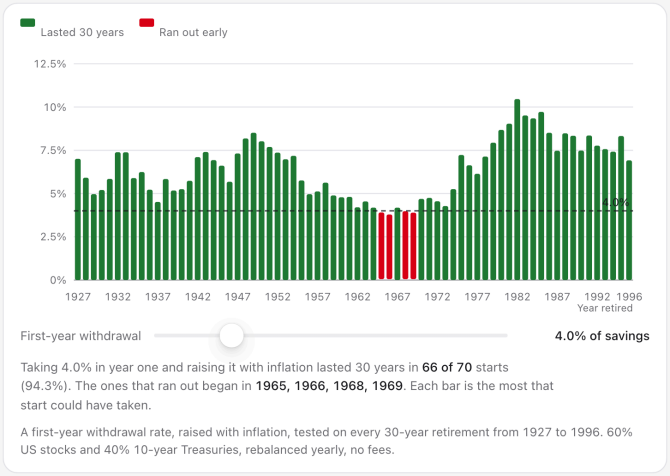
Eight tracks
Foundations (9 lessons), Saving and accounts (9), Investing and risk (10), Withdrawing (10), Taxes (8), Social Security and health care (7), Retiring early (4) and Big decisions (9). "Start here" opens with what a retirement plan answers.

In a lesson

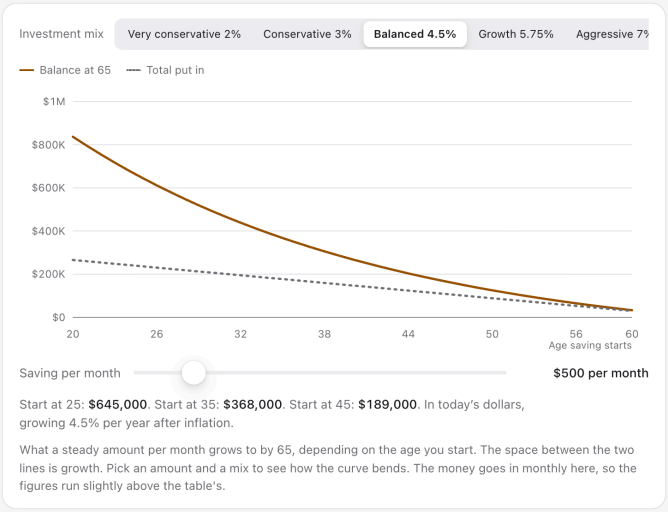
- The text beside **Your numbers**, **Key terms** and **Go deeper**
- **Try it**: the tool that applies the lesson, one click away
- **Mark as read** and **Next lesson**; progress follows a signed-in reader to every device

39 charts to try
One in each of 39 lessons. Every chart reads off its values on hover or touch.

Kept current
Each lesson shows the month it was last reviewed and is checked again every 90 days, and whenever a figure it quotes changes.



The 4% rule. The highest first-year rate that lasted 30 years, for every year retired from 1927 to 1996.



Compound growth. The balance at 65 by the age saving starts, for the investment mix picked.

ONE HOUSEHOLD, START TO FINISH

From a paycheck to a retirement that lasts.

A sample couple, carried through four tools by the tools' own buttons. Nothing was typed twice: each tool handed its answer to the next. Every figure on these two pages is from the site.

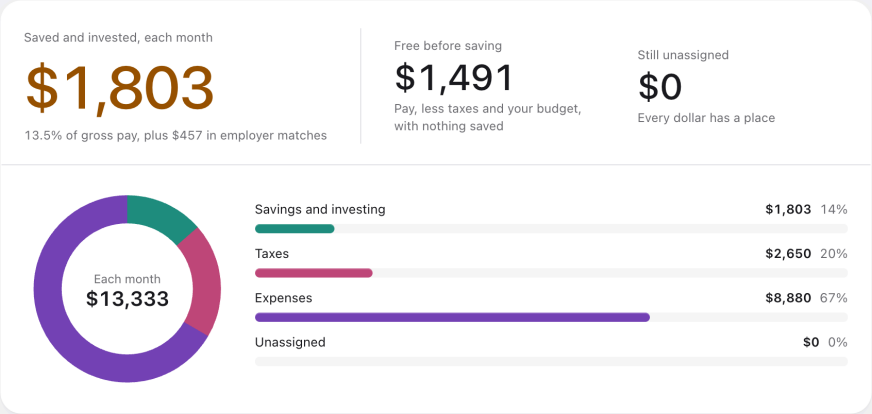
34 and 32	$160,000	$60,000	1	65
Alex and Jordan, in Illinois	$92,000 and $68,000 pay, per year	saved so far	child planned, in two years	when Alex retires, 31 years out

1 BUDGET

Where each paycheck goes

Their budget goes in line by line, then both paychecks on the Paycheck tab. One click on **The usual order** fills both 401(k) matches, an HSA, two Roth IRAs and more 401(k), and reruns their taxes as it goes.

They can save **$1,803 per month**, 13.5% of pay, plus $457 per month in employer matches.



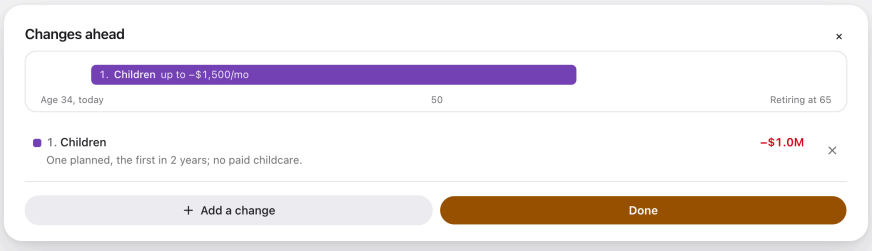
Copy to Advanced carries **$832.35 every two weeks**, split by account type: $316.92 traditional, $346.20 Roth and $169.23 HSA, plus the match.

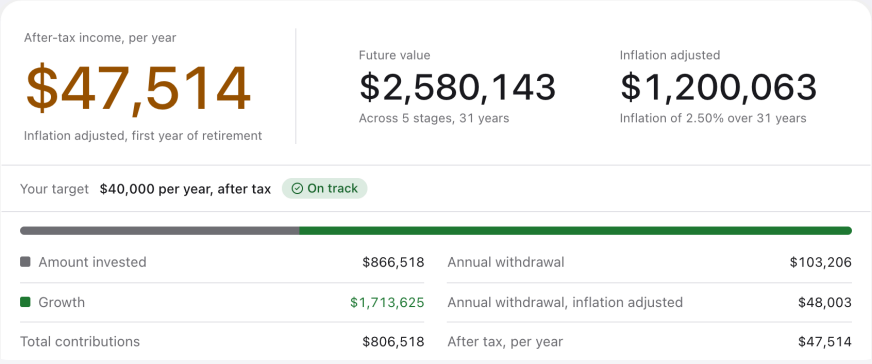
2 ADVANCED CALCULATOR

Saving around the child

Under **Changes ahead**, **Add a change** and **Children**: one child in two years, costing $1,300 to $1,500 per month as they grow. **How will you cover this?** Brokerage first, the recommended order.

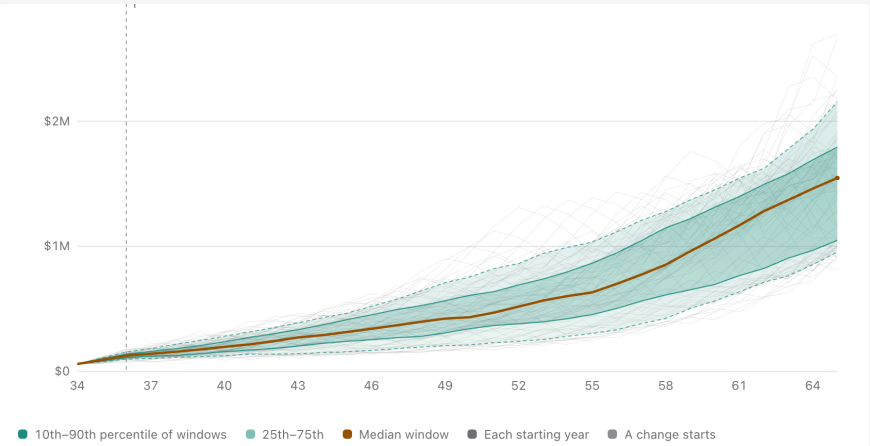
The child's costs come out of saving until the child is grown. At 65 that leaves **$1,200,063 in today's dollars**, about $47,514 per year after tax, against their $40,000 target: On track.





STEP 2, TESTED AGAINST HISTORY

The same plan run through all 829 31-year windows since 1926, with a dashed line where the child's costs begin. 83.4% of windows reach the $1,000,000 the target needs; the median finishes at **$1,546,552**, and the worst window, starting in August 1951, at $715,227.



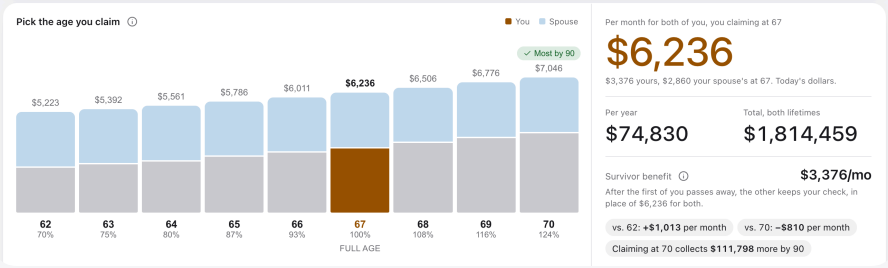
↓ Test withdrawals → Drawdown Simulator carries the portfolio: **$1,200,063**, the balance in today's dollars.

3 SOCIAL SECURITY CALCULATOR

When to claim

Born in 1992 and 1994, with the same pay. Claiming at 67, their full retirement age, brings **$6,236 per month** together: $3,376 for Alex and $2,860 for Jordan, in today's dollars.

Waiting until 70 would add $810 per month and more over a long life; the calculator shows by how much at every age.

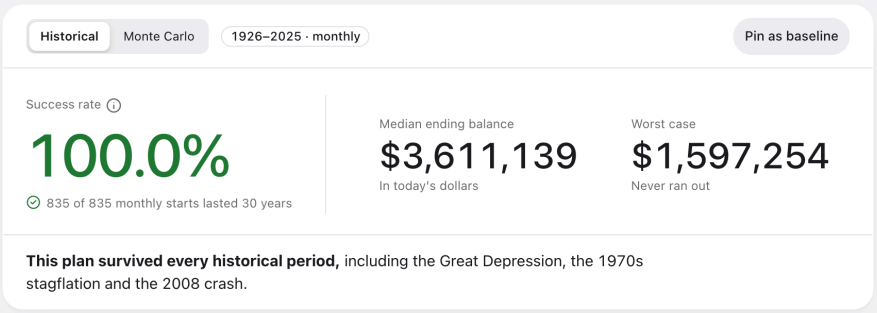


↓ Copy from Social Security Calculator carries both benefits into the Drawdown Simulator: **$74,829 per year** once both checks run.

4 DRAWDOWN SIMULATOR

Will it last?

The spending here is the whole household's: **$96,005 per year** ($8,000 per month), rising with inflation. Social Security pays $74,829 of it once both checks run; savings pay about $62,000 in the first two years and about $22,000 after. The plan is replayed through every monthly start since 1926, 835 of them, for 30 years each.



WHERE ALEX AND JORDAN LAND

$96,005

per year to spend in retirement, in today's dollars, with $74,829 of it from Social Security

835 of 835

historical retirements lasted the full 30 years, the Depression and the 1970s included

$3.6M

the median left after 30 years, in today's dollars; $1.6M in the worst start

A sample household, not advice. Returns before retirement are the plan's own assumptions; the historical tests use real monthly returns from 1926 to 2025.

PART 4

Under the hood.

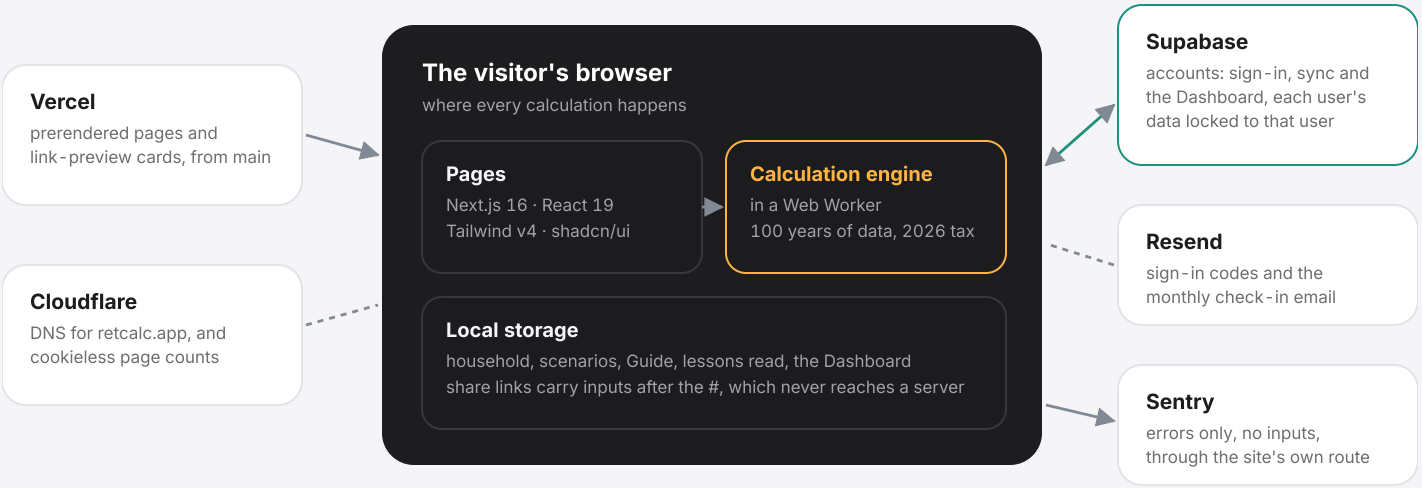
Next.js on Vercel, an engine that runs in the browser, accounts that sync and hold the Dashboard, and figures that each carry a source and a due date.

The stack	57
The calculation engine	58
Hosting and delivery	59
Accounts and privacy	60
Where the numbers come from	61

THE STACK

A static site that does its thinking in the browser.

Pages are prerendered and served from a CDN. Every number on them is worked out on the visitor's own machine, by a calculation engine that runs off the main thread. Accounts hold what people choose to save: their inputs, and My Dashboard, which needs one.



Solid lines carry data. Dashed lines are supporting services. No API routes: the site's own server receives only error reports, stripped of every input.

LAYER	WHAT IT USES
Framework	Next.js 16 (App Router), React 19, TypeScript 5
Look	Tailwind CSS v4 and shadcn/ui on Base UI primitives, Lucide icons; SF Pro on Apple devices and Inter, served by the site itself, elsewhere
Accounts	Supabase (Postgres with row-level security), reached from the browser with no app server between; Google or a 6-digit email code to sign in, no passwords
Engine	Plain JavaScript, type-checked through JSDoc, behind typed TypeScript bridges
Checks	ESLint with zero warnings allowed, tsc, @shadcn/lint, a calculation suite, unit tests, and end-to-end tests run by hand
Design	A written design system, Clear Glass: a light glass theme and a true-black night theme that follow the device, one amber accent, and one color for each kind of money

BY THE NUMBERS

53
page routes, 10 of them the Dashboard

~79,400
lines of app code

~8,100
lines in the engine and its data

66
lessons in Learn, plus 28 articles

The design rules are enforced, not just written: the linter rejects raw colors, one-off sizes and inline styles, and a new look has to be a component variant.

CLEAR GLASS, FROM THE DESIGN SYSTEM

Glass	Surface	Text	Amber
#f5f5f7	#ffffff	#1d1d1f	#975300

Night	Surface	Raised	Amber
#000000	#1c1c1e	#2c2c2e	#ffb340

Money map pay · tax · spending · saving · 401(k) · HSA · debt

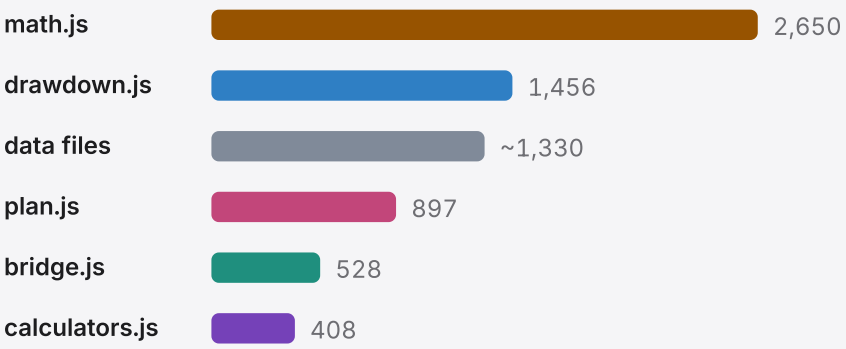
Type: **SF Pro · Inter**

THE CALCULATION ENGINE

About 6,100 lines of finance, and a test for every claim.

One engine serves every page: projections, Monte Carlo, federal and state income tax, FICA, Social Security, RMDs, Roth conversions, ACA subsidies, Medicare and IRMAA, mortgages, debt, rent against buy, college, FIRE, the 72(t) bridge and historical withdrawal backtests.

THE ENGINE, BY FILE (LINES)



Same seed, same chart

Every simulation shares one seeded random number generator (mulberry32), so a chart doesn't jump when an unrelated input changes. A **Re-roll** button moves the seed for all of them at once.

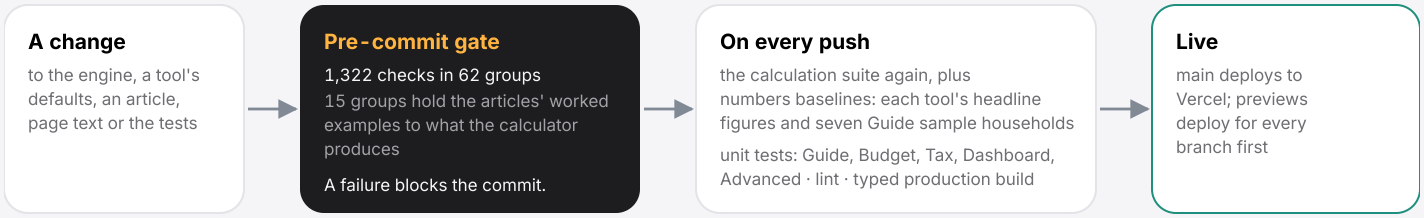
Off the main thread

A build step bundles the engine, its data and the Guide's planning jobs into one Web Worker, so a 300,000-retirement Optimizer run doesn't freeze the page.

THREE WAYS TO ASK "WHAT IF MARKETS DON'T COOPERATE?"

Historical replay	Every starting month (or January) since 1926, run forward through what actually happened. The Drawdown Simulator's 30-year test has 835 of them.
Monte Carlo, calculators	5,000 lognormal runs per redraw, set so the average matches the rate entered: with zero volatility it draws the plain projection exactly.
Monte Carlo, drawdown	A block bootstrap of real history from 1927. Whole runs of consecutive years are drawn together, so the 1930s and 1970s keep their streaks.

HOW A CHANGE GETS TO THE LIVE SITE



1,322 checks in the calculation suite, in 62 groups

15 of the groups tie the articles' worked examples to the engine

7 sample households whose Guide results are pinned

16 end-to-end browser specs of the public tools, run by hand

1,200 months of market history every engine walks

The Dashboard, too

33 unit tests in CI cover its changes, assumption ratings, bridge budget and cover rule; 13 recorded Dashboard plans are re-run against their saved results; 41 tests cover sync and merging, including a whole sync against a fake account.

HOSTING AND DELIVERY

Fast because there is almost nothing to serve.

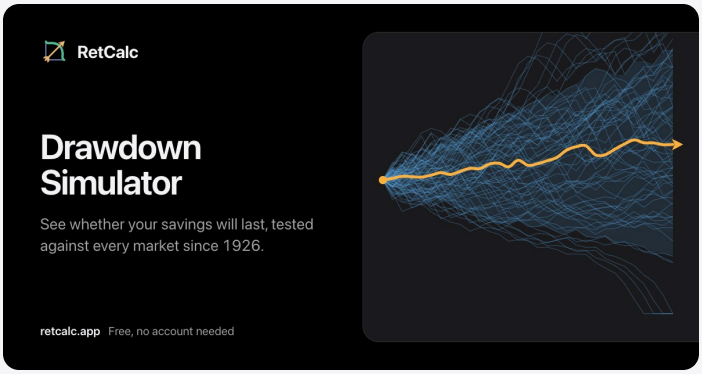
RetCalc runs on Vercel and deploys from the main branch. Pages are prerendered when the site is built, the homepage's and tool list's own figures included, so a visit is a handful of static files and a JavaScript engine that does the rest.

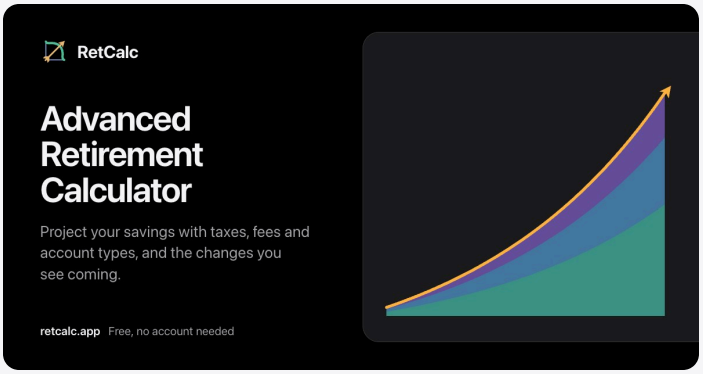
PIECE	HOW IT IS DONE
Hosting	Vercel. Production builds from main; every branch gets its own preview.
Rendering	Prerendered pages and articles. There are no API routes: all calculation happens in the browser, and the one thing the site's own server receives is error reports, stripped of inputs, which it passes on to Sentry.
Domain	retcalc.app, with DNS on Cloudflare; www redirects to the bare domain.
Analytics	Cloudflare and Vercel Web Analytics: cookieless page counts, production only. No ads and no ad trackers.
Email	Sign-in codes and the Dashboard's monthly check-in go out through Resend from the site's own domain. The check-in comes from a daily job on the account service and carries no figures.
Installable	A web app manifest and icons, so it can be added to a phone's home screen, with a bottom tab bar there.

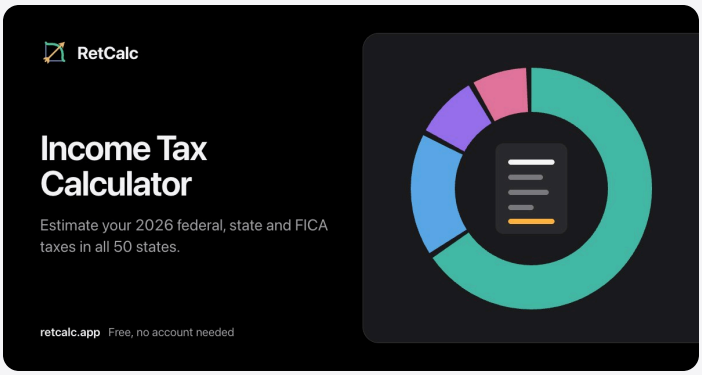
Error reports that can't leak a plan

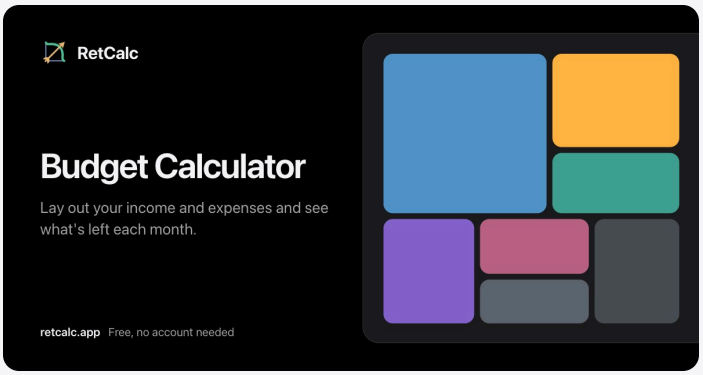
- Errors only: tracing, replay and debug code are compiled out
- Production only
- Addresses cut to the path, so the query and the # that holds a tool's inputs never leave the page
- No console breadcrumbs, user details or local variables
- Sent through the site's own route, so ad blockers don't drop them

LINK PREVIEWS, DRAWN FOR EVERY PAGE







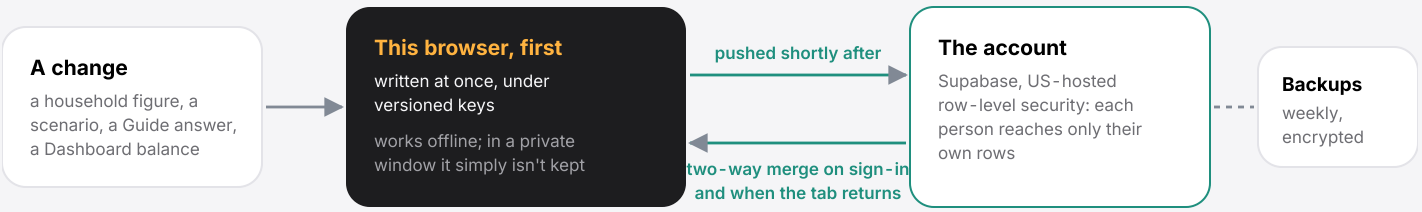


31 cards of 1200×630, one for each tool, guide and main page, in about 21 illustration types, so a link to any tool shows that tool.

ACCOUNTS AND PRIVACY

Local first. An account syncs, and holds the Dashboard.

The tools, both guides and Learn all work without signing in; a signed-in visitor gets the same site, with their household, scenarios, Guide and lessons read kept in step across devices. My Dashboard needs a free account, and it too is written to the browser first. Saving never waits on the network, and a sync never throws anything away.



A scenario changed on both sides is kept twice; a profile deleted on one device can't return. The merge is pure functions, unit-tested.

WHAT AN ACCOUNT STORES

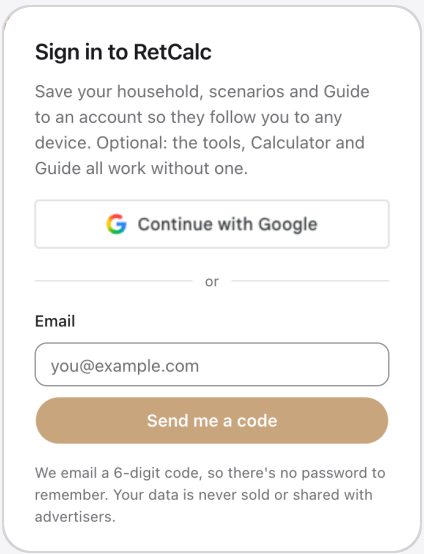
- Email, and a first name if given
- The household: ages, retirement age, state, savings, monthly saving, income, spending
- Saved scenarios, by tool
- Guide answers and plan, and the Paycheck plan
- The Dashboard, one or more profiles, and which lessons were read
- Whether the check-in email is off, and when each went out

WHAT IT NEVER STORES

- Numbers typed into a tool but not saved (they stay in the tab, and survive a reload)
- Browsing activity
- Bank logins: Dashboard balances are typed in by hand
- Anything for advertisers: there are none, and no data is sold or shared
- Share-link inputs, which ride after the # and never reach a server

SIGNING IN, AND OUT FOR GOOD

- A 6-digit code by email, or Continue with Google (whose script loads only when the sign-in box opens)
- Download everything, any time
- Delete the account: every device is signed out first, then the account and everything synced, the Dashboard included, is deleted. It can't be undone



Signing in, as it looks on the site.

No password to remember: a code by email, or Google. The dialog says up front that an account is optional and that data is never sold or shared.

The privacy page names every service that touches data (Supabase, Google, Resend, Vercel, Cloudflare and Sentry) and what each one sees. A dedicated test script checks the row-level security rules.

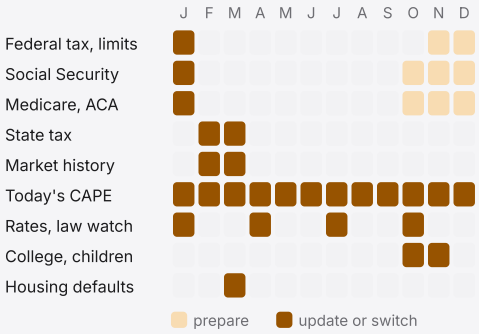
WHERE THE NUMBERS COME FROM

24 figures, each with a source and a due date.

Tax rules, Medicare and ACA prices, market history and sensible defaults all go stale. Each one is tracked in a list with its source, how often it changes, when it was last updated and when it is next due. Anything coming due is flagged at the start of every working session.

FIGURE	SOURCE	HOW OFTEN
TAXES		
Federal brackets, deductions	IRS Revenue Procedure (2025-32 for 2026)	Yearly
Contribution limits	IRS notices, incl. HSA limits	Yearly
Social Security figures	SSA's October fact sheet	Yearly
State income tax, 50 + DC	Tax Foundation's January survey, state revenue departments	Yearly
Local income tax	Tax Foundation; each state's revenue department	Yearly
State retirement rules	State revenue departments, checked against Tax Foundation and AARP	Yearly
RMD tables	IRS Publication 590-B	Yearly check
HEALTHCARE		
Medicare Part B, IRMAA	CMS fact sheet	Yearly
Part D, Medigap	CMS; KFF and insurer surveys	Yearly
ACA premiums, subsidies	KFF benchmark premiums, HHS poverty guidelines, IRS percentages	Yearly
MARKETS		
Stocks, bonds, inflation	Robert Shiller's data (Yale) via multpl; BLS CPI-U	Yearly
T-bills, small-cap value	Kenneth French Data Library (Dartmouth)	Yearly
Today's CAPE	multpl.com, Shiller PE	Monthly
HOUSING AND OTHER DEFAULTS		
Mortgage rate	Freddie Mac weekly survey	Quarterly
Home price and rent	NAR and Census, Zillow, Tax Foundation	Yearly
College and child costs	College Board; USDA, CPI-adjusted; Dept. of Labor childcare data	Yearly
72(t) rate, debt rates	IRS federal rates; Federal Reserve G.19	Quarterly, yearly

THE YEAR IN UPKEEP



THE RULES FOR EVERY UPDATE

- Figures are never invented. Each is cited in a comment beside it.
- A new tax year is added beside the old one, never over it, and Income Tax will then let you pick the year.
- After an update the whole check suite runs, and every expected move is reviewed before baselines are re-recorded.
- Articles and the 66 lessons are re-reviewed every 90 days and show the month.

RetCalc

Find out where you stand.

Free, private, and tested against every market since 1926.

retcalc.app

RetCalc is an educational tool, not financial, tax or legal advice. Past returns don't predict future ones, and tax figures are estimates. Screenshots are from the site as it stood on 11 November 2026, with sample households. Contact: contact@retcalc.app